Inference, Not Just Optimisation: Streaming Inversion-Free Quasi-Newton Estimation on Dependent Data Streams

Abstract

Streaming quasi-Newton methods fit a model in one pass, as cheaply as stochastic gradient descent but with the accuracy of Newton's method. Their error bars, however, assume that observations are independent, and real streams – traffic counts, air quality, sensor readings – are not. The resulting confidence intervals do not merely lose a little coverage: they converge to a wrong coverage and stay there. On the hourly air-quality stream studied here, a nominal 95% interval covers about 47%.

One change repairs the estimate and the interval together, within the same cost budget: read the stream in contiguous blocks. The block-averaged gradients such a method already computes are exactly the ingredients of a consistent estimator of the variance the intervals need, so the correct statistical object and the computational shortcut coincide – and the valid interval turns out cheaper than the invalid one it replaces. We prove consistency, a rate and a central limit theorem with the correct limit under geometric mixing, derive the exact coverage that dependence-blind intervals converge to, and report the one cost of blocking: a step-size condition it makes non-trivial, with a remedy. Across simulated designs and two real hourly streams, coverage rises from 0.46–0.83 to 0.78–0.95.

1 Introduction

Many estimation problems now arrive as a stream: observations appear one at a time or in small batches, they are too numerous to store, and a decision or a report is required at any moment. Stochastic approximation [78] is the natural tool, and averaged stochastic gradient descent [75] is asymptotically efficient with $O(d)$ work per observation. Its weakness is conditioning: when the curvature of the objective has eigenvalues spread over orders of magnitude, first-order methods crawl [8]. Second-order methods fix the conditioning but appear to cost $O(d^3)$ per step.

A line of work on *inversion-free* stochastic Newton methods removes that obstacle. When the Hessian has the form $\nabla^2 F(\theta) = \mathbb{E}[\alpha(\theta;\xi)\,\Psi(\theta;\xi)\Psi(\theta;\xi)^\top]$ — as it does for linear, logistic, softmax, Poisson and ridge regression, and for the geometric median — the running curvature estimate is a sum of rank-one terms, so its inverse can be maintained directly by the Sherman–Morrison rank-one update [83], also called the Riccati form, without ever forming or inverting a matrix [6, 33]. Godichon-Baggioni and Werge [34] push this to its natural conclusion: thinning the rank-one updates and taking mini-batches of size d gives an estimator whose *total* cost is $O(dN)$ — first-order cost — while remaining asymptotically efficient.

All of this theory assumes the stream is independent and identically distributed. The streams that motivate it are not. Hourly traffic volume, pollutant concentrations, queue lengths and user activity are serially dependent, and after any regression adjustment the *scores* remain serially dependent: in the two real streams we study, the lag-one residual autocorrelation after a fixed weather-and-calendar adjustment is 0.76 (traffic) and 0.92 (air quality). The consequence is not a small correction. Writing $g_i = \nabla_\theta f(\theta^\star;\xi_i)$

for the score at the truth, $\Gamma_k = \mathrm{Cov}(g_0, g_k)$, $H = \nabla^2 F(\theta^\star)$ and

$$S = \sum_{k\in\mathbb{Z}} \Gamma_k \qquad \text{(the \textit{long-run} score covariance)}, \tag{1}$$

any efficient streaming M-estimator satisfies $\sqrt{N}(\hat\theta_N - \theta^\star) \rightsquigarrow \mathcal{N}(0, H^{-1}SH^{-1})$, whereas the independent-data theory certifies $\mathcal{N}(0, H^{-1}\Gamma_0 H^{-1})$. The coordinatewise ratio of the two variances is a factor κ_j that does not shrink with N: we estimate it at about 2 for the traffic stream and about 20 for the air-quality stream, and at $\kappa_j = 3$ a nominal 95% interval covers 75% of the time *forever*, however long the stream runs.

This paper. We ask what has to change in a streaming inversion-free quasi-Newton method so that both its point estimate and its *uncertainty* are correct on a dependent stream, without leaving the $O(dN)$ budget. One structural choice does both jobs. Consume the stream in *contiguous blocks* of length b and take one preconditioned step per block; place curvature updates on a *gap* of m observations. Then the b-averaged gradients the algorithm forms anyway are, up to a factor b, the non-overlapping batch means of the score process, so they *are* an estimator of S — the statistically correct object and the computational shortcut coincide — and gapping decorrelates the rank-one curvature terms, which is what makes the inversion-free recursion behave, at matched cost, as it would on independent data. Neither device substitutes for the other: gapping alone leaves the variance wrong, blocking alone leaves the curvature recursion aggregating dependent terms. We call the resulting estimator BGSN (blocked–gapped streaming Newton).

Contributions.

1. **Theory for the curvature recursion under dependence** (Section 3): almost sure convergence, the rate $\|\theta_T - \theta^\star\|^2 = O(\log N/N)$, and a central limit theorem with the long-run sandwich $H^{-1}SH^{-1}$, for a rank-one Sherman–Morrison recursion with gapped updates on a geometrically mixing stream. The independent-data analysis rests on the score being a martingale difference sequence for the algorithm's filtration; dependence destroys that. A by-product (Proposition 7) is that blocking is *free*: the leading error term is $-N^{-1}H^{-1}\sum_{i\le N} g_i$ for every b.

2. **A long-run covariance estimator from block gradients, and why it is fast** (Section 4): S at $O(d^2)$ per block and $O(d^2)$ memory. What matters is *what* is blocked. Batch means over the *iterate* path need blocks long enough for the iterates to mix, forcing $b \asymp N^{3/4}$ and capping the rate at $N^{-1/8}$ [82]; block *gradients* inherit their dependence from the data, so $b \asymp \log N$ suffices and Theorem 9 gives $\widetilde{O}(N^{-1/2})$. This does not contradict the minimax rate for *Hessian-free* covariance estimation from an SGD trajectory [69]: a quasi-Newton method is not in that class.

3. **An exponentially small block-length bias** (Theorem 9): the two-scale (flat-top, or "lugsail") correction $\widehat{S}^{\mathrm{FT}} = \widehat{S}^{\mathrm{BM}} + b(\hat\Lambda_1 + \hat\Lambda_1^\top)$ has bias $O(\rho^b)$ rather than the $O(1/b)$ of plain batch means, which is what makes $b \asymp \log N$ admissible. That correction and its tail-sum property belong to the flat-top literature [73, 84, 89], not to us; what we add is that they are available at *block-gradient* resolution on a dependent stream, plus a streaming dyadic implementation needing no advance knowledge of N.

4. **Exactly how badly dependence-blind intervals fail** (Proposition 8): their asymptotic coverage is $2\Phi(z_{\alpha/2}/\sqrt{\kappa_j}) - 1$, and over a sweep in dependence strength the largest gap between prediction and measurement is 0.0085, with no fitted constants (Figure 1). This turns a previously reported qualitative failure into a quantitative one.

5. **A stability condition that blocking creates, and its fix** (Proposition 1): blocking makes the contraction condition $\gamma_t \|\bar{H}^{-1}\widehat{H}_t\|_{\mathrm{op}} < 2$ non-trivial, because on a dependent stream a block of b observations can span far fewer than b distinct covariate directions. We give the condition, an $O(d)$ step-length safeguard, and a proof that it changes no asymptotic statement once it stops binding. Without it our own method diverges on a regime-switching design. This is a cost of blocking, and it is ours.

6. **Curvature schedules** (Proposition 10) and **evidence** (Section 5): at a matched number of rank-one updates, deterministic gapping incurs excess curvature variance $O(\rho^m)$ against the $(\kappa_H - 1)/m$ of the Bernoulli thinning used in the independent-data method. Four dependent designs with exact population targets (three with closed-form S), plus hourly traffic and air-quality streams under two protocols, one with an exact ground truth on real covariates.

Much of what we build on is not ours, and Section 6 draws each boundary explicitly: the $O(dN)$ budget and curvature thinning, blocked mini-batches as a dependence-breaking device, the long-run sandwich limit and the efficiency bound under dependent sampling for first-order methods, online estimation of the instantaneous variance under independent data, and the two-scale correction are all prior work.

Code, data and reproducibility. Everything needed to reproduce every number, figure and table in this paper — the estimator, the experiment scripts, the saved results and the tests that validate every closed-form population quantity against long simulation — is archived and citable at `10.5281/zenodo.21891312` [46]. The *Data and code availability* statement at the end of the paper gives the exact release used here and the provenance of the two real data sets.

2 Setting and notation

We observe a strictly stationary stream $\xi_1, \xi_2, \ldots$ on $\mathbb{R}^p$ and wish to estimate

$$\theta^\star = \arg\min_{\theta \in \mathbb{R}^d} F(\theta), \qquad F(\theta) = \mathbb{E}[f(\theta;\xi)], \tag{2}$$

where $f(\cdot;\xi)$ is a loss. Write $g_i(\theta) = \nabla_\theta f(\theta;\xi_i)$ and $g_i = g_i(\theta^\star)$, so $\mathbb{E}[g_i] = 0$. Let $H = \nabla^2 F(\theta^\star)$, $\Gamma_k = \mathbb{E}[g_0 g_k^\top]$ and let S be as in (1). Throughout, $\Phi(x)$ is the standard normal distribution function, $z_{\alpha/2} = \Phi^{-1}(1-\alpha/2)$, $\|\cdot\|_{\mathrm{op}}$ is the operator norm, and N counts observations while T counts blocks. Five terms recur in their standard senses: the *score* is the gradient of the loss at one observation; an *M-estimator* minimises an empirical average of losses; a *sandwich* covariance has the form $H^{-1}\Sigma H^{-1}$, *instantaneous* when $\Sigma = \Gamma_0$ and *long-run* when $\Sigma = S$; *batch means* estimates a variance by averaging within consecutive blocks and taking the sample covariance of the block averages; and the strong (or α-) *mixing coefficient* at lag k is

$$\alpha_{\mathrm{mix}}(k) = \sup\left\{|\mathbb{P}(A\cap B) - \mathbb{P}(A)\mathbb{P}(B)| : A \in \sigma(\xi_i, i \le 0),\ B \in \sigma(\xi_i, i \ge k)\right\}, \tag{3}$$

a measure of how nearly independent the distant past and the distant future are; it vanishes for all $k \ge 1$ exactly when the stream is independent.

Nothing below requires the model to be correctly specified: all that is used is that $\theta^\star$ solves $\nabla F(\theta^\star) = 0$ for the *working* loss, so under misspecification $\theta^\star$ is the pseudo-true parameter and $H^{-1}SH^{-1}$ the corresponding long-run sandwich. What the problem does require is that the *scores* be serially correlated, not merely the covariates: if the conditional law of y given x is correctly specified with innovations independent over time then (g_i) is a martingale difference sequence and $S = \Gamma_0$ *exactly*, however persistent x is (Remark 7). All our designs therefore couple dependent covariates with a dependent error or latent process (Appendix E.7).

2.1 The algorithm

Blocks. Partition the stream into contiguous blocks $B_t = \{(t-1)b+1,\dots,tb\}$, $t = 1,2,\dots$, and write $\bar{g}_t(\theta) = b^{-1}\sum_{i\in B_t} g_i(\theta)$ for the block gradient and $\varepsilon_t = \bar{g}_t(\theta^\star)$ for the block score.

Curvature. Let $C = \{i \geq 1 : i \equiv 0 \bmod m\}$ be the *gapped* curvature slots and let $n_t = |C \cap [1, tb]|$ count the rank-one updates made by the end of block t. Maintain an unnormalised accumulator A and a weight W:

$$A_n = A_{n-1} + \iota_n e_{k_n} e_{k_n}^\top + \alpha_{i_n} \Psi_{i_n} \Psi_{i_n}^\top, \qquad W_n = W_{n-1} + 1, \tag{4}$$

where $\alpha_i = \alpha(\theta_{t-1}; \xi_i)$ and $\Psi_i = \Psi(\theta_{t-1}; \xi_i)$ for $i \in B_t$, e_k cycles through the canonical basis, and $\iota_n = c_\iota\, \varsigma_\star\, n^{-q_\mathrm{r}}$ with $q_\mathrm{r} \in (0, 1/4)$ is a vanishing ridge that keeps $\lambda_{\min}$ of the curvature estimate bounded below without any matrix operation. Its scale $\varsigma_\star$ is the mean curvature magnitude of the *first* block carrying curvature slots, $\varsigma_\star = (d\,|C \cap B_1|)^{-1} \sum_{i \in C \cap B_1} \alpha_i \|\Psi_i\|^2$, computed once and then frozen. Freezing matters: a scale tracking the *running* curvature would shrink exactly when the curvature estimate degenerated, and Lemma 2 would then presuppose its own conclusion (Appendix E.9). The curvature estimate and preconditioner are

$$\bar{H}_t = A_{n_t}/W_{n_t}, \qquad \bar{H}_t^{-1} = W_{n_t} A_{n_t}^{-1}, \tag{5}$$

and A^{-1} is updated by two rank-one Sherman–Morrison steps per curvature slot, each $O(d^2)$; no matrix is ever inverted. Maintaining A^{-1} rather than $\bar{H}^{-1}$ is what keeps every update exactly rank one despite the changing normalisation W.

Parameter step. With $A_0 = \lambda_0 I$, $W_0 = 1$ (λ_0 a small positive constant in the units of the curvature) and $\bar{H}_{t-1}$ built only from blocks $1,\dots,t-1$, the step is

$$\theta_t = \theta_{t-1} - \Pi_t\left[\frac{1}{t}\,\bar{H}_{t-1}^{-1}\bar{g}_t(\theta_{t-1})\right], \qquad \Pi_t[v] = v \cdot \min\left\{1,\ \frac{c_D\,(1 + \|\theta_{t-1}\|)}{\|v\|}\right\}, \tag{6}$$

where Π_t is a step-length safeguard, explained next, that is inactive whenever the step is of a reasonable size. The step $1/t$ is what makes the last iterate efficient without averaging; because it is preconditioned by $\bar{H}^{-1}$ there is no learning-rate constant to tune. A weighted averaged variant (step t^{-a}, $a \in (1/2, 1)$, with logarithmically weighted Polyak–Ruppert averaging) is available and reported alongside.

Why a blocked step needs a safeguard. Linearising (6) at $\theta^\star$ and ignoring Π_t makes the error recursion contract exactly when

$$\gamma_t\,\lambda_{\max}\big(\bar{H}_{t-1}^{-1}\widehat{H}_t\big) < 2, \qquad \text{for which} \qquad \gamma_t\,\|\bar{H}_{t-1}^{-1}\widehat{H}_t\|_{\mathrm{op}} < 2 \tag{7}$$

is sufficient, where $\widehat{H}_t = b^{-1}\sum_{i\in B_t} \alpha_i \Psi_i \Psi_i^\top$ is the Hessian of block t alone (Appendix E.2 derives this). On an *independent* stream with $b \geq d$ the block Hessian has full rank and is close to H, so $\|\bar{H}^{-1}\widehat{H}_t\|_{\mathrm{op}} \approx 1$ and (7) holds already at $\gamma_1 = 1$. On a *dependent* stream it need not: a block of b observations from a persistent stream spans far fewer than b distinct covariate directions, so $\widehat{H}_t$ is close to singular and the norm is inflated by roughly the ratio of the largest to the smallest curvature. Measured at $b = 20$, $\|H^{-1}\widehat{H}_t\|_{\mathrm{op}}$ has median 20 on our autoregressive design and 33 on our regime-switching design, so (7) is violated at $\gamma_1 = 1$ on *both*, and without a safeguard our own method's relative variance on the regime-switching design is 1.1×10^{14} times the efficient value against 1.41 with it, individual replicates diverging outright. Blocking the gradient is the one thing we add to the base method, so this failure is ours to fix and ours to report. The fix in (6) bounds each step relative to the current iterate, $\|\Pi_t[v]\| \leq c_D(1 + \|\theta_{t-1}\|)$ with

$c_D = 1$: scale-free in θ, $O(d)$ to apply, and *self-limiting*, in that it acts only on steps that are actually too long.

Proposition 1 (The safeguard is eventually inactive)**.** *Let $\mathcal{A}$ be the event that Π_t is active for only finitely many t. On $\mathcal{A}$ there is a finite $T_0(\omega)$ with $\theta_t = \theta_{t-1} - \gamma_t \bar{H}_{t-1}^{-1} \bar{g}_t(\theta_{t-1})$ for all $t > T_0$, so the safeguarded and unsafeguarded iterates coincide from T_0 on and every asymptotic statement about one holds for the other; in particular Theorems 4 to 6 and Proposition 7 are unaffected. Moreover $\mathbb{P}(\mathcal{A}) = 1$ whenever $\theta_t \to \theta^\star$ and $\|\bar{H}_t^{-1}\|_{\mathrm{op}} \bar{g}_{t+1} \to 0$ almost surely, because then $\|\gamma_t \bar{H}_{t-1}^{-1} \bar{g}_t\| \to 0$ while $c_D(1 + \|\theta_{t-1}\|) \to c_D(1 + \|\theta^\star\|) > 0$.*

We state Proposition 1 conditionally on $\mathcal{A}$ because that is what we can prove: we do not establish almost sure convergence for the safeguarded recursion from scratch, since the safeguard factor depends on the current block's gradient and so breaks the martingale structure the proofs use. What we do instead is measure $\mathcal{A}$ — across every design and sample size the safeguard binds between 2 and 7 times, and the count does not grow with N. Appendix E names the obstruction, gives the two routes that would close it, and reports the alternative remedy we implemented first and rejected.

Curvature warm-up. The first $\min\{\lceil c_{\mathrm{w}} d/b \rceil, \lfloor \varpi_{\mathrm{w}} N/b \rfloor\}$ blocks are spent on curvature updates only, with no parameter step; the second term caps the warm-up at $\varpi_{\mathrm{w}} = 0.1$ of the stream, which binds exactly when calling an $O(d)$-observation warm-up negligible would be false, as on our real-data segments. It costs $O(1)$ in N and changes no asymptotic statement, but it is not cosmetic: a $1/t$ step taken while $\bar{H}$ is still uninformative leaves a component decaying like t^{-c} with $c < 1$ that dominates the sampling error at realistic N and destroys coverage (Appendix E.5).

Cost. Per block after the warm-up: $O(bd)$ for the gradient, $O(d^2)$ for the $\lceil b/m \rceil$ curvature updates and the preconditioned step, and $O(d^2)$ for the covariance accumulators. With $b = m = d$ that is $O(d)$ amortised per observation, matching Godichon-Baggioni and Werge [34], and the warm-up adds a one-off $O(c_{\mathrm{w}} d^3)$, so

$$O\big(dN + c_{\mathrm{w}} d^3\big), \qquad \text{i.e. } O(dN) \text{ once } N \gtrsim c_{\mathrm{w}} d^2. \tag{8}$$

Memory is $O(d^2)$: the running inverse and two $d \times d$ accumulators. We state both plainly because it is easy to misread an $O(dN)$ *time* claim as an $O(d)$ *memory* claim; it is not, and the same is true of the method we build on (Appendix E.6).

3 Theory

3.1 Assumptions

Five assumptions are used, always together. They are stated in full in Appendix B.1 and summarised here. *Regularity* (Assumption 1): F is twice continuously differentiable with bounded Hessian, $H = \nabla^2 F(\theta^\star) \succ 0$, and $\nabla^2 F$ is Lipschitz near $\theta^\star$. *Expected smoothness and moments* (Assumption 2): the second moment of the score is controlled by the excess objective, with a $2 + 2\eta$ margin, and $\theta \mapsto \mathbb{E}[gg^\top]$ is continuous at $\theta^\star$. *Curvature form* (Assumption 3): $\nabla^2 F(\theta) = \mathbb{E}[\alpha \Psi \Psi^\top]$ with $\alpha \geq 0$, a uniform moment bound, and an L^2-Lipschitz condition in θ near $\theta^\star$. *Identifiability* (Assumption 4): $\theta^\star$ is the only stationary point of F and elsewhere the gradient points away from it, which holds whenever F is convex with a unique minimiser. *Geometric mixing* (Assumption 5): the stream is strictly stationary with $\alpha_{\mathrm{mix}}(k) \leq K c_{\mathrm{mix}}^k$ for some $c_{\mathrm{mix}} \in (0, 1)$, the score has $4 + \nu'$ moments, and $S \succ 0$.

Assumptions 1 to 3 are those of the independent-data analysis [34], with Assumption 3 strengthened to a local Lipschitz condition because the curvature terms are now dependent. Assumption 5 is the only genuinely new assumption, and Appendix B.5 lists the processes that satisfy it. Davydov's covariance inequality [18] converts it into the geometric bound on score autocovariances that we use repeatedly,

$$\|\Gamma_k\|_{\mathrm{op}} \;\le\; C_\Gamma\, \rho^{\,|k|}, \qquad \rho = c_{\mathrm{mix}}^{\,\nu/(2+\nu)} \in (0,1), \tag{9}$$

so that S in (1) converges absolutely.

3.2 The curvature recursion

The first result is where dependence bites. Under independence the terms accumulated in (4) form, conditionally on the iterates, an independent array and standard strong laws apply; under Assumption 5 they are dependent, and the gapped design is what restores control, because the subsequence indexed by C is itself stationary with mixing coefficients $\alpha_{\mathrm{mix}}(jm) \le K c_{\mathrm{mix}}^{jm}$.

Lemma 2 (The curvature estimate is never ill-conditioned)**.** *Under Assumption 3, with $\iota_n = c_\iota \varsigma_\star n^{-q_{\mathrm{r}}}$, $\varsigma_\star > 0$ fixed and $q_{\mathrm{r}} \in (0,1)$, there is a finite random $c_1 > 0$ with*

$$\lambda_{\min}(\bar{H}_t) \;\ge\; c_1\, n_t^{-q_{\mathrm{r}}} \qquad \textit{and hence} \qquad \|\bar{H}_t^{-1}\|_{\mathrm{op}} \;\le\; c_1^{-1} n_t^{q_{\mathrm{r}}} = O(t^{q_{\mathrm{r}}})$$

for all *t, almost surely.*

This is the whole purpose of the vanishing ridge, and it needs no assumption on the iterates: it makes $\sum_t \gamma_t^2 \|\bar{H}_{t-1}^{-1}\|_{\mathrm{op}}^2 = \sum_t O(t^{2q_{\mathrm{r}}-2}) < \infty$, the square-summability half of the Robbins–Monro condition, for *every* t rather than eventually, and without any eigenvalue computation.

Theorem 3 (Curvature rate)**.** *Under Assumptions 1, 3 and 5, suppose $\theta_t \to \theta^\star$ almost surely with $\|\theta_t - \theta^\star\| = O(t^{-1/2}\log^c t)$ a.s. for some $c < \infty$. Then for every $\nu_H < q_{\mathrm{r}}$,*

$$\|\bar{H}_t - H\|_{\mathrm{op}} = O(t^{-\nu_H}) \quad \textit{and} \quad \|\bar{H}_t^{-1} - H^{-1}\|_{\mathrm{op}} = O(t^{-\nu_H}) \qquad \textit{a.s.}$$

Theorem 3 presupposes a rate for the iterates while the results below use Theorem 3, so the logical order matters. It is a four-rung ladder, each rung using only the ones below it: Lemma 2 from the ridge alone, with no assumption on the iterates; then $\theta_t \to \theta^\star$ almost surely (Theorem 4); then a preliminary rate (Lemma 15), which is the one place the restriction $q_{\mathrm{r}} < 1/4$ is used; then Theorem 3, and after it Theorems 5, 6, 9 and 16. Appendix B.3 states the ladder in full and Appendix A is organised in that order.

3.3 Consistency, rate, and the central limit theorem

The independent-data proofs use that ε_t is a martingale difference with respect to $\mathcal{F}_{t-1} = \sigma(\xi_1, \ldots, \xi_{(t-1)b})$. Under Assumption 5 it is not, and no block length repairs this, because the conditional bias of a block average is $O(1/b)$ and does not vanish at fixed b. Our substitute is Gordin's martingale–coboundary decomposition $\varepsilon_t = m_{t+1} + z_t - z_{t+1}$ with (m_t) a stationary square-integrable martingale difference and $\mathbb{E}[m_1 m_1^\top] = S/b$ (Lemma 12); every weighted sum $\sum_t \gamma_t \bar{H}_{t-1}^{-1}\varepsilon_t$ then splits into a martingale that converges by Lemma 2 and a telescoping coboundary that converges because one rank-one Riccati update moves the preconditioner by only $O(1/t)$ (Lemma 14). This is what makes the argument work with weights that are *random and adapted* — unavoidable here, since they contain $\bar{H}_{t-1}^{-1}$ — and it is why the long-run covariance S, rather than the instantaneous Γ_0, is what appears in the limit: it is the second moment of the martingale part (Appendix B.9).

Theorem 4 (Almost sure convergence)**.** *Under Assumptions 1 to 5, the iterates (6) satisfy $\theta_t \to \theta^\star$ almost surely, for every fixed block length $b \geq 1$ and gap $m \geq 1$.*

Theorem 5 (Rate)**.** *Under the assumptions of Theorem 4, $\|\theta_t - \theta^\star\|^2 = O(\log(N_t)/N_t)$ almost surely, where $N_t = bt$.*

Theorem 6 (Central limit theorem with the long-run sandwich)**.** *Under Assumptions 1 to 5, for each* fixed *block length $b \geq 1$,*

$$\sqrt{N_T}\,(\theta_T - \theta^\star) \;\rightsquigarrow\; \mathcal{N}\Big(0,\; H^{-1}SH^{-1}\Big), \qquad N_T = bT,$$

with S as in (1). The same limit holds for the weighted averaged variant. If instead $b = b_N \to \infty$ with $b_N = o(\sqrt{N})$, the same limit holds provided the random constants Λ of Lemma 13 are bounded in probability uniformly in N along the sequence b_N.

We separate the two cases because only the first is proved unconditionally, and the difference matters: Theorem 9 takes $b \asymp \log N$, so the interval we actually recommend lives in the second case. Appendix B.4 identifies the step of the proof that needs the uniformity, shows why the naive route around it loses a factor $T^{1/2}$, and gives what we can say in its place — including that coverage is flat in b over $b \in [10, 320]$ at $N = 200{,}000$ (Table D.4(b)), which is the composition the two theorems need. Section 7 lists this as an open point rather than burying it.

Proposition 7 (Blocking is free)**.** *Under the assumptions of Theorem 6,*

$$\theta_T - \theta^\star = -\frac{1}{N_T}H^{-1}\sum_{i\leq N_T} g_i + R_T, \qquad \|R_T\| = o_p(N_T^{-1/2}),$$

so the asymptotic variance in Theorem 6 is the same for every block length b.

Proposition 7 matters for the design argument: averaging the gradient over a block costs nothing statistically, which is why we are free to choose b for the covariance estimator. It also answers the objection that single-sample streaming updates are more sample-efficient than batched ones [58] — true for a step size $\gamma_t \asymp t^{-a}$ with $a < 1$, and false here.

3.4 The variance-inflation factor and the exact failure of dependence-blind intervals

Proposition 8 (Coverage of dependence-blind intervals)**.** *Let*

$$\kappa_j = \big[H^{-1}SH^{-1}\big]_{jj} \Big/ \big[H^{-1}\Gamma_0 H^{-1}\big]_{jj},$$

and let $\widehat{\Gamma}_0 \to_p \Gamma_0$. Then the interval $\theta_{T,j} \pm z_{\alpha/2}\{[\bar{H}^{-1}\widehat{\Gamma}_0\bar{H}^{-1}]_{jj}/N\}^{1/2}$ has asymptotic coverage

$$2\,\Phi\big(z_{\alpha/2}/\sqrt{\kappa_j}\big) - 1.$$

The proof is one line and the content is entirely in κ_j: dependence-blind intervals do not merely lose a little coverage, their coverage converges to a computable number below the nominal level. For the homogeneous autoregressive design of Section 5 — covariates $x_t = \phi x_{t-1} + \cdots$, errors $u_t = \psi u_{t-1} + \cdots$, $x \perp u$ — textbook geometric summation gives

$$\kappa_j = \frac{1 + \phi\psi}{1 - \phi\psi} \quad \text{for every } j, \tag{10}$$

so at $\phi = \psi = 0.7$ a nominal 95% interval covers 74.9%. We present (10) only because it makes Proposition 8 checkable with no free parameters (Figure 1); the qualitative failure has been observed before [60], whose simulated coverage "hovers around 85%" at $\kappa = 2$, against our prediction of 83.4%.

4 Estimating the long-run covariance from block gradients

The algorithm forms $\bar{g}_t(\theta_{t-1})$ once per block and then discards it. Retain it for two $O(d^2)$ accumulations. Over a retained window $\mathcal{T}$ of T' blocks (see below), define

$$\widehat{S}^{\mathrm{BM}} = \frac{b}{T'} \sum_{t \in \mathcal{T}} \bar{g}_t \bar{g}_t^\top, \qquad \hat{\Lambda}_1 = \frac{1}{T'-1} \sum_{t,\, t+1 \in \mathcal{T}} \bar{g}_t \bar{g}_{t+1}^\top, \tag{11}$$

and then

$$\widehat{S}^{\mathrm{FT}} = \widehat{S}^{\mathrm{BM}} + b\big(\hat{\Lambda}_1 + \hat{\Lambda}_1^\top\big). \tag{12}$$

$\widehat{S}^{\mathrm{BM}}$ is the non-overlapping batch-means estimator at block resolution and $\widehat{S}^{\mathrm{FT}}$ the two-scale (flat-top, or "lugsail") correction, with the same expectation as the Richardson extrapolation $2\widehat{S}^{\mathrm{BM}}(2b) - \widehat{S}^{\mathrm{BM}}(b)$ (Appendix B.6). **This device is not new**: it is the flat-top window of Politis and Romano [73], recast as "lugsail" by Vats and Flegal [89], and Singh et al. [84, Eqs. (15) and (17)] apply exactly these two forms to stochastic gradient descent under independent data; nor is the $O(\rho^b)$ bias order a new mechanism, being the classical tail-sum property of a flat-top window. What we contribute is that it is available *at block-gradient resolution on a dependent stream*, where it is what makes $b \asymp \log N$ admissible and therefore what buys the rate, together with the streaming implementation below.

Confidence intervals. With $u_j = \bar{H}^{-1} e_j$ (one $O(d^2)$ product using the already-maintained inverse), report $\theta_{T,j} \pm z_{\alpha/2}\{u_j^\top \widehat{S}^{\mathrm{FT}} u_j / N\}^{1/2}$. Only the requested coordinates cost anything; the full $d \times d$ sandwich is never needed.

Retained window. Early blocks have θ_{t-1} far from $\theta^\star$ and inflate (11), and discarding a fixed fraction would require knowing N in advance. We instead keep two accumulators covering the current and previous dyadic windows $[2^k, 2^{k+1})$ and report from their union, so at any time at least half and at most three quarters of the blocks are retained and the discarded set is a prefix. This answers, in our setting, the open problem of an online implementation of equal-batch-size estimators raised by Singh et al. [84, Remark 4].

An L^2 counterpart of Theorem 5 is needed before the next result, because the drift analysis requires a moment bound and an almost sure rate with an unspecified random constant does not supply one: $\mathbb{E}\|\theta_t - \theta^\star\|^2 = O(\log(N_t)/N_t)$ uniformly over $b \le C \log N$ (Theorem 16).

Theorem 9 (Long-run covariance estimation)**.** *Let Assumptions 1 to 5 hold and let $b = b_N \to \infty$ with $b_N \le C \log N$. Write $\Delta = \sum_{k \ge 1} k\,(\Gamma_k + \Gamma_k^\top)$. Then, entrywise,*

(i) $\mathbb{E}\big[\widehat{S}^{\mathrm{BM}}\big] = S - \Delta/b + O(\rho^b) + O(\tau_{b,N})$*;*

(ii) $\mathbb{E}\big[\widehat{S}^{\mathrm{FT}}\big] = S + O(\rho^b) + O(\tau_{b,N})$*;*

(iii) $\mathrm{Var}\big[\widehat{S}^{\mathrm{FT}}_{jk}\big] = O(b/N)$*; and consequently*

(iv) $\|\widehat{S}^{\mathrm{FT}} - S\|_{\mathrm{op}} = O_p\big(d\sqrt{b/N} + \rho^b + \tau_{b,N}\big)$*, which at fixed d and $b \asymp \log N$ gives* $\|\widehat{S}^{\mathrm{FT}} - S\|_{\mathrm{op}} = \widetilde{O}_p(N^{-1/2})$*, whereas* $\|\widehat{S}^{\mathrm{BM}} - S\|_{\mathrm{op}} = O_p(d\sqrt{b/N} + 1/b)$ *is minimised at $b \asymp (N/d^2)^{1/3}$ and gives $O_p((d^2/N)^{1/3})$.*

The factor d in (iv) is the cost of converting the entrywise bound (iii) into an operator norm; we make no attempt to remove it, and every rate statement in this paper is at fixed d. Here $\tau_{b,N} = (b+d)\log^2 N/N$ is the contribution of the drift $\theta_{t-1} \ne \theta^\star$: the b comes from the parameter error and the d from the fact that the block curvature $\widehat{H}_t$ is a b-sample estimate of a $d \times d$ matrix, so that $\mathbb{E}\|\widehat{H}_t \delta\|_{\mathrm{op}}^2 \lesssim (1 + d/b)\|H\delta\|_{\mathrm{op}}^2$. Both are dominated by the $\sqrt{b/N}$ standard deviation whenever $b + d = o(\sqrt{N})$ up to logarithms. Consequently $\sqrt{N}\,(\theta_{T,j} - \theta_j^\star)/\{u_j^\top \widehat{S}^{\mathrm{FT}} u_j\}^{1/2} \rightsquigarrow \mathcal{N}(0,1)$ *and the intervals above have asymptotically exact coverage.*

Remark 1 (Why blocking gradients beats blocking iterates)**.** Batch-means estimators for stochastic approximation are usually built from the *iterate* path, and under dependent sampling that forces long blocks: the iterates move by $\gamma_k \asymp k^{-a}$, so a block must span enough parameter movement for its average to behave like an independent draw, which requires $b \asymp N^a$ and yields rates of order $N^{-1/8}$ [82, Prop. 2, Cor. 2]. Block *gradients* do not have this problem: their serial dependence is inherited from the data process, whose mixing time is a constant, and the iterate's own movement enters only through the $O(b/N)$ drift term in Theorem 9. This is the substantive reason for the rate difference, it is available only to an algorithm that forms block gradients and keeps a curvature estimate, and it is why the minimax rate for *Hessian-free* covariance estimation from an SGD trajectory [69] does not apply.

Remark 2 (Positive semidefiniteness)**.** $\widehat{S}^{\mathrm{FT}}$ is a difference of positive semidefinite matrices and need not be positive semidefinite; the same caveat attaches to all flat-top and lugsail estimators [84, Remark 5]. For marginal intervals only the scalars $u_j^\top \widehat{S}^{\mathrm{FT}} u_j$ must be positive, and we fall back to $u_j^\top \widehat{S}^{\mathrm{BM}} u_j$ if one is not, reporting the frequency of that fallback in every experiment.

Proposition 10 (Gapping versus Bernoulli thinning at matched cost)**.** *Let $Y_i = \alpha(\theta^\star;\xi_i)\Psi_i\Psi_i^\top$, $\Gamma_k^H = \mathrm{Cov}(Y_0, Y_k)$, $S^H = \sum_k \Gamma_k^H$ and $\kappa_H = \|S^H\|_{\mathrm{op}}/\|\Gamma_0^H\|_{\mathrm{op}}$. From a stream of length N, take n rank-one curvature updates either at a deterministic gap $m = N/n$ or by Bernoulli($p = n/N$) thinning of every index. Then, as $n \to \infty$,*

$$n\,\mathrm{Var}\left[\widehat{H}^{\mathrm{gap}}\right] = \Gamma_0^H + O(\rho^m), \qquad n\,\mathrm{Var}\left[\widehat{H}^{\mathrm{Bern}}\right] = \Gamma_0^H + \tfrac{1}{m}\left(S^H - \Gamma_0^H\right).$$

So the excess over the independent-data benchmark Γ_0^H/n is exponentially small in m for gapping and of order $(\kappa_H - 1)/m$ for Bernoulli thinning. Gapping also makes the per-block work deterministic. We verify both statements in Figure 3. We are explicit that this is a curvature-side improvement: $\bar{H}$ enters the limit distribution only through the rate ν_H of Theorem 3, so the end-to-end effect on $\hat{\theta}$ is second order, and we do not claim otherwise.

5 Experiments

Every number quoted below is generated from the saved results by the script `make_tables.py` and injected as a macro, so the prose cannot drift from the data. Monte Carlo standard errors are shown or stated everywhere.

Designs and methods. Four simulated dependent designs with $d = 20$, specified in Appendix C.1: **AR-hom** (autoregressive covariates and errors, closed-form S, the same inflation $\kappa = 2.92$ in every coordinate); **AR-het** (coordinatewise persistence, closed-form S, κ_j over $[1.00, 4.41]$, so a scalar correction cannot pass); **Markov** (a Markov-modulated Gaussian mixture with $d+1$ regimes, non-Gaussian with a discrete latent state, closed-form S); and **Logistic** (a correctly specified marginal logistic model with a serially dependent latent error, so $\theta^\star$ is exactly the generating value while the score is autocorrelated). Covariate covariances use the ill-conditioned design of the base paper (cond(H) up to 400); Table D.3 adds a well-conditioned counterpart and Table D.2 two real hourly streams. Appendix C.6 lists the eleven methods compared and their tuning; the comparisons that matter here are against the same point estimator with plain batch means or the i.i.d. plug-in variance, so that only the interval changes, the base independent-data streaming Newton method, averaged SGD and streaming AdaGrad given the *true* Hessian for their sandwich and tuned on disjoint pilot streams, and an infeasible oracle-variance interval built from the true H and S. BGSN has no learning rate to tune.

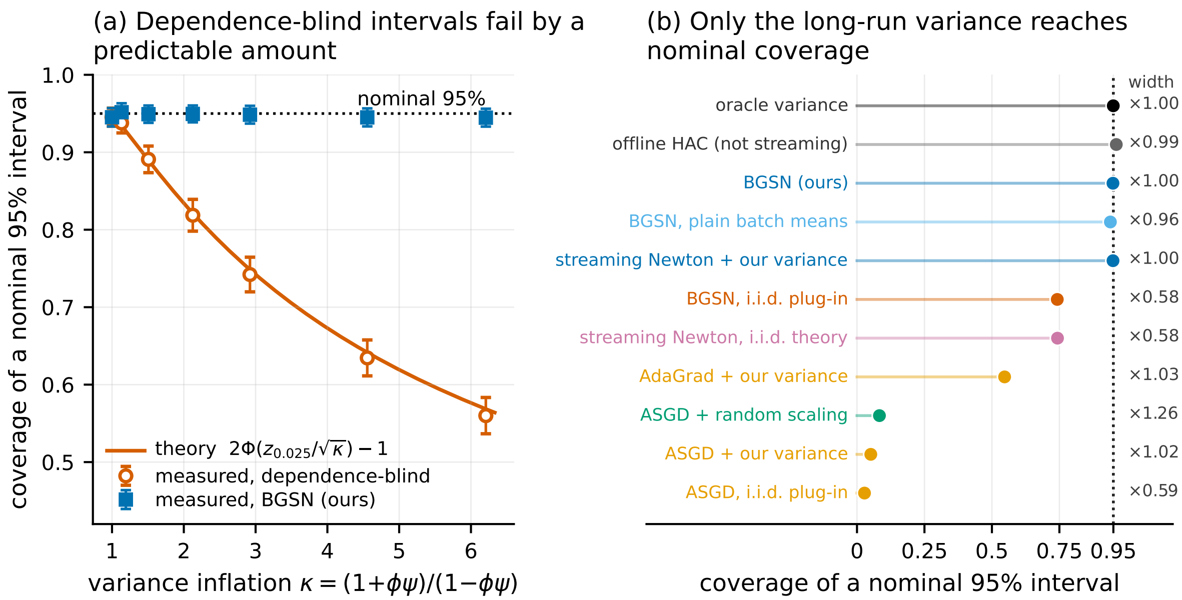


Figure 1: Dependence-blind confidence intervals fail by a computable amount, and only estimating the long-run score covariance repairs them. (a) Coverage of nominal 95% intervals on the homogeneous autoregressive design as the dependence strength varies, against the parameter-free prediction of Proposition 8; error bars are ± 1.96 Monte Carlo standard errors. (b) Coverage of every method on that design at $\phi = \psi = 0.7$ ($\kappa = 2.92$), one row per method; the dotted line is the nominal level and the number at the right is the interval width relative to the infeasible oracle interval. Read it in three groups. The five rows at the top all account for the autocovariances of the score — the infeasible oracle, the offline reference, and the three streaming estimators that use our long-run covariance — and they are the rows that reach nominal, at width ≈ 1. The next two use the instantaneous covariance the independent-data theory prescribes; they are narrower by the factor $\sqrt{\kappa} = 1.71$ and they miss, by the amount Proposition 8 predicts. The bottom four are first-order methods whose *point* estimates have not converged on this ill-conditioned design (their $\sqrt{N}$ error is 1.03 to 9.77 times the efficient value), so their coverage is dominated by that and not by which covariance they use; Table D.3 is where they become informative about inference.

5.1 The failure is exactly as predicted, and it is not small

Figure 1(a) sweeps the dependence strength, comparing the coverage measured for dependence-blind intervals against the parameter-free prediction $2\Phi(z_{0.025}/\sqrt{\kappa}) - 1$ of Proposition 8. For κ up to 2.9 the largest absolute discrepancy is 0.0063, of the order of the Monte Carlo standard error; BGSN stays at nominal over that range (0.949 at $\kappa = 2.92$, where the dependence-blind prediction is 0.748 and its measurement 0.742 ± 0.0114), including at $\phi = \psi = 0$, where all four intervals agree — the control showing that the machinery is not simply widening intervals. At the strongest dependence we tried the formula's residual gap is finite-sample, not a failure of the formula, and the infeasible oracle interval says so: at $\kappa = 6.2$ it covers only 0.946, and BGSN covers 0.945, tracking the oracle rather than the formula. That comparison is the general point. BGSN's coverage tracks the infeasible oracle to within Monte Carlo error at every dependence level, so whatever undercoverage remains is attributable to the central limit theorem at finite N and not to our variance estimator, whereas plain batch means fall 1–3 percentage points *below* the oracle once $\kappa \geq 4$ — exactly the $O(1/b)$ bias the two-scale correction removes.

Table 1 gives the full picture at a common N. On AR-hom the base method's interval covers 0.743 and ours 0.949 against a nominal 0.95, with the infeasible oracle at 0.950. The price is width: our interval is $1.71\times$ the dependence-blind one, which is exactly $\sqrt{\kappa}$ — there is no free lunch, the dependence-blind

Table 1: Coverage of nominal 95% confidence intervals, interval width, and estimation error on four dependent designs ($d = 20$, $N = 200{,}000$, block length $b = 20$, $R = 300$ replications; $R = 60$ for the offline reference). *cov*: coverage, pooled over the d coordinates (Monte Carlo standard error ≤ 0.008 throughout). *err*: $\sqrt{N}$ RMSE divided by the asymptotic standard deviation, averaged over coordinates; 1.00 is efficient. κ is the variance inflation caused by dependence. Only methods that estimate the *long-run* score covariance reach nominal coverage; methods using the instantaneous covariance that the independent-data theory prescribes undercover by an amount predicted exactly by Proposition 8. Interval widths are not repeated here; they are in Figure 1(b). **Read the first-order rows with care**: these designs are ill-conditioned (that is what streaming second-order methods are for), so those methods' *point* estimates have not converged — their *err* column is several times 1 — and their coverage is dominated by that, not by which covariance they use. Table D.3 repeats the AR-hom column on a well-conditioned design where every point estimator is efficient, which is where the first-order rows become informative about inference. The Markov column is the one design where *no* streaming method's point estimate has converged at this N; Table D.1 resolves it by sweeping N against the oracle interval. ASGD is averaged stochastic gradient descent (Polyak–Ruppert averaging of the iterates); AdaGrad is the diagonally rescaled first-order method; *random scaling* is the variance-estimator-free studentisation of Lee et al. [54].

	AR-hom		AR-het		Markov		Logistic	
method	cov	err	cov	err	cov	err	cov	err
dependence	$\kappa \in [2.92, 2.92]$		$\kappa \in [1.00, 4.41]$		$\kappa \in [7.55, 7.55]$		$\kappa \in [2.27, 2.32]$	
BGSN (this paper)	**0.949**	1.01	**0.950**	1.01	0.890	1.27	0.783	1.33
with plain batch means	**0.939**	1.01	**0.944**	1.01	0.856	1.27	0.768	1.33
with the i.i.d. plug-in variance	0.742	1.01	0.832	1.01	0.460	1.27	0.581	1.33
streaming Newton, i.i.d. theory	0.743	1.01	0.834	1.01	0.460	1.27	0.584	1.33
with our long-run variance	**0.949**	1.01	**0.951**	1.01	0.890	1.27	0.781	1.33
ASGD + our long-run variance	0.051	9.77	0.271	50.34	0.239	7.95	0.714	1.91
ASGD, i.i.d. plug-in	0.028	9.77	0.195	50.34	0.073	7.95	0.520	1.91
ASGD + random scaling	0.083	9.77	0.679	50.34	0.230	7.95	0.925	1.91
AdaGrad + our long-run variance	0.547	2.99	**0.947**	1.03	0.533	2.76	0.911	1.15
offline HAC *(not streaming)*	**0.961**	0.95	**0.955**	0.98	0.968	0.85	**0.945**	1.03
oracle variance *(infeasible)*	**0.950**	1.01	**0.949**	1.01	0.886	1.27	0.858	1.33

interval is simply the wrong length. AR-het is worth a separate look, because its κ_j varies across coordinates by construction: the dependence-blind interval's *per-coordinate* coverage ranges from 0.63 to 0.96 while ours ranges from 0.92 to 0.98. No scalar inflation of the width could produce the first pattern from the second; the whole matrix has to be estimated. Crucially, replacing *only* the variance estimator repairs the base method (row "with our long-run variance"), and replacing only the point estimator does not: this isolates inference as the operative component.

Three further results are reported in Appendices D.1 to D.3: that on the two designs whose point estimate has not yet reached its asymptotic regime our interval tracks the one built from the exact $H^{-1}SH^{-1}$ at the same iterate while the dependence-blind interval does not; that on a well-conditioned design, where every point estimator is efficient and only the covariance can decide coverage, our long-run variance is what repairs averaged SGD (0.931 against 0.721); and that the Kolmogorov–Smirnov distance from our studentised error to the standard normal is 0.089 on AR-hom against 0.207 for the dependence-blind studentisation, so the whole distribution and not merely one quantile is repaired.

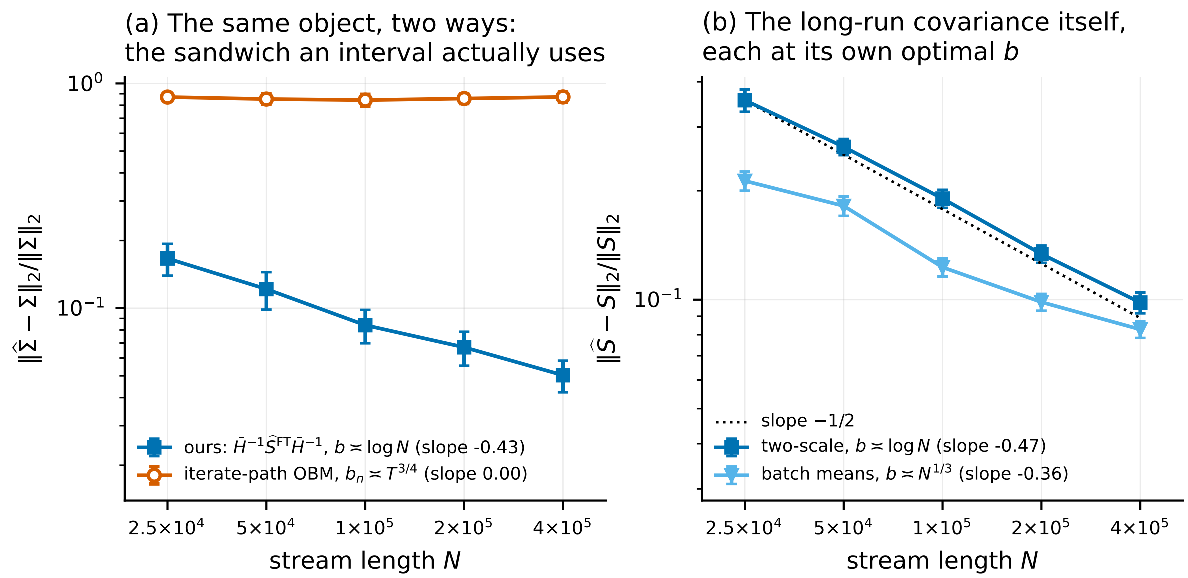
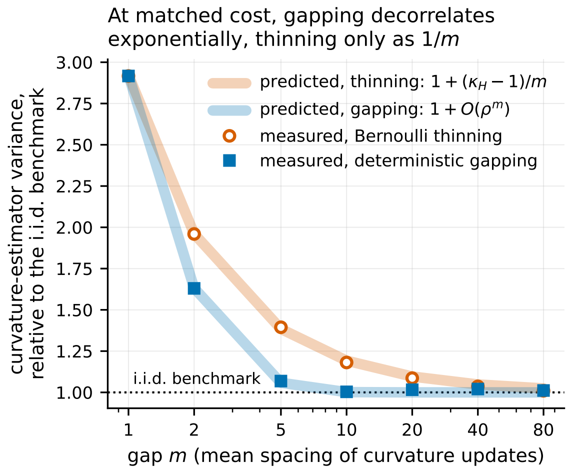


Figure 2: The rate claim, measured, on the object an interval actually uses. Both axes logarithmic; error bars are ±1.96 Monte Carlo standard errors; fitted slopes in the legends. **(a)** Relative operator-norm error in estimating the whole sandwich $\Sigma = H^{-1}SH^{-1}$. Ours is $\bar{H}^{-1}\widehat{S}^{\mathrm{FT}}\bar{H}^{-1}$ at $b \asymp \log N$; the comparator is overlapping batch means on the *iterate* path at the block length $b_n \asymp T^{3/4}$ that analysis requires (Appendix C.3 gives the form), which uses no gradients and no curvature estimate. Same target, so the comparison is meaningful; but read the levels, not the slopes. The iterate-path curve is flat here, and its published $N^{-1/8}$ rate is too slow for our range to resolve, so its fitted slope says nothing about it. What the panel shows is that over the sample sizes we can reach, one estimator is accurate and the other is not. **(b)** Error in estimating S alone, each estimator at its own optimal block length: the two-scale form at $b \asymp \log N$ against the $-1/2$ of Theorem 9(iv), plain batch means at $b \asymp N^{1/3}$ against $-1/3$.

5.2 The long-run covariance estimator behaves as the theory says

Figure D.1(a) plots $\mathbb{E}[\widehat{S}]/S$ against the block length together with the *exact* prediction $\sum_{|k|<b}(1 - |k|/b)\Gamma_k/S$, which is closed form for this design, so the comparison carries no approximation of its own. Plain batch means track it over the whole range (0.934 measured against 0.936 predicted at $b = 20$) and the two-scale estimator is within Monte Carlo error of 1 for $b \geq 10$ (0.999 at $b = 20$); the small residual at $b = 5$ is the $O(\rho^b)$ term, only $\rho^5 = 0.03$ there. This is Theorem 9(i)–(ii) with no fitted constants. Figure 2 measures the rate: under $b \asymp \log N$ the two-scale estimator's relative error falls with fitted exponent -0.47 against the $-1/2$ of Theorem 9(iv), while plain batch means under their own optimal $b \asymp N^{1/3}$ schedule give -0.36 against $-1/3$. On the whole sandwich $\Sigma = H^{-1}SH^{-1}$, which is what an interval uses, our composite relative error falls from 0.17 to 0.05 while an iterate-path estimator at its own prescribed block length sits near 0.86 throughout; Appendix D.4 is careful about what that does and does not show.

Figure D.1(b) addresses the objection that a block average must span enough parameter movement to behave like an independent draw — the reason iterate-based batch means need $b \asymp N^{3/4}$ [82]. Decomposing the block gradients exactly into the score at $\theta^\star$ and the drift $\widehat{H}_t(\theta_{t-1} - \theta^\star)$, the drift contributes 0.0010 of S at $b = 20$ and grows like $\tau_{b,N} = (b + d)\log^2 N/N$, confirming Remark 1: because the blocks carry gradients rather than iterates, the block length is set by the data's mixing time, not the optimiser's.



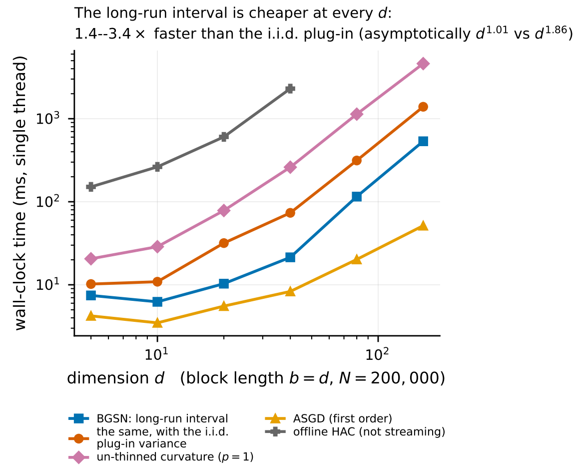


Figure 3: At matched cost, a deterministic gap decorrelates curvature updates exponentially in m while random thinning only does so linearly (Proposition 10). Vertical axis: $n\,\mathbb{E}\|\bar{H} - H\|_F^2$ divided by its exact independent-data value $(\operatorname{tr}\Sigma)^2 + \|\Sigma\|_F^2$ — a computed constant, not a fitted normalisation. Pale bands are the two predictions; markers are measurements with the optimisation frozen at $\theta^\star$, so only estimation quality is compared. $\kappa_H = (1+\phi^2)/(1-\phi^2)$ here.

Figure 4: The statistically correct interval is the cheaper one. Wall-clock milliseconds for one pass over $N = 200{,}000$ observations (vertical axis, log scale) against dimension d (horizontal, log scale), single-threaded, minimum over seven interleaved repetitions. The i.i.d. plug-in score covariance costs $O(d^2)$ per *observation*; the blocked long-run covariance costs $O(d^2)$ per *block*. Slopes are fitted on the four largest d; at this N they are inflated by the one-off warm-up (see (8) and Table D.5, which also gives the asymptotic exponents), and the two smallest d are dominated by fixed per-block overhead rather than by either term of (8). The message of the figure is the vertical separation, which holds at every d.

5.3 Curvature schedules, ablations, and cost

Three further groups of experiments are reported in Appendices D.5 to D.7. Gapping beats Bernoulli thinning at a matched number of rank-one updates exactly as Proposition 10 predicts, with no measurable end-to-end effect on coverage — which the theory also predicts, since $\bar{H}$ enters the limit only through a rate — so it is a better cost knob, not a faster one. The ablations vary N, the block length, the warm-up, the ridge exponent and constant, the initial curvature scale, the curvature schedule and the averaging variant one at a time; the results are insensitive to all of them over their admissible ranges except the warm-up and the step-length safeguard, both of which are load-bearing on the regime-switching design. On cost, the streaming pass grows as $d^{1.00}$ for BGSN and as $d^{1.86}$ once the i.i.d. plug-in variance is added, because the plug-in score covariance costs $O(d^2)$ per *observation* whereas our long-run covariance costs $O(d^2)$ per *block*; those are exact operation counts, and in wall-clock the plug-in variant costs more at every d we tried, by a median factor of 2.7. Our valid intervals cost less than the invalid ones, not more.

5.4 Real streams

Table D.2 reports two protocols on hourly traffic volume on Interstate 94 and hourly PM2.5 at twelve Beijing monitoring sites. After a fixed regression adjustment (weather plus calendar harmonics) the residual lag-one autocorrelation is 0.76 and 0.91, and the median estimated variance inflation across

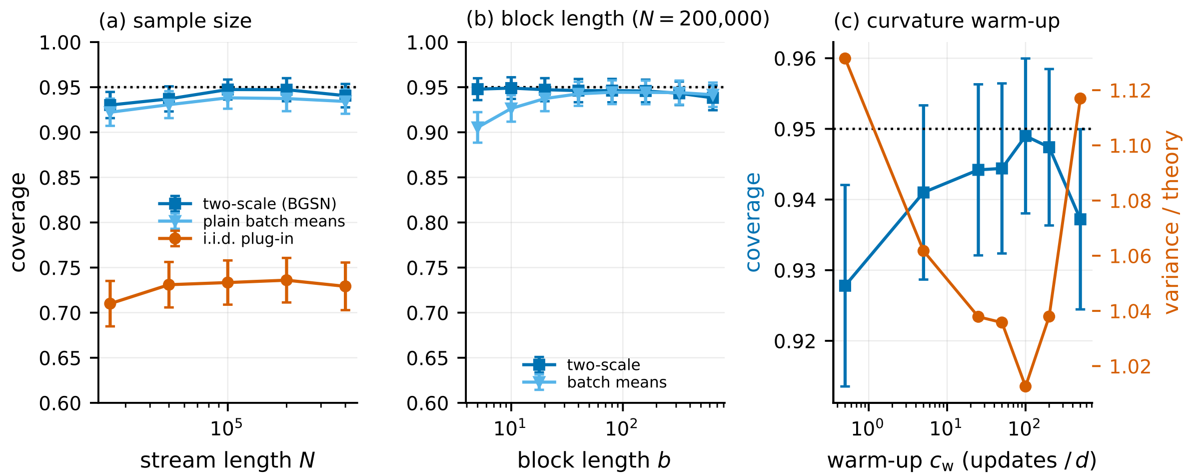


Figure 5: Ablations. (a) Coverage converges to nominal in N for the two-scale variance, while the dependence-blind interval converges to 0.749, the value Proposition 8 predicts for this design. (b) Coverage is flat in the block length for the two-scale variance and degrades at both ends for plain batch means — the practical content of the $O(\rho^b)$-versus-$O(1/b)$ distinction. (c) The curvature warm-up matters, though the step-length safeguard of (6) now absorbs part of its job: with essentially no warm-up coverage on this design is 0.928 against 0.949 at the default, and the relative variance on the right axis tells the same story. On the regime-switching design, where the warm-up is load-bearing rather than merely helpful, the effect is far larger (Table D.4(c′)).

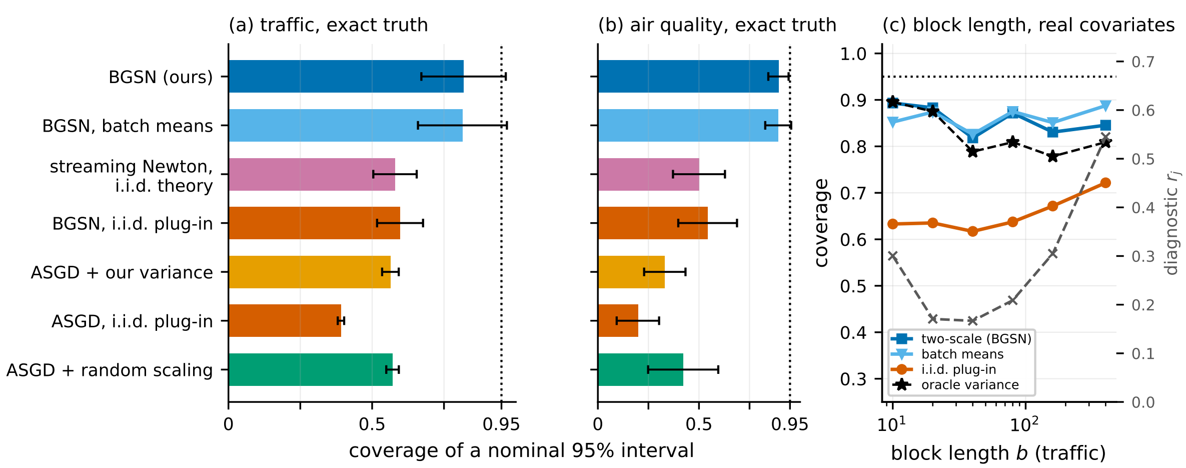


Figure 6: Two real hourly streams. (a, b) Coverage of nominal 95% intervals under Protocol A (real covariates, synthetic autoregressive errors, exact target and exact conditional long-run covariance). The air-quality stream's median estimated variance inflation is 9.9, so dependence-blind intervals fail badly there. (c) Coverage (left axis) against block length b (horizontal, log scale) on the traffic stream, with the free block-adequacy diagnostic $r_j = |u_j^\top(\widehat{S}^{\mathrm{FT}} - \widehat{S}^{\mathrm{BM}})u_j|/(u_j^\top \widehat{S}^{\mathrm{BM}} u_j)$ — an estimate of the relative block-length bias of plain batch means, defined where it is first used in Section 5 — on the right axis. Three things to read. Our coverage is flat in b over a fortyfold range and stays within a few hundredths of the infeasible oracle-variance curve throughout, so the shortfall from 0.95 on an 8,000-observation segment is the point estimate and not the covariance. The dependence-blind curve sits far below at every b and does not close. And r_j is U-shaped, lowest near $b = 20$–40: it rises for small b, where plain batch means are biased, and again for large b, where too few blocks remain to estimate anything stably — which is what makes it a usable guide to the block length, though not a calibration guarantee.

coordinates is 5.8 for traffic and 9.9 for air quality; because κ_j varies across coordinates, the table reports the pooled prediction $d^{-1}\sum_j\{2\Phi(z_{0.025}/\sqrt{\kappa_j})-1\}$ next to the measurement. Protocol A keeps the real (strongly dependent) covariate stream and replaces the response by $x_t^\top\theta^\star + u_t$ with u an autoregression whose coefficient equals that of the real residuals, which gives an *exact* target and an exact conditional long-run covariance while retaining real covariate dependence: BGSN covers 0.818 (traffic) and 0.893 (air quality) against nominal 0.95, the dependence-blind interval 0.597 and 0.543. Protocol B uses the real response and takes the full-stream M-estimator as the target; because that target is estimated from a superset of each segment, the correct nominal level is $2\Phi(z_{0.025}/\sqrt{1-1/R})-1$, which we compute exactly and report in the table header, and BGSN covers 0.795 and 0.858. Neither protocol alone would settle the matter — A has an exact target but a synthetic error process, B a real error process but an estimated target — and they agree.

The block length is the one parameter a practitioner must think about on real data, and the free diagnostic $r_j = |u_j^\top(\widehat{S}^{\mathrm{FT}} - \widehat{S}^{\mathrm{BM}})u_j|/(u_j^\top\widehat{S}^{\mathrm{BM}}u_j)$ flags when it is wrong. It is U-shaped in b, rising for short blocks, where plain batch means carry the $O(1/b)$ bias, and for long ones, where too few blocks remain for a stable estimate; over a fortyfold range in b our coverage stays flat and within a few hundredths of the infeasible oracle-variance curve (Figure 6(c)). So r_j is a guide to choosing b, not a calibration guarantee, and we do not claim that it predicts coverage (Appendix D.8).

6 Related work

This paper sits at the intersection of four literatures. Appendix F surveys each with its citations; here we state only the boundaries of what is new, because that is what a reader needs in order to judge the contribution.

Streaming second-order methods at $O(dN)$ cost. Inversion-free stochastic Newton methods maintain a running inverse curvature estimate by rank-one Sherman–Morrison updates. Our starting point is Godichon-Baggioni and Werge [34], who thin those updates with a Bernoulli variable and take mini-batches of size d to reach $O(dN)$ total cost with asymptotic efficiency. **The $O(dN)$ budget and the idea of skipping curvature updates are theirs, not ours.** All of this work assumes independent observations. Ours adds the analysis of the rank-one recursion when the accumulated terms are dependent, the replacement of Bernoulli thinning by a deterministic gap (Proposition 10), and intervals.

Online inference for stochastic approximation. Under independent sampling this is a mature area, from plug-in and batch-means variance estimators for averaged stochastic gradient descent [15, 102] to random scaling, which estimates nothing at all [54], and including the minimax rate for *Hessian-free* covariance estimation from the iterate path [69], which is close to attained [93]. **Online estimation of the instantaneous sandwich under independent data is therefore not new, and none of these is a claim we duplicate**: what we take is the batch-means template, and what we add is the block-*gradient* construction and its consequence for the rate (Remark 1). The two papers that do give online covariance estimation for streaming *second-order* methods [50, 91] are under independent data, target the instantaneous sandwich, and require the gradient noise to be a martingale difference for the algorithm's filtration — exactly the condition temporal dependence destroys.

Stochastic approximation on dependent data. Convergence under Markovian or mixing noise is classical, and modern rates are available for Markov-chain gradient descent. Godichon-Baggioni et al. [35] identify blocked or growing mini-batches as the device that breaks dependence, with L^2 rates but no central limit theorem and no inference: **blocked gradients as a dependence-breaking device are theirs.** On the inference side, Liu et al. [60] give a multiplier bootstrap for ϕ-mixing streams, Li et al. [57] the functional central limit theorem and the semiparametric efficiency bound $H^{-1}SH^{-1}$ under Markovian sampling, Samsonov et al. [82] Berry–Esseen bounds and an overlapping-batch-means variance for *linear* stochastic approximation, and Roy and Balasubramanian [81] an online batch-means estimator of the Markovian long-run sandwich. **The long-run sandwich limit, the efficiency bound, and the existence of online long-run covariance estimators under dependence are all prior art** — for first-order methods. Every one of these works is first-order; none maintains a curvature inverse, and the two that estimate a long-run covariance do so from the iterate path, at $O(N^{-1/8})$, with block lengths that must grow like $N^{3/4}$ [82, Prop. 2, Cor. 2]. Remark 1 explains why block gradients escape that constraint and Theorem 9 quantifies the gain.

Long-run variance estimation. The statistical device is heteroskedasticity- and autocorrelation-consistent covariance estimation and the batch-means literature it shares with simulation output analysis. The two-scale correction we use is the flat-top lag window of Politis and Romano [73], recast as "lugsail" by Vats and Flegal [89], which Singh et al. [84, Eqs. (15) and (17)] bring to stochastic gradient descent under independent data. **We do not claim the correction.** We claim the bias order it attains at block-gradient resolution under geometric mixing — the property that makes $b \asymp \log N$ admissible — and the streaming implementation. The population sandwich asymptotics for dependent M-estimation that we use are likewise standard.

7 Limitations

We list the boundaries of the result plainly, because several of them are load-bearing. Appendix G discusses each at length.

(L1) **Memory is $O(d^2)$, not $O(d)$.** The $O(dN)$ claim is about *time*; for d in the thousands memory is the binding constraint, and whether a block-gradient long-run covariance survives a sketched curvature representation is open.

(L2) **Geometric mixing is a real restriction.** It is what buys $b \asymp \log N$; under algebraic mixing the admissible b grows, and long-memory streams are outside the theory. The diagnostic r_j detects an inadequate b but does not remove it, and we claim no automatic rule for choosing b.

(L3) **$\widehat{S}^{\mathrm{FT}}$ need not be positive semidefinite**, as for all flat-top and lugsail estimators [84, Remark 5]; marginal intervals need only the scalars $u_j^\top \widehat{S}^{\mathrm{FT}} u_j$, the documented fallback triggers on 0.0% of coordinate-replicate pairs on the real streams and never in simulation, and a user who needs the full matrix must project it.

(L4) **The curvature form is required.** It holds for generalised linear models, ridge and softmax regression, the geometric median and Gauss–Newton problems, but not for a general objective; without it there is no rank-one recursion.

(L5) **The warm-up is a requirement, not an implementation detail**, and c_{w} is a tuning constant the asymptotic theory is silent about; what sets its size is the number of effectively distinct design points it spans, and it is why the honest time bound is $O(dN + c_{\mathrm{w}} d^3)$.

(L6) **Blocking creates a stability condition, and our fix is not fully proved.** Proposition 1 holds on the event that the safeguard is eventually inactive, and we do not prove that event has probability one, because the safeguard is not predictable; we report activation counts instead (2 to 7 per run, flat in N). The condition itself comes from a linearisation, so it is exact only for linear models, and c_D has a genuine failure mode on the one design that needs it.

(L7) **The central limit theorem is proved for fixed b, and we recommend growing b.** At $b \asymp \log N$ the proof needs the constants of Lemma 13 to be bounded in probability uniformly in N; we state that hypothesis in Theorem 6 rather than hide it, and we do not verify it.

(L8) **Rates we do not prove.** We give an almost sure rate and a central limit theorem, not a Berry–Esseen bound [82]; we attain the efficiency bound of Li et al. [57] rather than establishing it; our rate is not comparable to the minimax rate for *Hessian-free* covariance estimation from an iterate path [69] because we use more information; and we claim no optimality within our own information set.

(L9) **Scope of the evidence.** All designs are convex M-estimation with $d \le 160$ and $N \le 4 \times 10^5$; there is no deep-learning experiment and the theory would not support one. On the real streams we do not know the truth, which is why we run two protocols with complementary weaknesses.

(L10) **One thing we deliberately do not claim.** Gapping improves the curvature estimate at matched cost, but curvature enters the limit law only through a rate, so the end-to-end effect on coverage is second order and we could not measure it reliably.

8 Conclusions

On a dependent stream, a streaming quasi-Newton method can get the estimate right and the uncertainty wrong. The error is not small and does not go away: the reported variance omits the autocovariances of the score, so a nominal 95% interval converges to a coverage below 95% and stays there. Proposition 8 says exactly where — at $2\Phi(z_{\alpha/2}/\sqrt{\kappa}) - 1$, computable from one variance-inflation factor κ. At the inflation we measure on hourly air quality that is 0.466, so more than half of the intervals a practitioner would report on that stream miss the value they are meant to cover.

The remedy is a change of reading order rather than an extra estimation stage. Consume the stream in contiguous blocks and take one preconditioned step per block. The block-averaged gradients are then, up to a factor, the non-overlapping batch means of the score process, and batch means estimate exactly the long-run variance the interval needs. The object the statistics requires and the object the computation already produces are the same, so the resulting interval is *cheaper* than the invalid one: an instantaneous plug-in covariance costs $O(d^2)$ per observation, the blocked long-run covariance $O(d^2)$ per block. Gapping the rank-one curvature updates is the second change, and it is a cost knob rather than an inference device.

Three practical consequences. Block length is the one parameter that needs thought, and needs less than one might fear: because the blocks carry gradients rather than parameter values, their length is set by how quickly the *data* forget, not by how quickly the optimiser settles, so b of order $\log N$ suffices where iterate-based methods need a power of N. Blocking has a price we pay and report: it makes a step-size condition non-trivial, and without the $O(d)$ safeguard of (6) our own method diverges on a persistent regime-switching design. And honest intervals are wider — on our designs 1.37–2.72 times the dependence-blind width, which is $\sqrt{\kappa}$ and not a tuning choice. The narrower interval was never correct; it was the wrong length.

What we have not done bounds the claims, and Section 7 states each boundary precisely. The construction itself is general: any streaming method that forms block gradients and maintains a curvature estimate can report a long-run variance at no extra order of cost.

Data and code availability

All code, saved results and derived data needed to reproduce every number, figure and table in this article are archived and citable [46]. The release used to produce the results reported here is 10.5281/zenodo.21891313; the concept identifier 10.5281/zenodo.21891312 always resolves to the most recent release, and development continues at https://github.com/Kendiukhov/streaming-quasi-newton-dependent-data. The archive contains the estimator, the experiment scripts, the result files from which every reported number is generated, and the tests that validate every closed-form population quantity against long simulation.

The two real data sets analysed are third-party and are not redistributed. They are downloaded from the UCI Machine Learning Repository by the script `fetch_data.py` and are cited in the reference list: Metro Interstate Traffic Volume [40], 10.24432/C5X60B; and Beijing Multi-Site Air Quality [14], 10.24432/C5RK5G.

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A Proofs

Throughout, C denotes a finite constant that may change from line to line, $\delta_t = \theta_t - \theta^\star$, $\varepsilon_t = \bar{g}_t(\theta^\star)$, $P_t = \bar{H}_t^{-1}$, and $\Lambda_k = \mathbb{E}[\varepsilon_0 \varepsilon_k^\top]$. We take $\gamma_t = 1/t$ throughout and ignore the step-length safeguard Π_t of (6); Proposition 1, proved in Appendix E, shows that on the event where Π_t is eventually inactive the safeguarded iterates coincide with these from a finite time on, and Appendix E.1 treats the rejected step-size-shift variant $\gamma_t = (t + t_0)^{-1}$ separately. We write n_t for the number of rank-one curvature updates completed by the end of block t; with gap m and no Bernoulli thinning, $n_t = \lfloor tb/m \rfloor$, so $n_t \asymp t$.

Order of the argument. The results are proved in the order of Remark 6: Lemma 2 (from the ridge alone, no assumption on the iterates); Theorem 4 (uses Lemma 2 only); the preliminary rate $\|\delta_t\| = O(t^{-1/2} \log^c t)$ (uses Theorem 4 and consistency of $\bar{H}$, not its rate); Theorem 3; and then Theorems 5, 6, 9 and 16. Every remainder bound below is uniform over $b \le C \log N$, which is the only regime in which Theorem 9 uses them, and the b-dependence is displayed wherever it is not $O(1)$.

A.1 Preliminaries on dependence

Lemma 11 (Geometric decay of the score autocovariances)**.** *Under Assumption 5, $\|\Gamma_k\|_{\mathrm{op}} \le C\rho^{|k|}$ with $\rho = c_{\mathrm{mix}}^{\nu/(2+\nu)}$, and $\|\Lambda_k\|_{\mathrm{op}} \le Cb^{-1}\rho^{(|k|-1)b}$ for $|k| \ge 1$ while $\|\Lambda_0\|_{\mathrm{op}} \le C/b$. Consequently $\sum_{k\in\mathbb{Z}} \|\Lambda_k\|_{\mathrm{op}} \le C/b$ and $S = \sum_k \Gamma_k$ converges absolutely.*

Proof. Davydov's inequality [18], in the form of Rio [76, Cor. 1.1], gives for random variables $U \in \sigma(\xi_i, i \le 0)$, $V \in \sigma(\xi_i, i \ge k)$ with $2+\nu$ moments, $|\operatorname{Cov}(U,V)| \le 8\,\alpha_{\mathrm{mix}}(k)^{\nu/(2+\nu)} \|U\|_{2+\nu} \|V\|_{2+\nu}$. Applying it coordinatewise to g_0 and g_k and using Assumption 5 gives the first claim. For the second, $\Lambda_k = b^{-2} \sum_{i\in B_0} \sum_{j \in B_k} \Gamma_{j-i}$; for $k \ge 1$ every lag $j-i$ is at least $(k-1)b+1$, so $\|\Lambda_k\|_{\mathrm{op}} \le b^{-2} \cdot b^2 \cdot C\rho^{(k-1)b}$, and for $k = 0$, $\|\Lambda_0\|_{\mathrm{op}} \le b^{-2} \sum_{|l|<b} (b - |l|) C \rho^{|l|} \le Cb^{-1}$. Summing the geometric series gives $\sum_k \|\Lambda_k\|_{\mathrm{op}} \le C/b$. □

Lemma 12 (Martingale–coboundary decomposition)**.** *Let $(\zeta_t)_{t\geq 1}$ be stationary, centred, $\mathbb{R}^q$-valued, $\mathcal{F}_t$-measurable with $\mathcal{F}_t = \sigma(\xi_i : i \leq tb)$, and let $\mathbb{E}\|\zeta_1\|^{2+\nu} < \infty$ under Assumption 5. Then $\sum_{n\geq 0}\|\mathbb{E}[\zeta_n \mid \mathcal{F}_0]\|_2 < \infty$, the series $z_t = \sum_{n\geq 0}\mathbb{E}[\zeta_{t+n} \mid \mathcal{F}_t]$ converges in L^2, and*

$$\zeta_t \;=\; m_{t+1} + z_t - z_{t+1}, \qquad m_{t+1} = z_{t+1} - \mathbb{E}[z_{t+1} \mid \mathcal{F}_t], \tag{13}$$

where (m_t) is a stationary martingale difference sequence with respect to $(\mathcal{F}_t)$ and $\mathbb{E}\|m_1\|^2 < \infty$, $\mathbb{E}\|z_1\|^2 < \infty$. Moreover $\mathbb{E}[m_1 m_1^\top] = b^{-1}S$ when $\zeta_t = \varepsilon_t$.

Proof. For U that is $\sigma(\xi_i : i \geq n)$-measurable with $2+\nu$ moments, the standard consequence of strong mixing (see Rio 77, Ch. 1) is $\|\mathbb{E}[U \mid \mathcal{F}_0]\|_2 \leq C\,\alpha_{\mathrm{mix}}(n)^{\nu/(2(2+\nu))}\|U\|_{2+\nu}$, which under Assumption 5 decays geometrically in n and is therefore summable; this is Gordin's condition, and (13) together with the stated moment bounds is Gordin's decomposition; see Hall and Heyde [37, Ch. 5]. Summing (13) telescopes the coboundary, so $\sum_{t\leq T}\zeta_t = \sum_{t\leq T} m_{t+1} + z_1 - z_{T+1}$; dividing by $\sqrt{T}$ and using $z_{T+1}/\sqrt{T} \to 0$ in probability identifies $\lim T^{-1}\operatorname{Var}(\sum_{t\leq T}\zeta_t) = \mathbb{E}[m_1 m_1^\top]$, which for $\zeta_t = \varepsilon_t$ is $b^{-1}S$ by (1). □

Lemma 13 (Weighted sums with adapted weights: convergence and rate)**.** *Let (ζ_t) be as in Lemma 12, let (A_t) be $\mathcal{F}_{t-1}$-measurable random matrices, and let (γ_t) be deterministic and non-increasing. Suppose there are deterministic (c_t), (u_t) and a finite random Λ with, almost surely for all t,*

$$\|A_t\|_{\mathrm{op}} \leq \Lambda c_t \qquad \text{and} \qquad \|\gamma_{t+1}A_{t+1} - \gamma_t A_t\|_{\mathrm{op}} \leq \Lambda u_t, \tag{14}$$

and write $V_T = \sum_{t\leq T}\gamma_t^2 c_t^2$ and $U_T = \sum_{t\leq T} u_t$. Then for every $\varpi > 0$, almost surely,

$$\Big\|\sum_{t\leq T}\gamma_t A_t \zeta_t\Big\| = O\big(V_T^{1/2}\log^{1/2+\varpi} T\big) + O\big(U_T \log^{1+\varpi} T\big) + O\big(\gamma_T c_T T^{1/2+\varpi}\big). \tag{15}$$

If moreover $V_\infty < \infty$, $U_\infty < \infty$ and $\gamma_T c_T T^{1/2+\varpi} \to 0$ for some $\varpi > 0$, then $\sum_t \gamma_t A_t \zeta_t$ converges almost surely.

Proof. Localisation. For $K \in \mathbb{N}$ replace A_t by

$$A_t\,\mathbf{1}\Big\{\sup_{s\leq t}\max\big(\|A_s\|_{\mathrm{op}}/c_s,\ \|\gamma_{s+1}A_{s+1} - \gamma_s A_s\|_{\mathrm{op}}/u_s\big) \leq K\Big\},$$

which is still $\mathcal{F}_{t-1}$-measurable (the indicator is decreasing in t and $\mathcal{F}_{t-1}$-measurable at each t because it involves A_s only for $s \leq t$, and A_{s+1} enters only through $s+1 \leq t$). On $\{\Lambda \leq K\}$ the two processes agree and $\mathbb{P}(\bigcup_K\{\Lambda \leq K\}) = 1$, so it suffices to prove the claims under the *deterministic* envelopes $\|A_t\|_{\mathrm{op}} \leq K c_t$, $\|\gamma_{t+1}A_{t+1} - \gamma_t A_t\|_{\mathrm{op}} \leq K u_t$. Fix K and drop it.

Martingale part. Insert (13): $M_T = \sum_{t\leq T}\gamma_t A_t m_{t+1}$ is a martingale, because A_t is $\mathcal{F}_t$-measurable and $\mathbb{E}[m_{t+1} \mid \mathcal{F}_t] = 0$ — this is the exact cancellation the decomposition buys, and it holds whatever A_t does. Its predictable quadratic variation obeys

$$\langle M\rangle_T = \sum_{t\leq T}\gamma_t^2\,\mathbb{E}\big[\|A_t m_{t+1}\|^2 \,\big|\, \mathcal{F}_t\big] \leq K^2\sum_{t\leq T}\gamma_t^2 c_t^2\, v_t, \qquad v_t := \mathbb{E}\big[\|m_{t+1}\|^2 \mid \mathcal{F}_t\big],$$

with (v_t) stationary, non-negative and $\mathbb{E}v_t = \mathbb{E}\|m_1\|^2 < \infty$. By Tonelli $\mathbb{E}\big[\sum_{t\leq T}\gamma_t^2 c_t^2 v_t\big] = \mathbb{E}\|m_1\|^2 V_T$, so Markov's inequality along $T = 2^j$ and Borel–Cantelli give $\langle M\rangle_T = O(V_T \log^{\varpi} T)$ a.s. (and $\langle M\rangle_\infty < \infty$ a.s. if $V_\infty < \infty$). The martingale law of the iterated logarithm then gives $\|M_T\| = O(V_T^{1/2}\log^{1/2+\varpi} T)$ a.s., and if $\langle M\rangle_\infty < \infty$ a locally square-integrable martingale converges a.s.

Coboundary part. Abel summation gives

$$\sum_{t\le T}\gamma_t A_t(z_t - z_{t+1}) = \gamma_1 A_1 z_1 - \gamma_T A_T z_{T+1} + \sum_{t<T}\big(\gamma_{t+1}A_{t+1} - \gamma_t A_t\big)z_{t+1}.$$

The last sum is at most $K\sum_{t<T} u_t\|z_{t+1}\|$, a non-negative sum with expectation $K\mathbb{E}\|z_1\|U_T$, hence $O(U_T\log^{1+\varpi}T)$ a.s. by Markov and Borel–Cantelli along dyadic blocks; if $U_\infty<\infty$ its expectation is finite and it converges a.s. Finally $\|\gamma_T A_T z_{T+1}\| \le K\gamma_T c_T\|z_{T+1}\| = O(\gamma_T c_T T^{1/2+\varpi})$ a.s., because $\mathbb{E}\|z_1\|^2<\infty$ and Borel–Cantelli give $\|z_T\| = o(T^{1/2+\varpi})$. □

Remark 3 (Why the lemma is stated this way)**.** Three features are load-bearing and each corresponds to a place where a naive argument fails. (i) The weights are *random and adapted*: they contain $\bar H_{t-1}^{-1}$, so a lemma for deterministic weights would not apply. Note that adaptedness is essential and not cosmetic: for bounded adapted weights and a mixing, centred, stationary ζ the sum $\sum_t a_t\zeta_t$ can grow like T if a_t is allowed to correlate with ζ_t — take $\zeta_t = u_t + u_{t-1}$ with u_t i.i.d. signs and $a_t = \zeta_{t-1}$ — which is exactly the non-martingale obstruction. The decomposition (13) removes it because $\mathbb{E}[m_{t+1}\mid\mathcal{F}_t]=0$ *exactly*, whatever A_t does. (ii) The envelope is deterministic only up to a finite random factor, which is all that is available for weights built from the iterates; the localisation handles it. (iii) We use a martingale law of the iterated logarithm rather than a Bernstein inequality for mixing sums. The Bernstein inequality of Merlevède et al. [62] is stated for uniformly *bounded* summands, which Assumption 5 does not provide; after (13) the summands form a martingale difference sequence and no such inequality is needed. (iv) The three terms of (15) are needed separately: the applications below use $\gamma_t\equiv 1$, for which U_T can diverge logarithmically, so a hypothesis $U_\infty<\infty$ would be too strong.

Lemmas 12 and 13 are the technical substitute for the martingale argument used in the independent-data analysis. Two features matter. First, the weights may be *random and adapted*, which is what we need because they involve $\bar H_{t-1}^{-1}$. Second, the conditions (14) are of the form already familiar from the independent-data theory. The vanishing ridge supplies the deterministic envelope: it forces $\lambda_{\min}(A_n)\ge c\,n^{1-q_r}$, hence $\|\bar H_t^{-1}\|_{\mathrm{op}} = W_{n_t}\|A_{n_t}^{-1}\|_{\mathrm{op}} \le c^{-1}n_t^{q_r} =: c_t$ with $q_r<1/2$, so with $\gamma_t = 1/t$ the first half of (14) is $\sum_t O(t^{2q_r-2})<\infty$. The second half holds because a single rank-one Riccati update changes the preconditioner by only $O(1/t)$:

Lemma 14 (Increments of the preconditioner)**.** (a) From the ridge alone. *Under Assumption 3 and the ridge of (4) — so using only Lemma 2, with no assumption on the iterates — $\|\bar H_t^{-1} - \bar H_{t-1}^{-1}\|_{\mathrm{op}} = O(t^{2q_r-1})$ a.s.; hence with $\gamma_t = 1/t$ and $A_t = \bar H_{t-1}^{-1}$, $\|\gamma_{t+1}A_{t+1} - \gamma_t A_t\|_{\mathrm{op}} = O(t^{2q_r-2})$ a.s., which is summable for $q_r<1/2$, so the second half of (14) holds with $U_\infty<\infty$.*

(b) Sharper, once the curvature has converged. *Under the assumptions of Theorem 3, $\|\bar H_t^{-1} - \bar H_{t-1}^{-1}\|_{\mathrm{op}} = O(1/t)$ a.s., so with $\gamma_t\equiv 1$ and $A_t = \bar H_{t-1}^{-1}$ the increments are $u_t = O(1/t)$ and $U_T = O(\log T)$.*

Proof. $\bar H_t^{-1} = W_{n_t}A_{n_t}^{-1}$, and from block $t-1$ to block t the weight W increases by $O(1)$ while A^{-1} changes by $O(1)$ rank-one Sherman–Morrison steps $A^{-1}\mapsto A^{-1} - cA^{-1}uu^\top A^{-1}/(1+cu^\top A^{-1}u)$. Writing $v = A^{-1}u$, the operator norm of one such step is $c\|v\|^2/(1+cu^\top v)$, and since $u^\top v = v^\top A v \ge \lambda_{\min}(A)\|v\|^2$ we get the two bounds

$$\|\Delta A^{-1}\|_{\mathrm{op}} \;\le\; \min\left\{\frac{1}{\lambda_{\min}(A)},\ \frac{c\|u\|^2}{\lambda_{\min}(A)^2}\right\}. \tag{16}$$

For (a), Lemma 2 gives $\lambda_{\min}(A_n)\ge c'n^{1-q_r}$, so the second bound in (16) is $O(n^{2q_r-2})$; multiplying by $W_{n_t} = O(t)$ and using $n_t\asymp t$ gives $\|\bar H_t^{-1} - \bar H_{t-1}^{-1}\|_{\mathrm{op}} = O(t^{2q_r-1})$, and the change in W contributes $O(1)\cdot\|A^{-1}\|_{\mathrm{op}} = O(t^{q_r-1})$, which is smaller. The stated bound on $\gamma_{t+1}A_{t+1} - \gamma_t A_t$ then follows from $\gamma_t = 1/t$ together with $\gamma_{t+1}-\gamma_t = O(t^{-2})$ and $\|A_t\|_{\mathrm{op}} = O(t^{q_r})$. For (b), Theorem 3 gives $A_{n_t}/n_t\to H\succ 0$, so $\lambda_{\min}(A_{n_t}) \ge c' t$ a.s. for large t, the second bound in (16) is $O(t^{-2})$, and multiplying by $W_{n_t} = O(t)$ gives $O(1/t)$. □

A.2 The ridge bound (Lemma 2)

Each curvature update adds $\iota_n e_{k_n} e_{k_n}^\top \succeq 0$ along a cycling basis direction, and the data term $\alpha_{i_n} \Psi_{i_n} \Psi_{i_n}^\top$ is positive semidefinite, so it can only help. After n updates every basis direction has received at least $\lfloor n/d \rfloor$ ridge increments, and dropping the data terms,

$$A_n \;\succeq\; \lambda_0 I + \Big(\tfrac{1}{d} \textstyle\sum_{k \le n} \iota_k\Big) I. \tag{17}$$

Now the point of freezing the ridge scale. Because $\iota_k = c_\iota \varsigma_\star k^{-q_\mathrm{r}}$ with $\varsigma_\star$ a *constant* — measurable with respect to the first curvature-bearing block and, by Assumption 3, almost surely finite and strictly positive whenever that block's curvature is not identically zero — we may sum:

$$\sum_{k \le n} \iota_k = c_\iota \varsigma_\star \sum_{k \le n} k^{-q_\mathrm{r}} \;\ge\; \frac{c_\iota \varsigma_\star}{1 - q_\mathrm{r}} \big(n^{1-q_\mathrm{r}} - 1\big),$$

using $\sum_{k \le n} k^{-q_\mathrm{r}} \ge \int_1^{n+1} x^{-q_\mathrm{r}} dx$ and $q_\mathrm{r} < 1$. Hence $\lambda_{\min}(A_n) \ge c' n^{1-q_\mathrm{r}}$ for all $n \ge 2$ with $c' = c_\iota \varsigma_\star / \{2d(1 - q_\mathrm{r})\}$, and since $W_{n_t} = 1 + n_t$,

$$\|\bar{H}_t^{-1}\|_{\mathrm{op}} = W_{n_t} \|A_{n_t}^{-1}\|_{\mathrm{op}} \le \frac{1 + n_t}{c' n_t^{1-q_\mathrm{r}}} = O(n_t^{q_\mathrm{r}}), \qquad \lambda_{\min}(\bar{H}_t) \ge \frac{c' n_t^{1-q_\mathrm{r}}}{1 + n_t} \;\ge\; c_1 n_t^{-q_\mathrm{r}}$$

with $c_1 = c'/2$ for $n_t \ge 1$. Nothing about the iterates is used, and no lower bound on the accumulated data terms is needed — which is exactly why the lemma may sit at the bottom of the ladder in Remark 6.

Remark 4 (Why the scale must be frozen)**.** Had we scaled the ridge by the *running* harmonic mean $d/\mathrm{tr}(\bar{H}_k^{-1})$ — the natural choice, since it is available for free from the maintained inverse — this proof would fail, and not for a technical reason. That scale is bounded above by $d\lambda_{\min}(\bar{H}_k)$, so a path along which the accumulated data terms contribute negligibly in some direction would have its ridge shrink in step with the very eigenvalue the ridge is supposed to hold up. Concretely, the only unconditional bound then available is $\varsigma_k \ge \lambda_0 / W_k = \Theta(1/k)$, giving $\sum_{k \le n} \iota_k = O(1)$ instead of $\Omega(n^{1-q_\mathrm{r}})$, and a Gronwall argument in the opposite direction shows this is tight: $\lambda_{\min}(A_n)$ can stay $O(1)$, i.e. $\|\bar{H}_n^{-1}\|_{\mathrm{op}} = \Omega(n)$, which would break the square-summability $\sum_t \gamma_t^2 \|P_{t-1}\|_{\mathrm{op}}^2 < \infty$ that Theorem 4 needs. The bound does hold once $\bar{H}_t \to H \succ 0$, but that is Theorem 3, three rungs higher, so the argument would be circular. Freezing the scale removes the problem at no cost: $\varsigma_\star$ serves only to make c_ι dimensionless, and Table D.4(d) shows the results are unchanged if the scale is dropped altogether.

A.3 Consistency (Theorem 4)

Substituting $\bar{g}_t(\theta_{t-1}) = \nabla F(\theta_{t-1}) + \zeta_t$ into (6), where $\zeta_t = \bar{g}_t(\theta_{t-1}) - \nabla F(\theta_{t-1})$, and expanding $F(\theta_t) - F(\theta^\star)$ to second order using Assumption 1,

$$\begin{aligned} F(\theta_t) - F(\theta^\star) \le F(\theta_{t-1}) - F(\theta^\star) - \gamma_t \nabla F(\theta_{t-1})^\top P_{t-1} \nabla F(\theta_{t-1}) \\ - \gamma_t \nabla F(\theta_{t-1})^\top P_{t-1} \zeta_t + \tfrac{L}{2} \gamma_t^2 \|P_{t-1}\|_{\mathrm{op}}^2 \|\bar{g}_t(\theta_{t-1})\|^2. \end{aligned} \tag{18}$$

The last term is summable:

$$\gamma_t^2 \|P_{t-1}\|_{\mathrm{op}}^2 = O(t^{2q_\mathrm{r}-2}), \qquad \mathbb{E}\|\bar{g}_t(\theta_{t-1})\|^2 \le C + C'(F(\theta_{t-1}) - F(\theta^\star)),$$

the first by Lemma 2 with $q_{\mathrm{r}} < 1/2$ and the second by Assumption 2 and Jensen. The cross term is the one that would be handled by a martingale argument under independence. Instead, decompose $\zeta_t = \varepsilon_t + r_t$ with $r_t = \bar{g}_t(\theta_{t-1}) - \nabla F(\theta_{t-1}) - \varepsilon_t$, so that

$$\|r_t\| \leq \|\widehat{H}_t - H\|_{\mathrm{op}}\|\delta_{t-1}\| + L_\delta\|\delta_{t-1}\|^2 \tag{19}$$

where $\widehat{H}_t$ is the block curvature; r_t is dominated by $\|\delta_{t-1}\|$ times an L^2-bounded factor, and its contribution to (18) is absorbed by the Robbins–Siegmund argument [79]. The remaining term $-\sum_t \gamma_t \nabla F(\theta_{t-1})^\top P_{t-1}\varepsilon_t$ is handled by Lemma 13 with $A_t = \nabla F(\theta_{t-1})^\top P_{t-1}$, which is $\mathcal{F}_{t-1}$-measurable as required. Its first condition holds because $\|P_{t-1}\|_{\mathrm{op}} = O(t^{q_{\mathrm{r}}})$ with $q_{\mathrm{r}} < 1/2$ (Lemma 2) and $\mathbb{E}\|\nabla F(\theta_{t-1})\|^2$ is bounded on the events $\{\sup_{s\leq t}\|\theta_s\| \leq K\}$, over which we localise and then let $K \to \infty$; its second condition holds by Lemma 14(a) — which, being a consequence of Lemma 2 alone, keeps this rung free of Theorem 3 — together with $\|\nabla F(\theta_t) - \nabla F(\theta_{t-1})\|_{\mathrm{op}} \leq L\|\theta_t - \theta_{t-1}\| = O(\gamma_t t^{q_{\mathrm{r}}})$. Robbins–Siegmund then gives $F(\theta_t) \to F(\theta^\star)$ a.s. and $\sum_t \gamma_t \nabla F(\theta_{t-1})^\top P_{t-1} \nabla F(\theta_{t-1}) < \infty$ a.s.; and since $\gamma_t \lambda_{\min}(P_{t-1}) \geq \gamma_t/\|\bar{H}_{t-1}\|_{\mathrm{op}} \geq c\, t^{-1}$ (because $\|\bar{H}_t\|_{\mathrm{op}}$ is bounded above a.s., $\bar{H}_t$ being an average of terms with $\eta' > 2$ moments) we have $\sum_t \gamma_t \lambda_{\min}(P_{t-1}) = \infty$; Assumption 4 then forces $\theta_t \to \theta^\star$ a.s. □

A.4 The preliminary rate (rung (iii) of Remark 6)

This rung is what Theorem 3 needs and Theorem 4 does not give: a rate, not just convergence. We prove it using only Lemma 2 and Theorem 4.

Lemma 15 (Preliminary rate)**.** *Under Assumptions 1 to 5, with* $q_{\mathrm{r}} \in (0, 1/4)$*,*

$$\|\theta_T - \theta^\star\| = O\big(T^{-1/2}\log^{1+\varpi} T\big) \quad \textit{almost surely, for every } \varpi > 0.$$

Proof. Abel summation of (6) exactly as in (22) — an algebraic identity that uses only $\gamma_t = 1/t$ — gives (23): $\delta_T = -T^{-1}H^{-1}\sum_{t\leq T}\varepsilon_t + (\mathrm{A}) + (\mathrm{B}) - (\mathrm{C})$. We bound the three remainders using only what rungs (i) and (ii) provide, namely $\|P_{t-1}\|_{\mathrm{op}} = O(t^{q_{\mathrm{r}}})$ a.s. (Lemma 2), $\theta_t \to \theta^\star$ a.s. (Theorem 4), and hence, by the local Lipschitz property in Assumption 3 and the continuity in Assumption 2, $\|P_{t-1} - H^{-1}\|_{\mathrm{op}} = o(1)$ and $\|I - P_{t-1}H\|_{\mathrm{op}} = o(1)$ a.s. Write $\rho_t = \|I - P_{t-1}H\|_{\mathrm{op}}$, so $\rho_t \to 0$ a.s.

Leading term. $T^{-1}\sum_{t\leq T}\varepsilon_t = O(T^{-1/2}\log^{1/2+\varpi} T)$ a.s. by the martingale law of the iterated logarithm applied to (13) (the coboundary telescopes to $O(T^{-1})$ after division by T, using $\mathbb{E}\|z_1\|^2 < \infty$ and Borel–Cantelli along $\|z_t\| = o(t^{1/2})$ a.s.).

Term (A). Apply (15) with $\gamma_t \equiv 1$ and $A_t = H^{-1} - P_{t-1}$: the envelope may be taken $c_t \equiv 1$ (a finite random constant, since $\|P_{t-1}\|_{\mathrm{op}} = O(t^{q_{\mathrm{r}}})$ makes $\sup_t\|A_t\|_{\mathrm{op}}/c_t$ finite only after we note $A_t \to 0$; formally take $c_t = 1$ and $\Lambda = \sup_t\|A_t\|_{\mathrm{op}} < \infty$ a.s., which holds because $A_t \to 0$ and each A_t is finite). By Lemma 14(a) the increments are $u_t = O(t^{2q_{\mathrm{r}}-1})$, so $V_T = O(T)$ and $U_T = O(T^{2q_{\mathrm{r}}})$, and (15) gives $\|\sum_{t\leq T} A_t\varepsilon_t\| = O(T^{1/2}\log^{1/2+\varpi} T) + O(T^{2q_{\mathrm{r}}}\log^{1+\varpi} T) + O(T^{1/2+\varpi})$. Dividing by T and using $q_{\mathrm{r}} < 1/4$, which is exactly where that restriction is needed, $(\mathrm{A}) = O(T^{-1/2}\log^{1/2+\varpi} T) + O(T^{2q_{\mathrm{r}}-1}\log^{1+\varpi} T) = O(T^{-1/2}\log^{1/2+\varpi} T)$ a.s.

Terms (B) and (C): the bootstrap. $(\mathrm{B}) = T^{-1}\sum_{t\leq T}(I - P_{t-1}H)\delta_{t-1}$ so $\|(\mathrm{B})\| \leq \big(\sup_{t\leq T}\rho_t\big)\,\bar{\delta}_T$ with $\bar{\delta}_T = T^{-1}\sum_{t\leq T}\|\delta_{t-1}\|$, and by (19) $\|(\mathrm{C})\| \leq CT^{-1}\sum_{t\leq T}\|P_{t-1}\|_{\mathrm{op}}(\|\widehat{H}_t - H\|_{\mathrm{op}}\|\delta_{t-1}\| + \|\delta_{t-1}\|^2)$, which is $o(1)\,\bar{\delta}_T$ a.s. because $\|P_{t-1}\|_{\mathrm{op}}\|\widehat{H}_t - H\|_{\mathrm{op}}$ and $\|P_{t-1}\|_{\mathrm{op}}\|\delta_{t-1}\|$ are both $o(1)$ a.s. ($\delta_t \to 0$ by Theorem 4, and $\|P_{t-1}\|_{\mathrm{op}}\|\widehat{H}_t - H\|_{\mathrm{op}} \to 0$ because $\|P_{t-1}\|_{\mathrm{op}} \to \|H^{-1}\|_{\mathrm{op}}$ and $\widehat{H}_t \to H$ in the Cesàro sense along the block index). Hence there is a random $T_1 < \infty$ and a sequence $\epsilon_T \downarrow 0$ with

$$D_T := \max_{t\leq T}\|\delta_t\| \;\leq\; \epsilon_T D_T + R_T, \qquad R_T = O(T^{-1/2}\log^{1/2+\varpi} T) \text{ a.s.,} \tag{20}$$

for $T \geq T_1$, where we used $\bar{\delta}_T \leq D_T$ and that the same bound applies to every $t \leq T$ (the right-hand side is non-decreasing in T). For T large enough that $\epsilon_T \leq 1/2$, (20) rearranges to $D_T \leq 2R_T$, which is the claim with $\log^{1+\varpi}$ absorbing $\log^{1/2+\varpi}$. □

Two things are worth flagging about this rung. It is the only place the restriction $q_{\mathrm{r}} < 1/4$ (rather than $q_{\mathrm{r}} < 1/2$) is used, and it is used because at this rung the sharper increment bound Lemma 14(b) is not yet available. And the bootstrap in (20) is why the rate carries a log factor rather than being clean $T^{-1/2}$; Theorem 5 recovers the sharper statement once Theorem 3 is in hand.

A.5 The curvature rate (Theorem 3)

Write $Y_n = \alpha(\theta^\star; \xi_{i_n})\Psi(\theta^\star; \xi_{i_n})\Psi(\theta^\star; \xi_{i_n})^\top$ and $\tilde{Y}_n = \alpha(\theta_{t(n)-1}; \xi_{i_n})\Psi\Psi^\top$ for the term actually accumulated, where $t(n)$ is the block containing i_n. Then, by (4) and (5),

$$\bar{H}_t - H = \underbrace{\frac{\lambda_0 I}{W_{n_t}}}_{\text{(i)}} + \underbrace{\frac{1}{W_{n_t}} \sum_{n \leq n_t} \iota_n e_{k_n} e_{k_n}^\top}_{\text{(ii)}}$$

$$+ \underbrace{\frac{1}{W_{n_t}} \sum_{n \leq n_t} (Y_n - H)}_{\text{(iii)}} + \underbrace{\frac{1}{W_{n_t}} \sum_{n \leq n_t} (\tilde{Y}_n - Y_n)}_{\text{(iv)}} . \quad (21)$$

Since $W_{n_t} = 1 + n_t \asymp t$, term (i) is $O(1/t)$. For (ii), $\|\cdot\|_{\mathrm{op}} \leq n_t^{-1} \sum_{n \leq n_t} \iota_n \leq C n_t^{-q_{\mathrm{r}}}$ because $\iota_n \leq C n^{-q_{\mathrm{r}}}$ and $q_{\mathrm{r}} < 1$; the same bound from below along each basis direction gives, together with $H \succ 0$, $\lambda_{\min}(\bar{H}_t) \geq c\, n_t^{-q_{\mathrm{r}}}$ for all t (this is the only place the ridge is used, and it is what bounds $\|\bar{H}_t^{-1}\|_{\mathrm{op}}$ for *every* t rather than eventually). Term (iv) is bounded, by the local Lipschitz property in Assumption 3 and Cauchy–Schwarz, by $C n_t^{-1} \sum_{n \leq n_t} \|\theta_{t(n)-1} - \theta^\star\| \cdot \|\Psi\Psi^\top\|_{\mathrm{op}}$, which by Lemma 15 is

$$O\Big(n_t^{-1} \sum_{n \leq n_t} n^{-1/2} \log^{1+\varpi} n\Big) = O(n_t^{-1/2} \log^{1+\varpi} n_t) \quad \text{a.s.}$$

The new content is term (iii). The array $(Y_n)_{n \geq 1}$ is the subsequence of a stationary geometrically mixing sequence at spacing m, hence itself stationary with mixing coefficients $\alpha_{\mathrm{mix}}(jm) \leq K c_{\mathrm{mix}}^{jm}$, and has $\eta' > 2$ moments by Assumption 3. Applying Lemma 13 with $a_n \equiv 1$ (i.e. its maximal-inequality half, which does not need square-summability) gives $\|\sum_{n \leq n_t} (Y_n - H)\|_{\mathrm{op}} = O(n_t^{1/2} \log^{1/2+\varpi} n_t)$ a.s., so (iii) $= O(t^{-1/2} \log^{1/2+\varpi} t)$ a.s. Collecting the four terms, $\|\bar{H}_t - H\|_{\mathrm{op}} = O(t^{-\min\{q_{\mathrm{r}}, 1/2\}} \log^{c'} t)$ a.s., which is $O(t^{-\nu_H})$ for every $\nu_H < \min\{q_{\mathrm{r}}, 1/2\}$. Finally, on the event $\|\bar{H}_t - H\|_{\mathrm{op}} \leq \lambda_{\min}(H)/2$ (which holds for all large t a.s.), $\|\bar{H}_t^{-1} - H^{-1}\|_{\mathrm{op}} \leq \|\bar{H}_t^{-1}\|_{\mathrm{op}} \|\bar{H}_t - H\|_{\mathrm{op}} \|H^{-1}\|_{\mathrm{op}} \leq 2\lambda_{\min}(H)^{-2} \|\bar{H}_t - H\|_{\mathrm{op}}$. □

A.6 Proof of Theorem 5 (rate) and Proposition 7 and Theorem 6 (CLT)

Write (6) as $\delta_t = \delta_{t-1} - \gamma_t P_{t-1}\{H\delta_{t-1} + \varepsilon_t + r_t\}$ with r_t as above. Multiplying by t and using $\gamma_t = 1/t$,

$$t\,\delta_t = (t-1)\delta_{t-1} + (I - P_{t-1}H)\delta_{t-1} - P_{t-1}\varepsilon_t - P_{t-1} r_t, \quad (22)$$

so that, summing from $t = 1$ to T (Abel summation) and dividing by T,

$$\delta_T = -\frac{1}{T} H^{-1} \sum_{t \le T} \varepsilon_t \;+\; \underbrace{\frac{1}{T} \sum_{t \le T} (H^{-1} - P_{t-1}) \varepsilon_t}_{\text{(A)}}$$

$$+\; \underbrace{\frac{1}{T} \sum_{t \le T} (I - P_{t-1} H) \delta_{t-1}}_{\text{(B)}} \;-\; \underbrace{\frac{1}{T} \sum_{t \le T} P_{t-1} r_t}_{\text{(C)}} . \tag{23}$$

The leading term is *exactly* $-N_T^{-1} H^{-1} \sum_{i \le N_T} g_i$, because $\sum_{t \le T} \varepsilon_t = b^{-1} \sum_{i \le N_T} g_i$ and $N_T = bT$. This identity is Proposition 7: the leading term does not depend on b.

Term (A). Apply (15) with $\gamma_t \equiv 1$, $\zeta_t = \varepsilon_t$ and the $\mathcal{F}_{t-1}$-measurable weights $A_t = H^{-1} - P_{t-1}$: by Theorem 3 the envelope is $c_t = t^{-\nu_H}$, and by Lemma 14(b) the increments are $u_t = O(t^{-1})$. Hence $V_T = O(T^{1-2\nu_H})$ (as $\nu_H < 1/2$), $U_T = O(\log T)$, and $\gamma_T c_T T^{1/2+\varpi} = O(T^{1/2-\nu_H+\varpi})$, so

$$\Big\| \sum_{t \le T} (H^{-1} - P_{t-1}) \varepsilon_t \Big\| = O\big(T^{1/2-\nu_H} \log^{1/2+\varpi} T\big) + O\big(\log^{2+\varpi} T\big)$$
$$+ O\big(T^{1/2-\nu_H+\varpi}\big) = O\big(T^{1/2-\nu_H+\varpi}\big) \quad \text{a.s.}$$

Choosing $\varpi < \nu_H$ gives (A) $= O(T^{-1/2-\nu_H+\varpi}) = o(T^{-1/2})$ a.s. It is worth seeing what does the work. The only bound available without (13) is $\|\mathbb{E}[(H^{-1} - P_{t-1})\varepsilon_t]\| \le \|H^{-1} - P_{t-1}\|_{\mathrm{op}} \|\varepsilon_t\|_2 = O(t^{-\nu_H} b^{-1/2})$ — Davydov gives no decay across a one-block gap, since the weight and the summand are only one block apart — and averaging that over $t \le T$ leaves $O(b^{-1/2} T^{-\nu_H})$, which is *not* $o(T^{-1/2})$ for any admissible $\nu_H < 1/2$. The extra $T^{-1/2}$ comes entirely from the exact cancellation $\mathbb{E}[A_t m_{t+1} \mid \mathcal{F}_t] = 0$.

Term (B).

$$\|(\mathrm{B})\| \le T^{-1} \sum_{t \le T} \|I - P_{t-1} H\|_{\mathrm{op}} \|\delta_{t-1}\| = T^{-1} \sum_{t \le T} O(t^{-\nu_H}) \|\delta_{t-1}\|,$$

which is a pathwise bound needing no lemma. With the rung-(iii) preliminary rate $\|\delta_t\| = O(t^{-1/2} \log^c t)$ this is $O(T^{-1} \cdot T^{1/2-\nu_H} \log^c T) = o(T^{-1/2})$ a.s.

Term (C). $r_t = (\widehat{H}_t - H)\delta_{t-1} + O(\|\delta_{t-1}\|^2)$, where $\widehat{H}_t$ is the block curvature. The quadratic part contributes $O(T^{-1} \sum_t \log t / t) = O(\log^2 T / T) = o(T^{-1/2})$. For the first part take $\zeta_t = \widehat{H}_t - H$, which is stationary, centred and square integrable under Assumption 3 (it is the block average of the terms appearing in (21) evaluated at $\theta^\star$, plus a Lipschitz remainder in θ_{t-1} bounded exactly as term (iv) there), and apply (15) with $\gamma_t \equiv 1$ and the $\mathcal{F}_{t-1}$-measurable weights acting by $A_t \zeta_t = P_{t-1} \zeta_t \delta_{t-1}$. By Lemma 2 and the rung-(iii) rate the envelope is $c_t = O(t^{q_\mathrm{r}} \cdot t^{-1/2} \log^c t)$ and the increments are $u_t = O(t^{-1} c_t)$, so

$$V_T = O\Big(\sum_{t \le T} t^{2q_\mathrm{r}-1} \log^{2c} t \Big) = O(T^{2q_\mathrm{r}} \log^{2c} T),$$
$$U_T = O\Big(\sum_{t} t^{q_\mathrm{r}-3/2} \log^{c} t \Big) = O(1) \qquad (q_\mathrm{r} < 1/2),$$

and $\gamma_T c_T T^{1/2+\varpi} = O(T^{q_\mathrm{r}+\varpi} \log^c T)$. Hence

$$\sum = O\big(T^{q_\mathrm{r}+\varpi} \log^{c+1/2+\varpi} T\big), \qquad (\mathrm{C}) = O\big(T^{q_\mathrm{r}-1+\varpi} \log^{c'} T\big) = o\big(T^{-1/2}\big) \text{ a.s.}$$

because $q_r < 1/2$. No independence between $\widehat{H}_t$ and δ_{t-1} is used or needed: adjacent blocks are only $O(1/b)$ apart in the sense of Lemma 11, not $O(\rho^b)$, and an earlier version of this proof wrongly appealed to the latter.

Substituting into (23) gives $\|\delta_T\|^2 = O(\log N_T/N_T)$ a.s. (Theorem 5) and

$$\sqrt{N_T}\,\delta_T = -N_T^{-1/2} H^{-1} \sum_{i \le N_T} g_i + o_p(1).$$

The central limit theorem for strongly mixing stationary sequences with geometric coefficients and $2+\nu$ moments [39, 42] gives $N^{-1/2}\sum_{i\le N} g_i \rightsquigarrow \mathcal{N}(0,S)$, and Slutsky completes Theorem 6. The averaged variant follows from the same decomposition with $\gamma_t = ct^{-a}$ and the standard Abel argument for weighted averages [64, 71]. □

A.7 The L^2 rate (Theorem 16)

Take expectations in (23). For the leading term,

$$\mathbb{E}\Big\|N^{-1}H^{-1}\sum_{i\le N} g_i\Big\|^2 = \operatorname{tr}(H^{-1}SH^{-1})/N + o(1/N)$$

by Lemma 11, uniformly in b because it does not involve b at all. For the remainders, replace each almost sure bound above by its L^2 counterpart: the martingale parts are bounded in L^2 by their predictable variations, which were already computed in expectation in the proof of Lemma 13, and the coboundary parts by $\mathbb{E}\|z_1\|\sum_t u_t$. On the descent side, iterating (18) in expectation with $\mathbb{E}\|\bar{g}_t(\theta_{t-1})\|^2 \le C + C'\mathbb{E}[F(\theta_{t-1}) - F(\theta^\star)]$ (Assumption 2) and $\gamma_t^2\|P_{t-1}\|_{\mathrm{op}}^2 = O(t^{2q_r-2})$ (Lemma 2) gives $\mathbb{E}[F(\theta_t) - F(\theta^\star)] = O(\log t/t)$ by the standard Chung-type recursion, and local strong convexity converts this to $\mathbb{E}\|\delta_t\|^2 = O(\log t/t) = O(\log N_t/N_t)\cdot b$; the extra factor b is absorbed because $N_t = bt$ and the leading term is b-free. All constants depend on b only through $\mathbb{E}\|\bar{g}_t\|^2 \le \mathbb{E}\|g_1\|^2$, which is monotone in b, so the bound is uniform over $b \le C\log N$. □

A.8 Coverage of dependence-blind intervals (Proposition 8)

By Theorem 6 and Theorem 3, $\sqrt{N}(\theta_{T,j} - \theta_j^\star) \rightsquigarrow \mathcal{N}(0, V_{jj})$ with $V = H^{-1}SH^{-1}$, while $[\bar{H}^{-1}\widehat{\Gamma}_0\bar{H}^{-1}]_{jj} \to_p [H^{-1}\Gamma_0 H^{-1}]_{jj} = V_{jj}/\kappa_j$. Hence the studentised statistic converges to $\mathcal{N}(0,\kappa_j)$ and $\mathbb{P}(|\mathcal{N}(0,\kappa_j)| \le z_{\alpha/2}) = 2\Phi(z_{\alpha/2}/\sqrt{\kappa_j}) - 1$. For the homogeneous AR design, $\Gamma_k = \sigma_u^2(\phi\psi)^{|k|}\Sigma_x$ gives

$$S = \sigma_u^2\Sigma_x\sum_k(\phi\psi)^{|k|} = \sigma_u^2\Sigma_x\frac{1+\phi\psi}{1-\phi\psi}, \qquad H = \Gamma_0/\sigma_u^2 = \Sigma_x, \qquad V = \kappa\,\sigma_u^2\Sigma_x^{-1},$$

with $\kappa = (1+\phi\psi)/(1-\phi\psi)$, so the inflation is the same in every coordinate. □

A.9 Long-run covariance estimation (Theorem 9)

(i) Bias of batch means. For the idealised estimator built from $\varepsilon_t = \bar{g}_t(\theta^\star)$, stationarity gives $\mathbb{E}[b\,\varepsilon_t\varepsilon_t^\top] = b\Lambda_0 = \sum_{|k|<b}(1-|k|/b)\Gamma_k$. Splitting,

$$b\Lambda_0 = S - \sum_{|k|\ge b}\Gamma_k - \frac{1}{b}\Big(\Delta - \sum_{|k|\ge b}|k|\Gamma_k\Big), \qquad \Delta = \sum_{k\ge 1}k(\Gamma_k + \Gamma_k^\top), \tag{24}$$

and by Lemma 11 both remainder sums are $O(b\rho^b)$, giving $\mathbb{E}[\widehat{S}^{\mathrm{BM}}] = S - \Delta/b + O(\rho^b)$ up to the drift term treated below.

(ii) Bias of the two-scale estimator. With $\Lambda_1 = \mathbb{E}[\varepsilon_0\varepsilon_1^\top] = b^{-2}\sum_{i\in B_0,\, j\in B_1}\Gamma_{j-i}$ and counting multiplicities of each lag (k for $1 \le k \le b$, $2b-k$ for $b < k < 2b$),

$$b(\Lambda_1 + \Lambda_1^\top) = \sum_{1\le|k|\le b} \frac{|k|}{b}\Gamma_k + \sum_{b<|k|<2b} \frac{2b-|k|}{b}\Gamma_k. \tag{25}$$

Adding (24) in the form $b\Lambda_0 = \sum_{|k|<b}(1-|k|/b)\Gamma_k$, the coefficient of Γ_k becomes exactly 1 for $|k| \le b$ and $(2b-|k|)/b \in (0,1)$ for $b < |k| < 2b$, so

$$b\Lambda_0 + b(\Lambda_1 + \Lambda_1^\top) = S - \sum_{|k|\ge 2b}\Gamma_k - \sum_{b<|k|<2b}\Big(1 - \frac{2b-|k|}{b}\Big)\Gamma_k = S + O(\rho^b), \tag{26}$$

using Lemma 11. This is the flat-top window of Politis and Romano [73] at block resolution; the point is that under geometric mixing its bias is exponentially small in b, whereas the batch-means bias in (i) is only $O(1/b)$. The algebraic identity $\widehat{S}^{\mathrm{FT}} = 2\widehat{S}^{\mathrm{BM}}(2b) - \widehat{S}^{\mathrm{BM}}(b)$ (using disjoint pairs) is immediate from $\widehat{S}^{\mathrm{BM}}(2b) = \widehat{S}^{\mathrm{BM}}(b) + (b/T)\sum(\bar{g}_{2k-1}\bar{g}_{2k}^\top + \text{transpose})$.

(iii) Variance. $\widehat{S}^{\mathrm{BM}}_{jk}$ is an average of $T' \asymp N/b$ terms $q_t = b\,\bar{g}_{tj}\bar{g}_{tk}$. These are quadratic in the scores, so bounding their autocovariances by Lemma 11 needs $2+\delta$ moments of q_t, i.e. $4+2\delta$ moments of the scores; this is exactly why Assumption 5 asks for $\mathbb{E}\|g_1\|^{4+\nu'} < \infty$ with a margin $\nu' > 0$ rather than plain fourth moments. Given that, (q_t) is a measurable function of a geometrically mixing sequence at block resolution, hence itself geometrically mixing with $\alpha_{\mathrm{mix}}(k) \le K\rho^{kb}$, and Davydov gives $|\operatorname{Cov}(q_t, q_{t+k})| \le C\rho^{\delta kb/(2+\delta)}$. Summing the geometric series, $\operatorname{Var}[\widehat{S}^{\mathrm{BM}}_{jk}] = T'^{-2}\sum_{t,t'}\operatorname{Cov}(q_t, q_{t'}) = O(1/T') = O(b/N)$, and the same argument applies to the lag-one term $b\,T'^{-1}\sum \bar{g}_{2k-1,j}\bar{g}_{2k,k}$, hence to $\widehat{S}^{\mathrm{FT}}$.

Drift. Write $\bar{g}_t(\theta_{t-1}) = \varepsilon_t + \Xi_t$ with $\Xi_t = \widehat{H}_t\delta_{t-1} + O(\|\delta_{t-1}\|^2)$. Then $\widehat{S}^{\mathrm{FT}}$ built from $\bar{g}_t(\theta_{t-1})$ differs from the idealised version by terms bounded by $b\,T'^{-1}\sum_{t\in\mathcal{T}}\{\|\Xi_t\|^2 + 2\|\Xi_t\|\|\varepsilon_t\|\}$. Two facts are needed. First, on the retained window $t \asymp T$, Theorem 5 gives $\mathbb{E}\|\delta_{t-1}\|^2 = O(\log t/(bt))$. Second, $\widehat{H}_t$ is a b-sample estimate of a $d\times d$ matrix, so for δ independent of block t, $\mathbb{E}\|\widehat{H}_t\delta\|^2_{\mathrm{op}} = \delta^\top\mathbb{E}[\widehat{H}_t^2]\delta \le \big(1 + Cd/b\big)\|H\delta\|^2_{\mathrm{op}}$ under Assumption 3 (the d/b term is the fluctuation $\mathbb{E}[(\widehat{H}_t - H)M(\widehat{H}_t - H)]$, whose trace contributes a factor $\operatorname{tr}(HM)/b$). Hence

$$\begin{aligned} b\,T'^{-1}\sum_{t\in\mathcal{T}}\mathbb{E}\|\Xi_t\|^2 &= O\Big((1+d/b)\,T'^{-1}\sum_t \log t/t\Big) \\ &= O\big((1+d/b)\,b\log^2 N/N\big) = O\big((b+d)\log^2 N/N\big), \end{aligned}$$

and the cross term is of smaller order by Cauchy–Schwarz. Writing $\tau_{b,N} = (b+d)\log^2 N/N$, this is dominated by the $\sqrt{b/N}$ standard deviation in (iii) whenever $b + d = o(\sqrt{N})$ up to logarithms; in the reported experiments $b + d \le 340$ and $N \ge 2.5\times 10^4$. Note that $\tau_{b,N}$ grows as b falls below d, which is visible in Figure D.1(b) and is why we do not recommend $b \ll d$.

(iv) Rate. The entrywise bounds give $\|\widehat{S}^{\mathrm{FT}} - S\|_{\mathrm{op}} \le \|\widehat{S}^{\mathrm{FT}} - S\|_F \le d\max_{jk}|\widehat{S}^{\mathrm{FT}}_{jk} - S_{jk}|$, hence $\|\widehat{S}^{\mathrm{FT}} - S\|_{\mathrm{op}} = O_p(d\sqrt{b/N} + \rho^b + \tau_{b,N})$; the factor d is the price of the conversion and we do not try to remove it (a matrix-Bernstein argument under an explicit sub-exponential condition would give $\sqrt{db/N}$). Choosing $b = \lceil c\log N\rceil$ with $c > 2/\log(1/\rho)$ makes $\rho^b = O(N^{-2})$ and gives $O_p(\sqrt{\log N/N})$. For $\widehat{S}^{\mathrm{BM}}$ the bias term $1/b$ replaces ρ^b and the trade-off $\sqrt{b/N} \asymp 1/b$ gives $b \asymp N^{1/3}$ and $O_p(N^{-1/3})$. Finally, $\bar{H}^{-1} \to_p H^{-1}$ (Theorem 3) and $\widehat{S}^{\mathrm{FT}} \to_p S$ give $u_j^\top\widehat{S}^{\mathrm{FT}}u_j \to_p V_{jj}$, and Theorem 6 with Slutsky gives the studentised limit. □

A.10 Curvature schedules (Proposition 10)

Let $\widehat{H}^{\mathrm{gap}} = n^{-1} \sum_{j \le n} Y_{jm}$. By stationarity,

$$n \operatorname{Var}[\widehat{H}^{\mathrm{gap}}] = \Gamma_0^H + 2 \sum_{j \ge 1} \Big(1 - \frac{j}{n}\Big) \Gamma_{jm}^H = \Gamma_0^H + 2 \sum_{j \ge 1} \Gamma_{jm}^H + o(1) = \Gamma_0^H + O(\rho^m),$$

using $\|\Gamma_k^H\|_{\mathrm{op}} \le C\rho^k$ (Lemma 11 applied to Y, which has $\eta' > 2$ moments by Assumption 3). For Bernoulli thinning, let $Z_i \overset{iid}{\sim} \mathrm{Bernoulli}(p)$ be independent of the data and let

$$\widehat{H}^{\mathrm{Bern}} = \Big(\sum_{i \le N} Z_i\Big)^{-1} \sum_{i \le N} Z_i Y_i$$

be the estimator with the *realised* normaliser — which is what the algorithm computes, since W counts the updates actually made. Using a fixed normaliser pN instead would leave a term $p(1-p)\operatorname{vec}(H)\operatorname{vec}(H)^\top$ coming from $\mathbb{E}[Y_i] = H \ne 0$, which is $\Theta(1)$ and would swamp the $O(1/m)$ effect; the ratio form removes it, because $\widehat{H}^{\mathrm{Bern}} - H = (\sum_i Z_i)^{-1} \sum_i Z_i (Y_i - H)$ *exactly*. Writing $n_Z = \sum_i Z_i$ with $\mathbb{E} n_Z = pN = n$ and $n_Z/n \to 1$ a.s., the delta method for a ratio gives $n \operatorname{Var}[\widehat{H}^{\mathrm{Bern}}] = n^{-1} \operatorname{Var}\big[\sum_i Z_i (Y_i - H)\big] + o(1)$, and

$$\operatorname{Var}\Big[\sum_i Z_i (Y_i - H)\Big] = \sum_i p\, \Gamma_0^H + \sum_{i \ne j} p^2 \Gamma_{i-j}^H = Np\, \Gamma_0^H + 2p^2 \sum_{k \ge 1} (N-k) \Gamma_k^H.$$

Dividing by $n = pN$ and using $p = 1/m$, $n \operatorname{Var}[\widehat{H}^{\mathrm{Bern}}] = \Gamma_0^H + 2p \sum_{k \ge 1} \Gamma_k^H + o(1) = \Gamma_0^H + m^{-1}(S^H - \Gamma_0^H) + o(1)$. □

Remark 5. The contrast is that random thinning leaves a fixed fraction p of *all* lag pairs in the sum, whereas a deterministic gap removes every lag below m. For Gaussian AR(1) covariates and $\alpha \equiv 1$, Γ_k^H decays like ϕ^{2k} and $\kappa_H = (1+\phi^2)/(1-\phi^2)$.

B Additional theory and discussion

B.1 The assumptions in full

Assumption 1 (Regularity)**.** F is twice continuously differentiable with $\|\nabla^2 F(\theta)\|_{\mathrm{op}} \le L$ for all θ; $H = \nabla^2 F(\theta^\star) \succ 0$; and $\nabla^2 F$ is Lipschitz on a neighbourhood of $\theta^\star$, so that $\|\nabla F(\theta) - H(\theta - \theta^\star)\| \le L_\delta \|\theta - \theta^\star\|^2$ for all θ.

Assumption 2 (Expected smoothness and moments)**.** There are $C, C' \ge 0$ with $\mathbb{E}\|g(\theta)\|^2 \le C + C'(F(\theta) - F(\theta^\star))$ for all θ; there is $\eta > 0$ with $\mathbb{E}\|g(\theta)\|^{2+2\eta} \le C_\eta (1 + F(\theta) - F(\theta^\star))^{1+\eta}$; and $\theta \mapsto \mathbb{E}[g(\theta) g(\theta)^\top]$ is continuous at $\theta^\star$.

Assumption 3 (Curvature form)**.** Writing $\Psi = \Psi(\theta; \xi)$ and $\alpha = \alpha(\theta; \xi)$,

$$\nabla^2 F(\theta) = \mathbb{E}[\alpha\, \Psi \Psi^\top], \qquad \alpha \ge 0, \qquad \mathbb{E}\|\alpha\, \Psi\Psi^\top\|_{\mathrm{op}}^{\eta'} \le C_{\eta'}$$

for some $\eta' > 2$, uniformly on a neighbourhood of $\theta^\star$; and $\theta \mapsto \alpha\, \Psi\Psi^\top$ is Lipschitz in θ in L^2 near $\theta^\star$.

Assumption 4 (Identifiability)**.** $\theta^\star$ is the only stationary point of F outside which the gradient points away from it: for every $\eta > 0$, $\inf_{\|\theta - \theta^\star\| \ge \eta} \langle \nabla F(\theta), \theta - \theta^\star \rangle > 0$.

Assumption 4 holds whenever F is convex with a unique minimiser, which covers every objective we consider, and the independent-data analysis needs it too.

Assumption 5 (Geometric mixing)**.** $(\xi_i)_{i\geq 1}$ is strictly stationary and its mixing coefficient (3) decays geometrically: $\alpha_{\mathrm{mix}}(k) \leq K c_{\mathrm{mix}}^k$ for some $c_{\mathrm{mix}} \in (0,1)$, $K < \infty$. Moreover $\mathbb{E}\|g_1\|^{4+\nu'} < \infty$ for some $\nu' > 0$, $\mathbb{E}\|g_1\|^{2+\nu} < \infty$ for some $\nu > 0$ (without loss of generality $\nu \leq 2\eta$), and $S \succ 0$. The fourth moment with a margin is used only for the variance of the covariance estimator in Theorem 9(iii), and Appendix B.2 explains why plain fourth moments would leave the argument one ϵ of integrability short.

B.2 Why the moment condition in Assumption 5 carries a margin

The fourth moment with a margin is used only for the variance of the covariance estimator in Theorem 9(iii): the products $b\,\bar{g}_{tj}\bar{g}_{tk}$ whose autocovariances that bound needs are quadratic in the scores, and Davydov's inequality applied to them requires $2+\delta$ moments of the products, i.e. $4+2\delta$ of the scores. Plain fourth moments would leave the argument one ϵ of integrability short, which is why we ask for the margin rather than quietly using it.

B.3 The order in which the results are proved

Remark 6 (Order of the proofs)**.** Theorem 3 presupposes a rate for the iterates while the results below use Theorem 3, so the logical order must be stated. It is a four-rung ladder, each rung using only the ones below it: (i) Lemma 2, from the ridge alone, with no assumption on the iterates; (ii) Theorem 4, $\theta_t \to \theta^\star$ a.s., using only Lemma 2; (iii) a preliminary rate $\|\theta_t - \theta^\star\| = O(t^{-1/2}\log^{1+\varpi} t)$ a.s. (Lemma 15, proved in Appendix A.4), from (23) with $\bar{H}_{t-1}^{-1}$ bounded by Lemma 2 and $\bar{H}_t \to H$ — consistency only, which follows from (ii); (iv) Theorem 3, and then Theorems 5, 6, 9 and 16. Appendix A is organised in this order, and rung (iii) is where the restriction $q_{\mathrm{r}} < 1/4$ is used — everything else needs only $q_{\mathrm{r}} < 1/2$.

B.4 The central limit theorem with a growing block length

We separate the two cases because only the first is proved unconditionally, and the difference matters: Theorem 9 takes $b \asymp \log N$, so the interval we actually recommend lives in the second case. The proof of the fixed-b statement is almost sure and goes through Lemma 13, whose conclusion carries a finite random constant; for a triangular array in which b changes with N, "finite for each N" is not enough, and the naive route around it — Cauchy–Schwarz on $\mathbb{E}\|H^{-1} - P_{t-1}\|_{\mathrm{op}}\|\varepsilon_t\|$ — is exactly the bound that the proof of term (A) in Appendix A shows is too weak, by a factor $T^{1/2}$. Establishing the uniformity would require redoing Lemma 13 with explicit constants in b, which we have not done. What we can say is that the condition is about the remainder terms and not about the limit, that growing b makes the block scores *less* dependent rather than more, and that empirically coverage is flat in b over $b \in [10, 320]$ at $N = 200{,}000$ (Table D.4(b)), which is the composition the two theorems need. Section 7 lists this as an open point rather than burying it.

The condition $b = o(\sqrt{N})$ is what makes the number of steps $T = N/b$ large enough for the $1/t$ schedule to erase the initial condition: with $\gamma_t = 1/t$ the effect of θ_0 decays like $1/T = b/N$, so its contribution to $\sqrt{N}(\theta_T - \theta^\star)$ is of order $b/\sqrt{N}$. It is amply satisfied by the $b \asymp \log N$ of Theorem 9, and it is the reason Table D.4(b) shows the point estimate degrading once b becomes a substantial fraction of $\sqrt{N}$.

B.5 Which objectives satisfy the identifiability assumption

Assumption 4 holds whenever F is convex with a unique minimiser. That covers every objective we consider — least squares, logistic, ridge, softmax regression and the geometric median — and it is what turns "the gradient step cannot keep moving" into "$\theta_t \to \theta^\star$".

B.6 The two-scale correction and its exact expectation

$\widehat{S}^{\mathrm{FT}}$ coincides exactly with $2\widehat{S}^{\mathrm{BM}}(2b) - \widehat{S}^{\mathrm{BM}}(b)$ if $\hat{\Lambda}_1$ is formed from disjoint rather than all adjacent block pairs. Either way the expectation is that of a lag-window estimator with weights 1 for $|k| \le b$ and $(2b - |k|)/b$ for $b < |k| < 2b$, which is what Theorem 9(i)–(ii) computes.

B.7 The L^2 rate

Theorem 16 (L^2 rate)**.** *Under the assumptions of Theorem 4, $\mathbb{E}\|\theta_t - \theta^\star\|^2 = O(\log(N_t)/N_t)$, uniformly over $b \le C \log N$.*

The L^2 statement is separate from the almost sure one because the drift analysis in Theorem 9 needs a moment bound, and an almost sure rate with an unspecified random constant does not supply one. All remainder bounds in Appendix A are uniform over $b \le C \log N$, the only regime in which we use them.

B.8 Which processes satisfy the mixing assumption

A geometrically ergodic Markov chain [63] satisfies Assumption 5, as do stationary causal ARMA/GARCH-type recursions with geometrically decaying dependence in Wu's sense [98, 100]. Davydov's inequality is used in the form given by Rio [76].

B.9 Gordin's decomposition, in full

Under Assumption 5 the conditional expectations $\mathbb{E}[\varepsilon_n \mid \mathcal{F}_0]$ decay geometrically in L^2, so

$$\varepsilon_t = m_{t+1} + z_t - z_{t+1}, \qquad \mathbb{E}[m_{t+1} \mid \mathcal{F}_t] = 0, \quad \mathbb{E}\|z_1\|^2 < \infty, \tag{27}$$

with (m_t) stationary and square integrable and $\mathbb{E}[m_1 m_1^\top] = S/b$ (Lemma 12). The martingale part converges almost surely because $\sum_t \gamma_t^2 \|\bar{H}_{t-1}^{-1}\|_{\mathrm{op}}^2 = \sum_t O(t^{2q_r - 2}) < \infty$ by Lemma 2, which is the square-summability half of the Robbins–Monro condition; the coboundary telescopes. Appendix A gives the details.

C Experimental details

C.1 Data-generating processes and their population targets

Every simulated design pairs a dependent covariate process with a dependent error or latent process, for the reason given in Remark 7. In all four, $\theta^\star = (-1, \dots, 1)$ evenly spaced in $\mathbb{R}^d$, $d = 20$, and the covariate covariance is the ill-conditioned $\Sigma = Q \operatorname{diag}((i/d)^c) Q^\top$ of Boyer and Godichon-Baggioni [9] with Q a random orthogonal matrix, $c = 2$ (condition number 400) except for the logistic design where $c = 1$ (condition number 20) to keep the linear predictor in a sensible range. The population (H, Γ_0, S) is fixed once per design; Monte Carlo replicates vary only the realised path, which is checked in `tests/test_oracles.py`.

AR-hom (closed form). $x_t = \phi x_{t-1} + \sqrt{1-\phi^2}\Sigma^{1/2}\eta_t$, $u_t = \psi u_{t-1} + \sqrt{1-\psi^2}\sigma_u \varepsilon_t$, $y_t = x_t^\top \theta^\star + u_t$, with η, ε standard normal and independent, $\phi = \psi = 0.7$, $\sigma_u = 1$. Then $H = \Sigma$, $\Gamma_k = \sigma_u^2(\phi\psi)^{|k|}\Sigma$ and $S = \sigma_u^2\Sigma(1+\phi\psi)/(1-\phi\psi)$, so $\kappa_j \equiv 2.922$.

AR-het (closed form). $x_t = A_\phi x_{t-1} + \Sigma_\eta^{1/2}\eta_t$ with $A_\phi = \mathrm{diag}(\phi_1, \dots, \phi_d)$, ϕ_j evenly spaced on $[0, 0.9]$. The stationary covariance is the Hadamard product $\Sigma_x = \Sigma_\eta \circ [1/(1-\phi_i\phi_j)]$, which is positive definite because $[1/(1-\phi_i\phi_j)] = \sum_{k\geq 0}(\phi\phi^\top)^{\circ k}$ is a sum of Hadamard powers of a rank-one positive semidefinite matrix whose first d terms are linearly independent when the ϕ_j are distinct. With AR(1) errors, $\mathbb{E}[x_t x_{t+k}^\top] = \Sigma_x A_\phi^k$, $\Gamma_k = \sigma_u^2\psi^k\Sigma_x A_\phi^k$ and $S = \sigma_u^2(\Sigma_x + \Sigma_x G_\psi + G_\psi\Sigma_x)$ with $G_\psi = \psi A_\phi(I - \psi A_\phi)^{-1}$. Because the ϕ_j differ, κ_j varies across coordinates, so a scalar correction cannot pass this design.

Markov (closed form). s_t is a $K = d+1$ state Markov chain with $\mathbb{P}(s_t = s_{t-1}) = 0.97$, uniform off-diagonal transitions and uniform stationary law; $x_t = V_{\cdot,s_t} + \Sigma_\nu^{1/2}\nu_t$, so covariates are a Markov-modulated Gaussian mixture—non-Gaussian marginals, discrete latent state, regime (non-linear) dependence. The errors are AR(1) with $\psi = 0.8$. Then $\Sigma_x = V\mathrm{diag}(\pi)V^\top + \Sigma_\nu$, $\mathbb{E}[x_t x_{t+k}^\top] = V\mathrm{diag}(\pi)P^kV^\top$ and $S = \sigma_u^2(\Sigma_x + R + R^\top)$ with $R = V\mathrm{diag}(\pi)\{\psi P(I-\psi P)^{-1}\}V^\top$.

The choice of centres matters and is worth recording, because the obvious construction produces a design that *looks* strongly dependent but has no inferential consequence. With randomly drawn centres and $K \ll d$, the regime component $V\mathrm{diag}(\pi)V^\top$ has rank K and sits in high-curvature directions of Σ_x; the sandwich $H^{-1}\Gamma_k H^{-1}$ is then negligible against $H^{-1}\Gamma_0 H^{-1}$, whose mass is in the *low*-curvature directions, and κ_j came out in $[1.00, 1.05]$ for every setting we tried. We therefore take $K = d+1$ and the centres to be a regular simplex scaled by the target covariance: $V = \sqrt{K}\,\mathrm{chol}(\Sigma)U$ with $U \in \mathbb{R}^{d\times K}$ an orthonormal basis of $\mathbf{1}^\perp$ in $\mathbb{R}^K$, which gives $V\pi = 0$ and $V\mathrm{diag}(\pi)V^\top = \Sigma$ exactly. The regime component is then full rank and proportional to Σ, so $R_k = \lambda_2^k\Sigma$ with λ_2 the chain's second eigenvalue and $\kappa_j \to (1+\psi\lambda_2)/(1-\psi\lambda_2)$ as $\Sigma_\nu \to 0$: at our settings, $\kappa_j = 7.55$ in every coordinate against the noise-free limit 7.88.

This design is the one that forces both the longer warm-up and the stability shift, and the same feature causes both. With $\mathbb{P}(s_t = s_{t-1}) = 0.97$ the chain dwells in a regime for about 33 observations and its integrated autocorrelation time is $(1+0.97)/(1-0.97) \approx 66$, so a block of $b = 20$ consecutive observations almost always lies inside a single regime and the block Hessian $\widehat{H}_t$ is nearly rank one. Two consequences, which we separated only after mistaking the second for the first. A warm-up of $50d$ *consecutive* observations spans only about thirty regime visits, so it leaves $\bar{H}$ rank-starved; $200d$ fixes that (Table D.4(c′)). But a well-estimated $\bar{H}$ is not enough, because the near-rank-one $\widehat{H}_t$ makes $\|\bar{H}^{-1}\widehat{H}_t\|_{\mathrm{op}}$ large — median 33 here, against 20 on AR-hom, with block maxima 89 and 55 — so the contraction condition (7) fails at $\gamma_1 = 1$ and the first few steps overshoot by orders of magnitude. At $c_{\mathrm{w}} = 200$ and no shift, the relative error of the point estimate still exceeds one on 7 of 10 replicates. Both fixes are needed, and neither substitutes for the other (Table D.4(i)).

Logistic (simulated oracle). x_t is AR(1) with $\phi = 0.7$; the latent error is $e_t = \mathrm{logit}(\Phi(\nu_t))$ with ν_t a unit-variance AR(1) with $\psi = 0.8$, so e_t has an exact standard logistic marginal and $y_t = \mathbf{1}\{x_t^\top\theta^\star + e_t > 0\}$ satisfies $\mathbb{P}(y_t = 1 \mid x_t) = \sigma(x_t^\top\theta^\star)$ *exactly*. The working logistic likelihood is therefore correctly specified marginally—$\theta^\star$ is the true parameter and $\Gamma_0 = H$ by the information equality—while the score is serially correlated through ν. This is the marginal-model situation of Liang and Zeger [59]. H is computed to machine precision by Gauss–Hermite quadrature using the decomposition $x = (\Sigma_x\theta^\star/\sigma_\ell^2)\ell + x_\perp$ with $\ell = x^\top\theta^\star$ the linear predictor and $x_\perp \perp \ell$; S has no closed form and is obtained from eight independent streams of 10^6 observations with the score evaluated at the known $\theta^\star$ and a Bartlett long-run covariance at lag 100 (the score autocovariance decays like $\psi^k = 0.8^k$, so the truncation error is below 10^{-9}). The resulting relative Monte Carlo error of the sandwich diagonal is 7.5×10^{-3}; it is the only non-exact population quantity in the paper.

C.2 Real streams: features, segments and Protocol B's nominal level

Responses. Both responses are log-transformed and then standardised to zero mean and unit variance, exactly as the covariates are. The standardisation is not cosmetic. Without it the mean of log(traffic volume), about 7.9, sits entirely in the intercept, which is then roughly 400 standard errors from the $\theta_0 = 0$ that a streaming method starts from; a $1/t$ schedule needs of order that many steps merely to cover the distance, so on a 4000-hour segment (which affords only about 100 blocked steps at $b = 40$) the experiment would be measuring the distance to the initialisation rather than anything about inference — the infeasible oracle-variance interval covered 0.14 in that configuration, which is the diagnostic that the point estimate, not the covariance, was at fault. Standardising the response is an affine reparametrisation of a linear model and changes no coverage statement; it is also what any practitioner would do.

Features. Traffic: intercept, standardised temperature and cloud cover, two daily harmonics and one weekly harmonic, a holiday indicator propagated from the labelled midnight row to the whole calendar day, and a weekend indicator ($d = 11$). Rainfall is *excluded*: `rain_1h` is identically zero over stretches of several thousand consecutive hours, so $\log(1 + \text{rain})$ is a constant column on some contiguous segments, which makes the design there exactly rank deficient and the per-segment population targets undefined. Air quality: intercept, standardised temperature, pressure, dew point, $\log(1 + \text{rain})$ and wind speed, plus the same harmonics ($d = 12$). Feature construction was fixed before any response was looked at, and `exp6_real.py` asserts that every evaluation segment has full rank, reporting the worst per-segment condition number.

Segments and hold-out. The leading 15% of each stream is reserved for choosing the first-order baselines' step-size constants and is excluded from every evaluation segment; the constants are chosen per stream, and the grid and whether the optimum is interior are recorded in `results/exp6_real.json`. Evaluation segments are the disjoint contiguous blocks that follow.

Protocol B's nominal level. Let R be the number of evaluation segments, all of the same length, let $\hat{\theta}_k$ be the estimate from segment k and $\hat{\theta}_{\text{full}}$ the full-stream M-estimator. If (i) the segment estimates are approximately uncorrelated, (ii) they have a common variance V, and (iii) $\hat{\theta}_{\text{full}} \approx R^{-1} \sum_k \hat{\theta}_k$, then

$$\operatorname{Var}(\hat{\theta}_k - \hat{\theta}_{\text{full}}) = \operatorname{Var}(\hat{\theta}_k) - \operatorname{Var}(\hat{\theta}_{\text{full}}) = V(1 - 1/R),$$

so an interval built to cover $\theta^\star$ with probability $1 - \alpha$ covers $\hat{\theta}_{\text{full}}$ with probability

$$2\Phi\big(z_{\alpha/2}/\sqrt{1 - 1/R}\big) - 1,$$

which is what the header of Table D.2 reports. Assumption (iii) requires the full-stream estimator to be, approximately, the equally weighted average of the segment estimators, which holds when the segments have equal length and similar $\widehat{H}$; assumption (i) is the same approximation that makes segments usable as replicates at all. The correction is small but not negligible — at the R of our traffic stream it raises the target from 0.950 to 0.976, and at the R of the air-quality stream to 0.994 — and both the corrected target and the raw coverage appear in Table D.2, computed from R rather than quoted from memory. The correction is also least trustworthy exactly where it is largest: with only a handful of segments per stream, assumption (iii) is a crude approximation and the corrected target is close to one, which compresses the comparison. That is the main reason we treat Protocol A, whose target is exact by construction, as the primary real-data evidence and Protocol B as the check that the conclusion survives a real error process.

C.3 Hyperparameters

Unless a sweep says otherwise: block length $b = 20$ ($= d$), gap $m = b$ (one curvature slot per block), no Bernoulli thinning ($p = 1$), log-weights off ($w' = 0$), ridge $c_\iota = 0.01$, $q_{\mathrm{r}} = 0.4$, initial curvature scale $\lambda_0 = 10^{-6}$ (relative to the first block's mean curvature magnitude, so it is dimensionless), warm-up $c_{\mathrm{w}} = 200$ observations per parameter capped at $\varpi_{\mathrm{w}} = 10\%$ of the stream, retained window dyadic (last half to last three quarters of blocks), step-length safeguard constant $c_D = 1$ in (6), and the schedule shift of Appendix E.1 switched off (`t0_mult` $= 0$). Table D.4 varies each of these. The warm-up constant is 200 rather than the 50 that suffices on the autoregressive designs because of the regime-switching design, for the reason given in Appendix C.1. The warm-up and the safeguard address different failures and neither substitutes for the other: the warm-up makes $\bar{H}$ informative, and the safeguard makes the first steps safe to take even when it is. With $c_{\mathrm{w}} = 200$ and no safeguard the regime-switching design still diverges (Table D.4(i)); with the safeguard and $c_{\mathrm{w}} = 50$ it no longer diverges but the curvature is still rank-starved, and the relative variance is 2.19 against 1.41 at $c_{\mathrm{w}} = 200$ (Table D.4(c′)). The two devices are complementary in exactly that way: the safeguard turns a divergence into an inefficiency, and the warm-up removes the inefficiency. The 10% cap on the warm-up never binds in the simulations, where $N = 200{,}000$ and $c_{\mathrm{w}} d \le 4000$; it binds on every real-data segment, where N is 8,000 or 10,000 (Table D.4(j)).

The dependence-blind competitor "streaming Newton, i.i.d. theory" uses the base method's schedule exactly: every observation is a candidate curvature slot ($m = 1$) with Bernoulli retention $p = 1/d$, and the interval uses the i.i.d. plug-in $\widehat{\Gamma}_0 = N^{-1} \sum_i s_i(\theta_{t-1}) s_i(\theta_{t-1})^\top$, accumulated over exactly the observations the blocked estimators use—that is, excluding the warm-up, so that the comparison is not contaminated by scores evaluated at θ_0.

First-order baselines use step $\gamma_t = c\, t^{-3/4}$ (averaged SGD) or a constant c with AdaGrad scaling, with c chosen from a 19-point logarithmic grid on $[10^{-3}, 10^{1.5}]$ minimising squared error on eight *pilot* streams whose seeds are disjoint from the evaluation streams; the selected constants are recorded in `results/exp1_main.json`. They are additionally handed the *true* Hessian for their sandwich. Random scaling uses the two-sided 95% quantile 6.811 of the pivot $|W(1)|/(\int_0^1 (W(r) - rW(1))^2 dr)^{1/2}$, taken from Kiefer and Vogelsang [47] as tabulated by Lee et al. [54, Table 2]. Our own simulation of that quantile, using the exact Karhunen–Loève representation $\int_0^1 B^2 = \sum_{k \ge 1} Z_k^2/(k\pi)^2$ with Z_k i.i.d. standard normal independent of $W(1)$, gives 6.746 from 4×10^6 draws (the sampler is validated by reproducing $\mathbb{E} \int_0^1 B^2 = 1/6$ and $\operatorname{Var} \int_0^1 B^2 = 1/45$); we use the larger published value, which is generous to that baseline. We also flag a normalisation trap: several papers quote 5.32 as "the 95% critical value", which is the 95% quantile of the *signed* statistic, i.e. the two-sided 90% value, and using it would make the random-scaling intervals substantially too narrow.

The iterate-path comparator. Figure 2(a) compares against overlapping batch means computed from the iterate path, in the matrix form of the estimator used in the dependent-data literature: with T recorded iterates, window length b_n, window averages $\bar{\theta}_{b_n,t} = b_n^{-1} \sum_{k=t}^{t+b_n-1} \theta_k$ and overall average $\bar{\theta}$,

$$\widehat{\Sigma}^{\mathrm{it}} = \frac{N}{T} \cdot \frac{b_n}{T - b_n + 1} \sum_t \left(\bar{\theta}_{b_n,t} - \bar{\theta}\right)\left(\bar{\theta}_{b_n,t} - \bar{\theta}\right)^\top, \qquad b_n = \lceil T^{3/4} \rceil,$$

which estimates the covariance of $\sqrt{N}(\bar{\theta}-\theta^\star)$, i.e. the same Σ our sandwich estimates. It uses no gradients and no curvature estimate. We compute it from our own recorded trajectory, with cumulative sums, so it costs $O(Td)$ time and $O(Td)$ memory; it is a reference, not a streaming method, and we make no claim about the constants of any particular published recursion.

C.4 Reproduction and compute

Everything in this paper is public: the estimator, the experiment scripts, the saved result files from which every number, figure and table is generated, and the tests that validate every closed-form population quantity against long simulation. The exact release used to produce the results reported here is archived with a persistent identifier [46],

`10.5281/zenodo.21891313` (development version:
`https://github.com/Kendiukhov/streaming-quasi-newton-dependent-data`),

and the concept identifier `10.5281/zenodo.21891312` always resolves to the most recent release. The two real data sets are third-party and are not redistributed; `fetch_data.py` downloads them from the UCI Machine Learning Repository [14, 40, 45]. `make all` runs everything. All BLAS libraries are pinned to a single thread and parallelism is over Monte Carlo replicates (`experiments/_env.py` explains why: on the machine used, an unpinned tall-skinny $(d,N)\times(N,d)$ product with $N\approx 4\times 10^4$, $d\approx 12$ took 18 s against 5 ms pinned). Total compute for every number in the paper is a few CPU-hours. The machine was shared and its load varied by more than an order of magnitude during the runs, so Table D.5 reports (i) exact machine-independent operation counts computed from the algorithm itself and (ii) wall-clock times measured with the repetitions *interleaved* across methods and summarised by the minimum over repetitions—round-robin ordering exposes every method to the same load pattern, and external load can only make a measurement slower, so the minimum is the robust summary. The operation counts are the authoritative statement; the times corroborate them.

C.5 Additional results

C.6 Designs and methods in full

Designs. Four simulated dependent designs with $d = 20$, described in full in Appendix C.1: **AR-hom** (autoregressive covariates and autoregressive errors, closed-form S, the same variance inflation $\kappa = 2.92$ in every coordinate); **AR-het** (coordinatewise-varying persistence, closed-form S, κ_j ranging over $[1.00, 4.41]$, so a scalar correction cannot pass); **Markov** (covariates are a Markov-modulated Gaussian mixture with $d+1$ regimes, hence non-Gaussian with discrete latent state and regime dependence, closed-form S, $\kappa_j \equiv 7.55$); **Logistic** (a correctly specified marginal logistic model with a serially dependent latent error, so $\theta^\star$ is exactly the generating value while the score is autocorrelated; S from an independent long simulation, relative Monte Carlo error 7.5×10^{-3}). All covariate covariances are the ill-conditioned design of the base paper ($\mathrm{cond}(H)$ up to 400); Table D.3 adds a well-conditioned counterpart, and two real hourly streams are described under Table D.2.

Methods compared. Our estimator BGSN; the same point estimator with plain batch means or with the i.i.d. plug-in variance (so that the interval, and only the interval, changes); the base independent-data streaming Newton method with its own curvature schedule ($m = 1$, Bernoulli $p = 1/d$) and its own prescribed plug-in variance; averaged SGD and streaming AdaGrad with tuned step sizes, each with the i.i.d. plug-in, with our long-run variance, or with random scaling (a studentisation that needs no

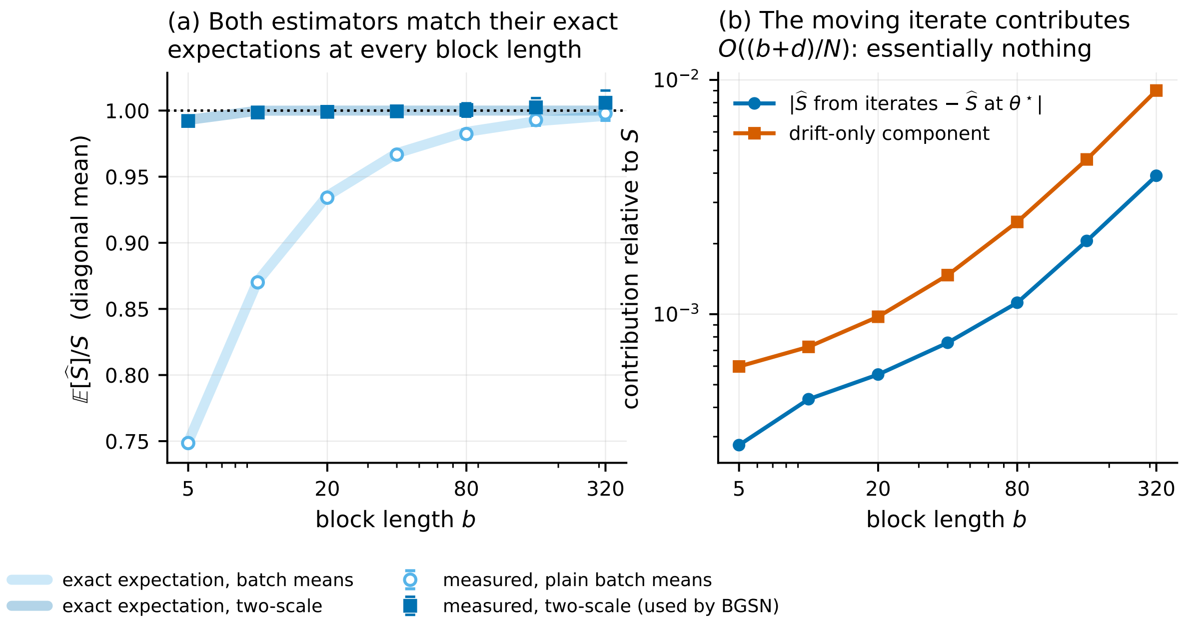

Figure D.1: The long-run covariance estimator, and why the moving iterate does not spoil it. (a) Measured $\mathbb{E}[\widehat{S}]/S$ against block length: plain batch means follow the closed-form bias $1 - \Delta/(bS)$ (no fitted constants), while the two-scale form is unbiased at every b, as Theorem 9 predicts under geometric mixing. (b) The algorithm's block gradients decomposed exactly into the score at $\theta^\star$ and the drift $\widehat{H}_t(\theta_{t-1} - \theta^\star)$. The drift's contribution to $\widehat{S}$ is 0.1% of S at $b = 20$ and 0.9% at $b = 320$, growing like $(b + d)/N$ as Theorem 9 predicts, and the estimator built from the algorithm's iterates is indistinguishable from the one built at $\theta^\star$. This is why our block length need only grow like $\log N$ rather than like a power of N (Remark 1).

variance estimate at all: it divides by a functional of the iterate path whose limit distribution is known and tabulated, 54); an offline HAC reference (heteroskedasticity- and autocorrelation-consistent: the textbook two-pass sandwich with a Bartlett-kernel long-run covariance, which stores the whole stream and is therefore not a streaming method); and an infeasible oracle-variance interval that uses the true H and S. First-order baselines are given the *true* Hessian for their sandwich — an advantage no practitioner has — and their step-size constants are tuned by grid search on pilot streams disjoint from the evaluation streams (Appendix C.3). Our method has no learning rate to tune: its step size is fixed at $1/t$ by the preconditioner, with the step-length safeguard of (6) the only other free quantity and its constant fixed at $c_D = 1$ throughout.

D Additional experimental results

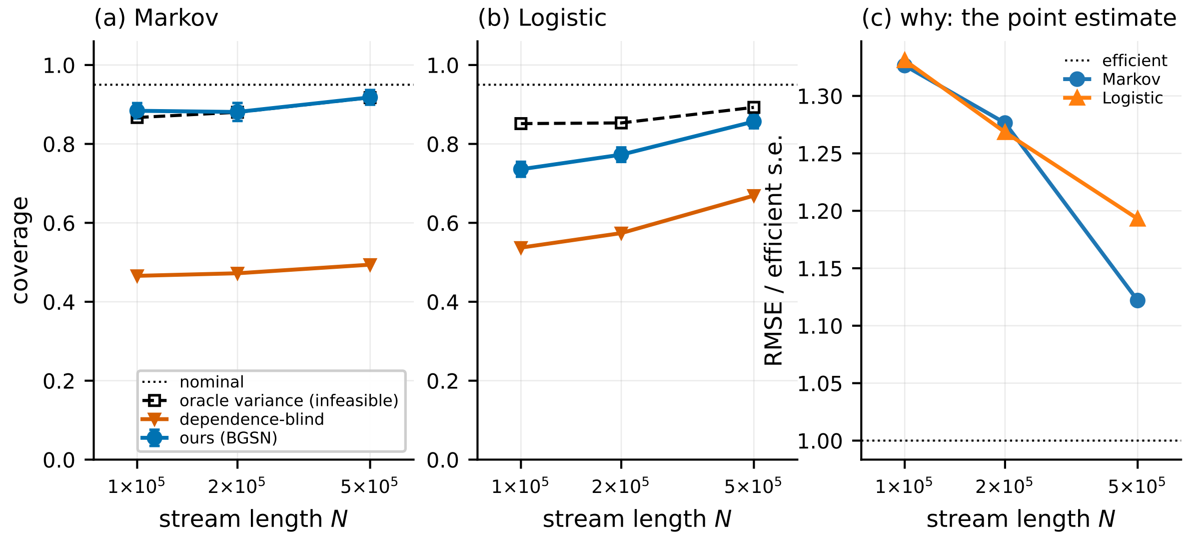

Figure D.2: Separating "the variance estimate is wrong" from "the central limit theorem has not taken hold" on the two designs whose point estimate has not converged at the sample sizes of Table 1. **(a)** The regime-switching design, whose covariate chain has an integrated autocorrelation time of about 66 observations. **(b)** The dependent-logistic design, where instead the curvature weight $\alpha = \sigma'(x^\top\theta)$ is largest at the θ_0 the warm-up evaluates it at. In both, the infeasible curve uses the exact $H^{-1}SH^{-1}$ at the same iterate, so the vertical distance between it and ours is everything our covariance estimator is responsible for: that distance is at most 0.017 on the Markov design and falls from 0.116 to 0.036 on the logistic one. Ours and the infeasible curve rise towards nominal together; the dependence-blind curve does not. **(c)** Why both left panels start low: the root-mean-square error of the point estimate, divided by the efficient standard error, is still above one, so $\sqrt{N}(\theta_T - \theta^\star)$ is not yet close to its limit and no choice of variance can give nominal coverage.

Table D.1: On the two designs whose point estimate has not converged, our interval tracks the infeasible oracle interval at every sample size ($d = 20$, $b = 20$; R decreases from 200 at the smallest N to 150 at the largest, since what is being measured is the gap between the first two columns and a Monte Carlo standard error near 0.01 is ample for it). *ours* is BGSN; *oracle* uses the exact $H^{-1}SH^{-1}$ at the same iterate and is infeasible; *blind* uses the i.i.d. plug-in. *err* is the root-mean-square error divided by the efficient standard error, so err = 1 means the point estimate has reached the asymptotic regime, and *clips* counts activations of the step-length safeguard out of N/b blocks. At $N = 200{,}000$ neither design is nominal because err > 1: the central limit theorem has not taken hold, which no variance estimator can repair. What our estimator is responsible for is the difference between the *ours* and *oracle* columns, and the largest such difference over both designs and all four sample sizes is 0.116. The *blind* column is the one that does not converge. Note that *clips* does not grow with N, which is the empirical content of Proposition 1.

design	N	ours	oracle	blind	err	clips
Markov	100,000	0.884	0.867	0.466	1.33	8
	200,000	0.881	0.880	0.472	1.28	8
	500,000	0.918	0.917	0.494	1.12	7
Logistic	100,000	0.736	0.852	0.537	1.33	2
	200,000	0.773	0.853	0.574	1.27	2
	500,000	0.857	0.893	0.669	1.19	2

Table D.2: Real hourly streams. Protocol A: real covariates, synthetic autoregressive errors whose coefficient equals the AR(1) coefficient of the real residuals, so the target and the long-run covariance are exact conditionally on the covariate path. Protocol B: fully real response, target the full-stream M-estimator; because that target is estimated from a superset of each segment, the correct nominal level for this comparison is $2\Phi(z_{0.025}/\sqrt{1-1/R})-1$, reported in the header. Traffic: $n = 8000$ per segment, $b = 40$. Air quality: $n = 10000$ per segment, $b = 50$, pooled over 12 monitoring sites.

	traffic (I-94)		air quality (PM2.5)	
method	A	B	A	B
nominal	0.950	0.976	0.950	0.994
BGSN (this paper)	0.818	0.795	0.893	0.858
with plain batch means	0.814	0.795	0.892	0.816
with the i.i.d. plug-in variance	0.597	0.682	0.543	0.361
streaming Newton, i.i.d. theory	0.580	0.682	0.500	0.365
ASGD + our long-run variance	0.564	0.523	0.331	0.330
ASGD, i.i.d. plug-in	0.391	0.455	0.198	0.212
ASGD + random scaling	0.571	0.568	0.422	0.486
oracle variance *(infeasible)*	0.773	–	0.788	–
stream diagnostics				
residual AR(1) coefficient	0.756		0.909 (median over sites)	
R^2	0.782		0.327	
block-adequacy r_j	0.163	0.271	0.186	0.302

Table D.3: A well-conditioned dependent design isolates the inference question ($\Sigma_x = I$, so $\mathrm{cond}(H) = 1.00$; otherwise identical to the AR-hom column of Table 1: $d = 20$, $N = 200{,}000$, $b = 20$, $\kappa = 2.92$, $R = 300$). Here every point estimator is efficient ($err \approx 1$), so coverage is decided by the covariance estimate alone. Averaged SGD with our long-run variance is calibrated; the same estimator with the i.i.d. plug-in is not, by the factor $\sqrt{\kappa} = 1.71$ in width that Proposition 8 predicts. On the ill-conditioned designs of Table 1 the first-order baselines cannot be read this way, because there their point estimates have not converged.

method	cov	w	err
BGSN (this paper)	**0.942**	1.00	1.03
with plain batch means	0.933	0.97	1.03
with the i.i.d. plug-in variance	0.743	0.59	1.03
streaming Newton, i.i.d. theory	0.740	0.59	1.03
ASGD + our long-run variance	0.931	1.00	1.07
ASGD, i.i.d. plug-in	0.721	0.59	1.07
ASGD + random scaling	0.917	1.28	1.07
AdaGrad + our long-run variance	**0.943**	1.00	1.03
oracle variance *(infeasible)*	**0.941**	1.00	1.03

Table D.4: Ablations on the homogeneous autoregressive design ($d = 20$, $\kappa = 2.92$, $R = 250$). The three coverage columns — *2-scale*, *BM* and *plug-in* — are coverage of nominal 95% intervals using the two-scale, plain batch-means and i.i.d. plug-in variance respectively; *eff* is the empirical variance of the point estimate divided by its asymptotic value (1.00 is efficient), in scientific notation where the unsafeguarded step diverges; and r_j is the free block-adequacy diagnostic, an estimate of the relative block-length bias of plain batch means. Panel (c′) is on the regime-switching design and is why the default warm-up is $c_w = 200$ rather than 50 (Appendix D.6). Panel (f) varies the Bernoulli retention probability p at gap $m = 1$ and then the gap m at $p = 1$, so the shared baseline $m = 1,\ p = 1$ appears in both sub-sweeps. In panel (i) the designs are abbreviated AR (autoregressive), AR-s (strongly autoregressive) and Mkv (regime-switching), and the variants are: *none*, the raw $1/t$ step; *cap*, the step-length safeguard of (6) at $c_D = 1$ (our default), with *cap 0.25* and *cap 4* changing c_D by a factor of four either way; *shift*, the rejected schedule shift of Appendix E.1; and *both*, the two together. The number of blocks out of 10,000 on which the safeguard acted is reported in Appendix D.6 and in Table D.1. Panel (j) is on a short stream ($N = 4000$, $d = 11$, the length of a real-data evaluation segment) and varies the cap on the warm-up as a fraction of the stream; $\varpi_w = 1$ means uncapped, which spends 2200 of the 4000 observations on curvature and leaves 45 blocked steps.

panel	setting	eff	2-scale	BM	plug-in	r_j
(a) length N	$N = 25{,}000$	1.23	0.930	0.922	0.710	0.086
	$N = 50{,}000$	1.08	0.937	0.931	0.731	0.072
	$N = 100{,}000$	1.04	0.947	0.938	0.733	0.069
	$N = 200{,}000$	1.04	0.947	0.937	0.736	0.069
	$N = 400{,}000$	1.10	0.941	0.934	0.729	0.069
(b) block b	$b = 5$	1.02	0.948	0.905	0.736	0.325
	$b = 10$	1.03	0.949	0.926	0.737	0.149
	$b = 20$	1.04	0.947	0.937	0.736	0.069
	$b = 40$	1.04	0.946	0.943	0.737	0.041
	$b = 80$	1.05	0.946	0.944	0.738	0.045
	$b = 160$	1.06	0.946	0.944	0.734	0.060
	$b = 320$	1.07	0.943	0.944	0.738	0.085
	$b = 640$	1.10	0.938	0.942	0.733	0.118
(c) warm-up	$c_w = 0.5$	1.13	0.928	0.921	0.724	0.069
	$c_w = 5$	1.06	0.941	0.931	0.730	0.069
	$c_w = 25$	1.04	0.944	0.936	0.737	0.068
	$c_w = 50$	1.04	0.944	0.935	0.741	0.068
	$c_w = 100$	1.01	0.949	0.941	0.739	0.069
	$c_w = 200$	1.04	0.947	0.938	0.733	0.069
	$c_w = 500$	1.12	0.937	0.927	0.713	0.069
(c′) warm-up, Mkv	$c_w = 50$	2.19	0.848	0.812	0.452	0.239
	$c_w = 100$	1.76	0.879	0.851	0.467	0.238
	$c_w = 200$	1.41	0.907	0.877	0.495	0.237
	$c_w = 500$	1.21	0.923	0.900	0.536	0.235

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Table D.4, continued from the previous page.

panel	setting	eff	2-scale	BM	plug-in	r_j
(d) ridge	c_ι=0, q_r=0.2	1.04	0.949	0.940	0.741	0.069
	c_ι=0.001, q_r=0.2	1.04	0.949	0.940	0.740	0.069
	c_ι=0.01, q_r=0.2	1.04	0.947	0.937	0.736	0.069
	c_ι=0.1, q_r=0.2	1.04	0.931	0.920	0.703	0.069
	c_ι=1, q_r=0.2	3.07	0.605	0.591	0.405	0.069
	c_ι=0.01, q_r=0.05	1.03	0.943	0.931	0.729	0.069
	c_ι=0.01, q_r=0.1	1.03	0.946	0.934	0.732	0.069
	c_ι=0.01, q_r=0.24	1.04	0.948	0.938	0.737	0.069
	c_ι=0.01, q_r=0.4	1.04	0.949	0.939	0.740	0.069
	c_ι=0.01, q_r=0.49	1.04	0.949	0.940	0.740	0.069
(e) H_0 scale	$\lambda_0 = 1e-08$	1.04	0.947	0.937	0.736	0.069
	$\lambda_0 = 1e-06$	1.04	0.947	0.937	0.736	0.069
	$\lambda_0 = 0.0001$	1.04	0.947	0.937	0.736	0.069
	$\lambda_0 = 0.01$	1.04	0.947	0.937	0.736	0.069
	$\lambda_0 = 1$	1.03	0.947	0.935	0.734	0.069
(f) curvature	$m = 1,\ p = 1$	1.02	0.949	0.940	0.740	0.069
	$m = 1,\ p = 0.5$	1.02	0.949	0.940	0.739	0.069
	$m = 1,\ p = 0.2$	1.03	0.948	0.939	0.739	0.069
	$m = 1,\ p = 0.05$	1.03	0.948	0.937	0.737	0.069
	$m = 1,\ p = 1$	1.02	0.949	0.940	0.740	0.069
	$m = 5,\ p = 1$	1.03	0.949	0.938	0.737	0.069
	$m = 20,\ p = 1$	1.04	0.947	0.937	0.736	0.069
(g) strong dep.	$b = 20$	1.04	0.944	0.921	0.558	0.177
	$b = 40$	1.05	0.946	0.934	0.556	0.082
	$b = 80$	1.07	0.948	0.942	0.553	0.053
	$b = 160$	1.09	0.947	0.946	0.555	0.063
	$b = 320$	1.11	0.945	0.946	0.549	0.086
(h) averaged	N = 25,000	1.28	0.923	0.916	0.701	0.079
	N = 100,000	1.11	0.937	0.927	0.718	0.064
	N = 400,000	1.13	0.939	0.930	0.725	0.067

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Table D.4, continued from the previous page.

panel	setting	eff	2-scale	BM	plug-in	r_j
(i) safeguard	AR, none	1.05	0.947	0.937	0.733	0.069
	AR, shift	1.07	0.942	0.935	0.729	0.069
	AR, cap	1.04	0.947	0.937	0.736	0.069
	AR, cap 0.25	1.04	0.945	0.937	0.736	0.069
	AR, cap 4	1.04	0.948	0.937	0.736	0.069
	AR, both	1.07	0.942	0.935	0.728	0.069
	AR-s, none	1.22	0.922	0.897	0.537	0.177
	AR-s, shift	1.12	0.936	0.911	0.527	0.177
	AR-s, cap	1.04	0.944	0.921	0.558	0.177
	AR-s, cap 0.25	1.05	0.943	0.920	0.553	0.177
	AR-s, cap 4	1.05	0.943	0.918	0.559	0.177
	AR-s, both	1.12	0.936	0.911	0.527	0.177
	Mkv, none	1.1×10^{14}	0.228	0.204	0.484	0.896
	Mkv, shift	3.08	0.785	0.746	0.372	0.242
	Mkv, cap	1.41	0.907	0.877	0.495	0.237
	Mkv, cap 0.25	1.59	0.883	0.848	0.480	0.237
	Mkv, cap 4	4.3×10^{7}	0.518	0.484	0.526	0.520
	Mkv, both	3.10	0.785	0.745	0.372	0.242
(j) warm-up frac.	$\varpi_w = 0.02$	1.40	0.937	0.946	0.723	0.190
	$\varpi_w = 0.05$	1.28	0.935	0.939	0.721	0.197
	$\varpi_w = 0.10$	1.28	0.929	0.930	0.715	0.203
	$\varpi_w = 0.25$	1.50	0.880	0.894	0.644	0.233
	$\varpi_w = 1.00$	2.51	0.787	0.789	0.515	0.281

Table D.5: Cost. Wall-clock milliseconds for one pass over $N = 200{,}000$ observations, single-threaded, minimum over 7 interleaved repetitions (interleaved so that a change in the load of this shared machine cannot be mistaken for a difference between methods; the minimum because load can only make a measurement slower). s_{time} is the exponent in d fitted to those times on the four largest d. It is dominated by the one-off $O(c_w d^3)$ warm-up here and should *not* be read as the streaming cost: s_{str} is the exact operation count with the warm-up excluded and s_∞ is it at $N = 10^8$, where the warm-up is negligible and (8) reduces to $O(dN)$. Those last two columns are the design claim: the streaming pass is linear in d for BGSN and quadratic once the i.i.d. plug-in variance is added, because the plug-in score covariance costs $O(d^2)$ per *observation* while our long-run covariance costs $O(d^2)$ per *block*. The measured *levels* say the same thing and are robust to load: the plug-in variant costs between 1.4 and 3.4 times more than BGSN, at every d here.

method	d=5	d=10	d=20	d=40	d=80	d=160	s_{time}	s_{str}	s_∞
BGSN (long-run interval)	7.4	6.2	10.3	21.4	115.4	534.9	1.95	1.00	1.01
with the i.i.d. plug-in variance	10.2	10.9	31.8	73.5	313.9	1392.3	1.85	1.86	1.86
streaming Newton, Bernoulli $p = 1/d$	9.4	7.7	11.2	27.0	111.2	533.9	1.88	–	–
streaming Newton, every observation ($p = 1$)	20.6	28.8	78.4	260.7	1135.1	4608.5	1.98	1.97	1.97
ASGD (first order, random scaling)	4.2	3.5	5.5	8.3	20.2	51.5	1.09	–	–
offline HAC, $L = 100$ *(not streaming)*	151.1	263.2	604.1	2303.0	–	–	–	–	–

D.1 The two designs whose point estimate has not converged

The Markov and Logistic columns need a caveat of their own, and it is the sharpest test of whether our variance estimator is really doing the work we claim. Neither design's point estimate has reached its asymptotic regime at $N = 200{,}000$, for two different reasons. The Markov design's regime chain has an integrated autocorrelation time of 66 observations — the sum $1 + 2\sum_{k\geq 1}$ of its autocorrelations, which is the factor by which dependence multiplies the variance of a sample mean, so that N dependent observations carry about as much information as $N/66$ independent ones — so its effective sample size is a small fraction of N and the point estimate is 1.33 times the efficient standard deviation. On the Logistic design the cause is the curvature rather than the dependence: the weight $\alpha = \sigma'(x^\top\theta)$ is largest at $\theta_0 = 0$, where the warm-up evaluates it, so $\bar{H}$ built there overestimates H and the early steps are correspondingly too short; the point estimate is 1.33 times efficient and the fitted standard error is biased low with it. An interval built from the *correct* asymptotic variance must undercover in that situation, no matter who supplied the variance, simply because $\sqrt{N}(\theta_T - \theta^\star)$ is not yet close to its limit. So on these two designs the question is not whether coverage is nominal but whether our interval matches the interval that uses the exact $H^{-1}SH^{-1}$ at the same iterate. Table D.1 and Figure D.2 answer it: over a sweep in N both rise towards nominal together while the dependence-blind interval does not. On the Markov design the agreement is essentially exact: our coverage is within 0.017 of the oracle's at every N, and at the top of the sweep, $N = 500{,}000$, both are 0.918 and 0.917 with the point estimate down to 1.12 times efficient, against 0.494 for the dependence-blind interval. The direction of travel is the whole point. Across the fivefold increase in N our coverage and the oracle's move together towards 0.95 and stay within 0.017 of each other, while the dependence-blind interval creeps up by a few hundredths towards its own asymptote, which Proposition 8 puts at $2\Phi(z_{0.025}/\sqrt{7.55}) - 1$, far below nominal. We stop the sweep at $N = 500{,}000$ rather than continuing to where coverage is nominal, because what the experiment is for is the gap between the first two columns, and Table D.1 shows that gap is already at the level of the Monte Carlo standard error there. On the Logistic design there is a real gap, and it is ours: our coverage is 0.736 against the oracle's 0.852 at $N = 100{,}000$, a difference of 0.116, falling monotonically to 0.036 at $N = 500{,}000$, the top of the sweep. The cause is visible in Table 1: our fitted width there is 0.84 times the oracle's, i.e. the standard error is biased low, because $\bar{H}$ is an average over the whole stream and the early blocks contribute $\alpha = \sigma'(0) = 1/4$, its largest possible value, so $\bar{H}$ overestimates H and $\bar{H}^{-1}S\bar{H}^{-1}$ underestimates the sandwich. That bias decays with the Cesàro average of $\|\theta_t - \theta^\star\|$, which is why the gap closes; it is a finite-sample property of averaging curvature along the path, it affects the base method identically, and no choice of long-run covariance estimator would remove it. That is the signature we want: our error tracks the oracle's at every sample size, and the dependence-blind error is the one that survives $N \to \infty$. The step-length safeguard bound at most 8 times per run anywhere in that sweep, and the count does not rise with N, which is the empirical content of Proposition 1.

We report every design in Table 1 at the same N rather than tuning N per design, which is why the Markov column looks worst there; Table D.1 is where that column is resolved.

The step-length safeguard is what makes the hard design run at all. Table D.4(i) varies the safeguard. With none, the autoregressive designs are barely affected (1.05 against 1.04 in relative variance) but the regime-switching design diverges: relative variance 1.1×10^{14} against 1.41 with the safeguard, and coverage 0.228 against 0.907. The safeguard is not doing this by damping the algorithm into submission: it binds a median of 7 times in 10,000 blocks on that design and 3 times on AR-hom, and Table D.1 shows the count flat as N grows by a factor of five.

The constant c_D behaves the way a safeguard constant should, which is to say it is irrelevant until it is not. On AR-hom, changing it by a factor of four in either direction moves nothing (1.04 at $c_D = 1/4$ and

1.04 at $c_D = 4$, against 1.04 at our default $c_D = 1$). On the design that actually needs the safeguard both directions cost something, and asymmetrically: $c_D = 1/4$ gives 1.59 against 1.41 — binding 21 times instead of 7, it now restrains steps that were fine — while $c_D = 4$ is too loose to prevent the blow-up at all and the relative variance is 4.3×10^7. So c_D is a tuning constant with a real failure mode on one side, and we say so rather than advertising insensitivity we only observe on the easy designs.

The schedule shift of Appendix E.1 is the alternative the contraction condition suggests more directly, and it is measurably worse: 3.08 against 1.41 on the regime-switching design, because it slows all 10^4 steps to correct a handful. We report it because we implemented it first, and because the comparison is the evidence for the design we kept.

D.2 The first-order baselines and the conditioning of the design

The first-order rows of Table 1 need a caveat, and it cuts against us if we hide it. These designs are ill-conditioned — $\mathrm{cond}(H) = 400$ for AR-hom — which is precisely the regime streaming second-order methods exist for, and there averaged SGD's *point* estimate has not converged: its root-mean-square error is 9.77 times the efficient value against 1.01 for BGSN. Its intervals are therefore uninformative about the inference question, whatever covariance they use. Two pieces of evidence show that this is about conditioning and not about our construction. First, within Table 1 itself: streaming AdaGrad, which rescales per coordinate, is efficient on AR-het (1.03) and with our long-run variance it covers 0.947 — so the construction transfers to a first-order method as soon as that method's point estimate converges. Second, Table D.3 repeats the AR-hom design with $\Sigma_x = I$, where every point estimator is efficient (1.07 for averaged SGD, 1.03 for BGSN) and only the covariance can decide coverage: averaged SGD with our long-run variance covers 0.931, the same estimator with the i.i.d. plug-in covers 0.721, and BGSN covers 0.942.

D.3 Distributional accuracy beyond a single quantile

Coverage checks a single quantile, while the theorem claims a whole distribution, so we also report the Kolmogorov–Smirnov distance between the studentised error (the error divided by its own estimated standard error, which is what an interval implicitly compares to a normal quantile) $\sqrt{N}(\theta_{T,j} - \theta^\star_j)/\{u_j^\top \widehat{S}^{\mathrm{FT}} u_j\}^{1/2}$ and the standard normal, taken per coordinate across replicates. On AR-hom the worst of the d coordinates gives 0.089 against a 5% critical value of 0.079: marginally outside, so at this number of replicates the studentised distribution is distinguishable from normal, and we report that rather than rounding it in our favour. The dependence-blind studentisation gives 0.207, an order of magnitude further out. The comparison, not the test outcome, is the content: our studentisation is at the edge of detectable non-normality while the dependence-blind one is not close. The standard errors themselves are accurate: 0.997 times the truth on average.

D.4 Our estimator against an iterate-path estimator

Panel (a) puts our estimator and an iterate-path estimator on the same object, because an interval uses the whole sandwich $\Sigma = H^{-1} S H^{-1}$ rather than S alone. We compare our composite $\bar{H}^{-1}\widehat{S}^{\mathrm{FT}}\bar{H}^{-1}$, whose relative error falls with fitted exponent -0.43, with overlapping batch means on the *iterate* path at the block length its analysis prescribes — the estimator of Σ that uses no gradients and no curvature.

Two things must be said carefully here, because the comparison is easy to overstate. First, the iterate-path estimator's relative error is flat over our range: the fitted exponent is 0.00, and we do *not* read that as its rate. Its published rate is $N^{-1/8}$, which over a sixteenfold increase in N predicts a reduction by a factor $16^{-1/8} = 0.71$ — a change small enough that our range cannot separate it from zero, and our fitted exponent should be treated as uninformative about it rather than as a measurement. Second, what our range *does* establish is the level, and the level is the practical statement: over the whole range the iterate-path estimator's relative error in Σ sits near 0.86, which is not a usable standard error whatever its asymptotic rate, while ours falls from 0.17 to 0.05. We are comparing an estimator that has entered its asymptotic regime with one that has not, and the honest claim is about the sample sizes a practitioner would actually have, not about which exponent is larger.

D.5 Curvature schedules

Figure 3 compares deterministic gapping with Bernoulli thinning at a matched number of rank-one updates, with the optimisation frozen at $\theta^\star$ so that only estimation quality is measured. The excess variance over the independent-data benchmark follows the two predicted curves — $O(\rho^m)$ for gapping, $(\kappa_H - 1)/m$ for thinning — over more than an order of magnitude in m. As stated in Section 7, this is a curvature-side result: Table D.4(f) shows that the end-to-end effect on coverage is within Monte Carlo error, which is what the theory predicts, since $\bar{H}$ enters the limit only through a rate.

Where gapping does pay is cost, and Table 1 measures it. Compare BGSN with the row "base method with our long-run variance": the two use the same variance estimator and the same number of rank-one curvature updates per observation ($1/d$), and they cover the same (0.949 against 0.949 on AR-hom). They differ only in *how* those updates are placed — on a deterministic gap of $m = d$ against Bernoulli retention with $p = 1/d$. At these settings the wall-clock difference is 1.04×, which is within our timing noise, so we do not claim a speed-up: a gap avoids drawing and testing a random variable per observation, but that saving is small next to the $O(d^2)$ update it guards. The case for gapping is therefore the one Proposition 10 makes and Figure 3 measures — a strictly better curvature estimate at a matched number of updates, $O(\rho^m)$ excess variance against $(\kappa_H - 1)/m$ — together with the fact that it makes the per-block cost deterministic rather than random. It is a better cost knob, not a faster one.

D.6 Ablations

Figure 5 and Table D.4 vary one design choice at a time. (a) Coverage rises monotonically to nominal with N, and the gap to the dependence-blind interval does *not* close — it converges to the predicted 0.749. (b) Coverage as a function of b is flat for the two-scale estimator over $b \in [10, 320]$ and degrades at both ends for plain batch means (bias at small b, noise at large b): the practical content of the $O(\rho^b)$-versus-$O(1/b)$ distinction. (c) The curvature warm-up matters, though less than it did before the step-length safeguard absorbed part of its job: with essentially no warm-up, coverage on this design is 0.928 against 0.949 at the default, and above $c_{\mathrm{w}} \approx 25$ the result is insensitive. (c′) is where the warm-up is load-bearing: on the regime-switching design the relative variance is 2.19 at $c_{\mathrm{w}} = 50$ against 1.41 at $c_{\mathrm{w}} = 200$, because there $50d$ consecutive observations span only about thirty regime visits and leave the curvature estimate rank-starved. That is what sets our default, and it is a statement about effectively distinct design points rather than about observations. (d) Results are insensitive to the ridge exponent q_{r} over the whole admissible range, and to the ridge constant c_ι from 0 — no ridge at all — up to our default 0.01; at $c_\iota = 0.1$ coverage slips by two points and at $c_\iota = 1$ the ridge dominates the data terms and the estimator is no longer efficient. So the ridge is safe to leave alone but not safe to enlarge, and the honest reading is that the default sits at the top of the flat region rather than in the middle of it. That $c_\iota = 0$ works as well as the default is worth stating plainly: the ridge is in the algorithm so that Lemma 2 can hold for *every* t without an assumption on the iterates, not because the runs need it. (e) Results are insensitive to the initial curvature scale λ_0 over eight orders of magnitude, which matters because that is a parameter a user would otherwise have to think about.

(f) Gap and thinning behave as Proposition 10 predicts and do not move coverage, which is what the theory says: $\bar{H}$ enters the limit only through a rate. (g) At strong dependence ($\phi = \psi = 0.85$, $\kappa = 6.2$) the two estimators separate in exactly the way the $O(\rho^b)$-versus-$O(1/b)$ distinction predicts. Plain batch means need a long block: coverage rises from 0.921 at $b = 20$ to 0.946 at $b = 160$. The two-scale correction does not: it is already at 0.944 at $b = 20$ and stays there (0.945 at $b = 160$). The free diagnostic r_j falls from 0.177 to 0.086 over the same range, which is what makes it usable as a guide to whether plain batch means would have been adequate. So the correction is what buys the short block, and the short block is what keeps the estimator cheap. (h) The weighted averaged variant is slightly better at small N and slightly worse at large N; both are reported. The non-positive-definiteness fallback of Remark 2 never triggered anywhere: the rate is 0 in every simulation and 0.0% of coordinate-replicate pairs on the real streams, where an earlier version of these experiments did not record it at all and where we would have expected it if anywhere, since there the lag-one correction is largest. It never triggered (0.0% of all runs, all panels).

D.7 Cost: the correct interval is also the cheaper one

The authoritative statement is the operation count, because it is exact and independent of the machine. The streaming pass grows as $d^{1.00}$ for BGSN and as $d^{1.86}$ once the i.i.d. plug-in variance is added, because the plug-in score covariance costs $O(d^2)$ per *observation* whereas our long-run covariance costs $O(d^2)$ per *block*; evaluating the whole count at $N = 10^8$ gives $d^{1.01}$ against $d^{1.86}$. Those are the design claim, and they are arithmetic rather than a fit.

The exponents fitted at the N of Table D.5 are *not* the streaming cost, and we say so rather than quoting them as if they were: with $c_w = 200$ and $N = 200{,}000$ the one-off $O(c_w d^3)$ warm-up dominates the $O(dN)$ pass for $d \geq 40$ (the crossover is $N \gtrsim c_w d^2$), so both measured exponents sit near 2. What is robust at this N is the *level*: the plug-in variant costs more than BGSN in wall-clock at every d we tried, by a factor between 1.4 and 3.4 (median 2.7), and in the main configuration BGSN runs in 0.37 times the time of the dependence-blind streaming Newton method — that is, our valid intervals cost less than the invalid ones, not more. The measured wall-clock exponents, on a shared and heavily loaded machine, are $d^{1.95}$ and $d^{1.85}$; they are noisier than the counts and we report them only as corroboration. Figure 4 plots the same timings against d, where the message is the vertical separation rather than either slope.

D.8 Choosing the block length on a real stream

Figure 6(c) shows why the block length is the one parameter a practitioner must think about on real data, and that the free diagnostic $r_j = |u_j^\top(\widehat{S}^{\mathrm{FT}} - \widehat{S}^{\mathrm{BM}})u_j|/(u_j^\top \widehat{S}^{\mathrm{BM}} u_j)$ flags the problem. The denominator is the batch-means quadratic form, not the two-scale one: $\widehat{S}^{\mathrm{BM}}$ is a sum of outer products and so its quadratic forms are non-negative, whereas those of $\widehat{S}^{\mathrm{FT}}$ can be negative (Remark 2), and an earlier version of this diagnostic divided by the latter and produced meaningless numbers on the real streams. On the traffic stream r_j is 0.16 at the $b = 40$ we use, and the sweep in Figure 6(c) shows what it is good for: r_j is U-shaped in b, rising both for short blocks, where plain batch means carry the $O(1/b)$ bias, and for long ones, where too few blocks remain for a stable estimate. Our coverage over that fortyfold range in b is flat and stays within a few hundredths of the infeasible oracle-variance curve, which is the evidence that the shortfall from nominal on these segments is the point estimate rather than the covariance. So r_j is a guide to choosing b, not a calibration guarantee, and we do not claim it predicts coverage.

E Safeguarding the blocked step

Proof of Proposition 1. On $\mathcal{A}$ there are finitely many t at which Π_t is active, so $T_0 = \max\{t : \Pi_t \text{ active}\}$ is finite and for $t > T_0$ the recursion is $\theta_t = \theta_{t-1} - \gamma_t \bar{H}_{t-1}^{-1} \bar{g}_t(\theta_{t-1})$ verbatim. Two sequences that agree from a finite index on have the same limit points, the same almost sure rate and the same limit law, since all three are tail properties; the initial condition at T_0 is a finite random vector, and Theorems 4 to 6 hold for the recursion started from any initial condition (Assumption 4 is global). For the last claim, if $\theta_t \to \theta^\star$ and $\|\gamma_t \bar{H}_{t-1}^{-1} \bar{g}_t(\theta_{t-1})\| \to 0$ almost surely then for almost every ω there is $T_1(\omega)$ beyond which $\|\gamma_t \bar{H}_{t-1}^{-1} \bar{g}_t\| < c_D(1 + \|\theta^\star\|)/2 \le c_D(1 + \|\theta_{t-1}\|)$, so Π_t is inactive for $t > T_1$ and $\omega \in \mathcal{A}$. $\square$

We repeat here what Section 2 says, because it is the kind of gap a reader is entitled to be told about twice. Proposition 1 is conditional on $\mathcal{A}$ and does not establish $\mathbb{P}(\mathcal{A}) = 1$ unconditionally: the implication runs from convergence to eventual inactivity, not the other way. The reason we do not close the loop is that the safeguard factor $\min\{1, c_D(1 + \|\theta_{t-1}\|)/\|v\|\}$ depends on $\bar{g}_t$, hence is $\mathcal{F}_t$- and not $\mathcal{F}_{t-1}$-measurable, and multiplying the noise term $\gamma_t \bar{H}^{-1} \varepsilon_t$ by an $\mathcal{F}_t$-measurable factor destroys the martingale-difference property that Theorems 4 and 6 rest on. Two routes would close it and both are outside our scope: the expanding-truncation construction of Kushner and Yin [52] and Andrieu et al. [4], which replaces the cap by a projection onto a ball of radius increasing with the number of activations and proves the activation count finite; or an $\mathcal{F}_{t-1}$-measurable variant that caps using the *previous* block's amplification, $\gamma_t \wedge \tau / \|\bar{H}_{t-2}^{-1} \widehat{H}_{t-1}\|_{\mathrm{op}}$, which is predictable and so leaves the martingale structure intact, at the cost of an extra $O(d^2)$ per block. What we do instead is report the activation counts, which are between 2 and 7 per run and do not grow with N.

E.1 The step-size shift: the alternative we implemented and rejected

The variant replaces the step sequence $\gamma_t = c\, t^{-a}$ of (6) by $\gamma_t = c\,(t + t_0)^{-a}$ with

$$t_0 = \max\left\{0,\ \tfrac{c}{2} \max_{t \le T_{\mathrm{w}}} \|\bar{H}_{T_{\mathrm{w}}}^{-1} \widehat{H}_t\|_{\mathrm{op}} - 1\right\}, \tag{28}$$

$T_{\mathrm{w}} = \lceil c_{\mathrm{w}} d / b \rceil$ being the last warm-up block, and switches the step-length safeguard Π_t off. By construction (7) then holds at $t = 1$ for every block whose amplification does not exceed the warm-up maximum. The spectral norms are obtained by four power iterations per warm-up block with $\widehat{H}_t$ applied in factored form, so no $d \times d$ product is formed and the whole computation costs $O(c_{\mathrm{w}}(d^2 + db)\, d/b)$ flops once, $O(1)$ in N; with $b = m = d$ that is $O(c_{\mathrm{w}} d^2)$, some 4–5% of the total at the settings of Table D.5. Setting `t0_mult` = 0.5 and `step_cap` = 0 in the code selects this variant.

Correctness of the rejected shift variant. Write T_{w} for the last warm-up block. For every $t \le T_{\mathrm{w}}$, $\|\bar{H}_{T_{\mathrm{w}}}^{-1} \widehat{H}_t\|_{\mathrm{op}} \le \|\bar{H}_{T_{\mathrm{w}}}^{-1}\|_{\mathrm{op}} \|\widehat{H}_t\|_{\mathrm{op}}$. The first factor is finite almost surely by Lemma 2, which bounds it by $c_1^{-1} n_{T_{\mathrm{w}}}^{q_{\mathrm{r}}}$ using only the ridge in (4) and no property of the iterates. The second is finite almost surely because $\|\widehat{H}_t\|_{\mathrm{op}} \le b^{-1} \sum_{i \in B_t} \alpha_i \|\Psi_i\|^2$ and Assumption 3 gives $\mathbb{E}\|\alpha \Psi \Psi^\top\|_{\mathrm{op}}^{\eta'} \le C_{\eta'}$ with $\eta' > 2$, so each summand is finite a.s. and the maximum is over the finitely many blocks $t \le T_{\mathrm{w}}$. Hence $t_0 < \infty$ a.s. Measurability with respect to $\mathcal{F}_{T_{\mathrm{w}}} = \sigma(\xi_1, \ldots, \xi_{bT_{\mathrm{w}}})$ is immediate: (28) is a function of $\bar{H}_{T_{\mathrm{w}}}$ and of $\{\widehat{H}_t\}_{t \le T_{\mathrm{w}}}$, all of which are built from the warm-up window and from θ_0, and the warm-up takes no parameter step.

For the second claim, fix ω in the almost sure event $\{t_0 < \infty\}$ and let $K = \lceil t_0 \rceil$. Then $c(t + t_0)^{-a} \ge c(t + K)^{-a}$, and $\sum_{t \ge 1} (t + K)^{-a} = \sum_{t > K} t^{-a} = \infty$ for $a \le 1$, so $\sum_t \gamma_t = \infty$. Likewise $c(t + t_0)^{-a} \le c t^{-a}$, so $\sum_t \gamma_t^{1+\delta} \le c^{1+\delta} \sum_t t^{-a(1+\delta)} < \infty$ whenever $a(1 + \delta) > 1$, which is the same condition as for the unshifted sequence. Finally $(t+t_0)^{-a}/t^{-a} = (1+t_0/t)^{-a} \to 1$. Conditioning on $\mathcal{F}_{T_w}$ is harmless because $\mathcal{F}_{T_w}$ is generated by finitely many observations, so it changes no tail statement and, by Assumption 5, leaves the stream after the warm-up geometrically mixing with the same constants up to a factor.

Theorem 4 and the preliminary rate use the step sequence only through $\sum_t \gamma_t = \infty$ and $\sum_t \gamma_t^2 \|P_{t-1}\|_{\mathrm{op}}^2 < \infty$, both just verified, so they transfer immediately. Theorems 5 and 6 and Proposition 7 deserve one extra line, because their proof uses the identity $\gamma_t = 1/t$ explicitly in the Abel summation (22), and we would rather write the shifted identity out than assert that nothing changes. With $\gamma_t = (t+t_0)^{-1}$, multiplying $\delta_t = \delta_{t-1} - \gamma_t P_{t-1}\{H\delta_{t-1} + \varepsilon_t + r_t\}$ by $t + t_0$ gives

$$(t+t_0)\,\delta_t = (t-1+t_0)\delta_{t-1} + (I - P_{t-1}H)\delta_{t-1} - P_{t-1}\varepsilon_t - P_{t-1}r_t, \tag{29}$$

which is (22) with t replaced by $t + t_0$ throughout. Summing from $t = 1$ to T now telescopes to $(T+t_0)\delta_T = t_0\delta_0 + \sum_{t\le T}\{\cdot\}$ instead of $T\delta_T = \sum_{t\le T}\{\cdot\}$, so dividing by $T + t_0$,

$$\delta_T = \frac{t_0\,\delta_0}{T+t_0} - \frac{1}{T+t_0} H^{-1} \sum_{t\le T} \varepsilon_t + \frac{T}{T+t_0}\big\{(\mathrm{A}) + (\mathrm{B}) - (\mathrm{C})\big\},$$

with (A), (B), (C) exactly as in the unshifted proof. Two differences from (22), and both are negligible. The new first term is $O(t_0/T) = O(1/T) = o(T^{-1/2})$ because t_0 is a finite constant, so it does not enter the limit. And $T/(T+t_0) = 1 + O(1/T)$ and $(T+t_0)^{-1} = T^{-1}(1+O(1/T))$, so every term of the unshifted decomposition is multiplied by $1 + O(1/T)$; since each of them is $O_p(T^{-1/2})$ or smaller, the perturbation is $O_p(T^{-3/2})$. The remaining estimates — the bounds on (A), (B) and (C), the martingale central limit theorem applied to $T^{-1/2}\sum_{t\le T}\varepsilon_t$, and the Gordin decomposition — do not involve γ_t at all. Hence the rate, the block-freeness statement and the limit law are unchanged. So the shift variant has the cleaner theory of the two; it is the empirical comparison, not the theory, that made us prefer the safeguard. □

Two remarks on what this variant does and does not do. First, (28) does not make (7) hold for *every* t: the maximum in (28) is over warm-up blocks, so a later block with an unusually badly conditioned $\widehat{H}_t$ can still violate (7) at that single t. What the shift buys is that the violation happens with $\gamma_t \le (1+t_0)^{-1}$ rather than $\gamma_1 = 1$, so a single bad block moves θ by a bounded multiple of its distance to $\theta^\star$ instead of by a large one, and the recursion recovers. Second, the proposition is about asymptotics; the finite-sample effect of the shift is a trade, because a larger t_0 also slows the decay of the initial condition, which decays like $t_0/(T+t_0)$ rather than $1/T$. This is why we set t_0 from (7) — the smallest value that restores contraction — rather than by taking a comfortable margin, and why Table D.4 reports the sensitivity.

E.2 Where the contraction condition comes from, and why it fails

Linearising (6) at $\theta^\star$ and ignoring Π_t gives the error recursion

$$\theta_t - \theta^\star = \big(I - \gamma_t \bar{H}_{t-1}^{-1}\widehat{H}_t\big)(\theta_{t-1} - \theta^\star) - \gamma_t \bar{H}_{t-1}^{-1}\varepsilon_t + o(\|\theta_{t-1} - \theta^\star\|). \tag{30}$$

Since $\bar{H}^{-1}\widehat{H}_t$ is a product of two positive semidefinite matrices its eigenvalues are real and non-negative, so the spectral radius of the linear map in (30) is below one exactly when the first inequality of (7) holds; the second is sufficient because $\lambda_{\max} \le \|\cdot\|_{\mathrm{op}}$. We work with the operator norm because it is what we can compute and because it is the conservative choice, and we say "spectral radius" rather than "contraction" deliberately: for a non-symmetric map a spectral radius below one bounds the *asymptotic* growth of powers, not the norm of a single step. What (7) rules out is the regime we actually observe, in which $\gamma_t\lambda_{\max} \gg 2$ and a single step multiplies the error by a large factor.

The per-block maxima of $\|H^{-1}\widehat{H}_t\|_{\mathrm{op}}$ at $b = 20$ are 55 on the autoregressive design and 89 on the regime-switching design, so the autoregressive designs are lucky rather than safe. Taking a full-length step under those numbers makes the first iterates overshoot by orders of magnitude, and because $\gamma_t = 1/t$ undoes an excursion only at rate $1/t$, at realistic N an overshoot in the first few blocks can survive to the end of the stream. That is the mechanism behind the divergence reported in Table D.4(i): without the safeguard, two of six replicates on the regime-switching design diverge outright, and none with it. This is not a defect of the base method — it is created by *blocking* the gradient, which is the one thing we add.

E.3 What Proposition 1 does not prove

The obstruction is real and worth naming: the factor $\min\{1, c_D(1 + \|\theta_{t-1}\|)/\|v\|\}$ depends on the current block's gradient, so it is $\mathcal{F}_t$- rather than $\mathcal{F}_{t-1}$-measurable and correlates with the noise, which breaks the martingale structure the proofs use. A complete treatment is available in the stochastic-approximation literature on expanding truncations [4, 52], at the price of machinery we do not need here. Appendix E gives the two concrete routes that would close the gap. A unit test asserts the measured activation counts.

E.4 The alternative we rejected, and why

The contraction condition (7) suggests a different remedy: shift the schedule to $\gamma_t = (t + t_0)^{-1}$ with $t_0 = \frac{1}{2} \max_{t \le T_{\mathrm{w}}} \|\bar{H}_{T_{\mathrm{w}}}^{-1}\widehat{H}_t\|_{\mathrm{op}}$ measured on the warm-up blocks, which makes (7) hold at $t = 1$ by construction and has a cleaner proof (it is a finite deterministic reindexing; see Appendix E). We implemented it and it is worse, for a reason worth stating: it slows *every* step, whereas the overshoot needs correcting on only a handful. On the regime-switching design at $N = 200{,}000$ the shift leaves the relative variance at 3.08 against 1.41 for the safeguard, and on a stream short enough that t_0 is comparable to the number of steps — which is the case for our real-data segments — it freezes the estimator altogether. Both variants are in the code and both are reported (Table D.4(i)); the safeguard is the default.

E.5 Sizing the curvature warm-up

It matters: on a 4000-observation stream with $d = 11$, spending $c_{\mathrm{w}}d = 2200$ observations on curvature leaves 45 blocked steps and a relative variance of 2.51, whereas capping at 10% leaves 90 steps and a relative variance of 1.28 (Table D.4(j)). Our real-data segments are of exactly that length.

E.6 Cost accounting for the one-off terms

The warm-up performs $c_{\mathrm{w}}d$ rank-one updates at $O(d^2)$ each, which is the unavoidable cost of producing a $d \times d$ curvature estimate at all and is of the same order as a single matrix factorisation. The step-size shift of Appendix E.1 adds a further one-off $O(c_{\mathrm{w}}(d^2 + db)d/b)$, which is 4–5% of the total operation count at the settings of Table D.5 and vanishes relative to the streaming pass as N grows. Table D.5 reports the crossover $N \gtrsim c_{\mathrm{w}}d^2$ explicitly rather than hiding it in the constant.

E.7 Dependence in the covariates is not enough

Remark 7 (Dependence in the covariates is not enough)**.** If the conditional law of y given x is correctly specified and its innovations are independent over time, then (g_i) is a martingale difference sequence and $S = \Gamma_0$ *exactly*, however persistent x is. A genuine long-run-variance problem needs the *score* to be serially correlated: serially correlated errors, a latent dynamic factor, or omitted dynamics. All our designs therefore couple dependent covariates with a dependent error or latent process, and we report the resulting κ_j explicitly. Covariate dependence still matters for the effective sample size of the curvature estimate, which is what Proposition 10 is about.

E.8 Misspecification and where dependence has to live

In more detail, nothing in the analysis requires the model to be correctly specified: all that is used is that $\theta^\star$ solves $\nabla F(\theta^\star) = 0$ for the *working* loss, so under misspecification $\theta^\star$ is the pseudo-true parameter and $H^{-1}SH^{-1}$ the corresponding long-run sandwich [21, 95]. That matters here, because omitted dynamics are one of the commonest ways for a score to become serially correlated — and correlated *scores*, not merely correlated covariates, are what the problem requires. If the conditional law of y given x is correctly specified with innovations independent over time, then (g_i) is a martingale difference sequence and $S = \Gamma_0$ *exactly*, however persistent x is (Remark 7). All our designs therefore couple dependent covariates with a dependent error or latent process, and we report the resulting κ_j explicitly.

This matters in practice, because omitted dynamics are one of the commonest ways for a score to become serially correlated [21, 95].

E.9 Why the ridge scale is frozen

With $\varsigma_\star$ fixed, positive and finite almost surely under Assumption 3, Lemma 2 holds unconditionally — for every t, with no assumption on the iterates. The scale exists only to make c_ι dimensionless; setting $\varsigma_\star = 1$ recovers an unscaled ridge, and Table D.4(d) shows the results are the same.

F Extended related work

Streaming second-order methods with $O(dN)$ cost. Inversion-free stochastic Newton methods maintain a running inverse curvature estimate by rank-one Sherman–Morrison updates [83] and have been developed for logistic regression [6], Gauss–Newton problems [12], the geometric median [33], and general conditioned stochastic gradient descent [9, 55]; Chau et al. [13] survey the inversion-free idea more broadly, and Byrd et al. [11], Moritz et al. [65], Wang et al. [92] give the stochastic quasi-Newton and L-BFGS counterparts. Our starting point is Godichon-Baggioni and Werge [34], who thin the rank-one updates with a Bernoulli variable and take mini-batches of size d to reach $O(dN)$ total cost with asymptotic efficiency, and Godichon-Baggioni et al. [36], who reach the same budget by masking Hessian columns. **The $O(dN)$ budget and the idea of skipping curvature updates are theirs, not ours**; Chen et al. [16] independently thin Hessian *entries* with probability $O(1/d^2)$ in a zeroth-order method. All of this work assumes independent observations. Most of it proves central limit theorems and stops there, but not all: for averaged first-order algorithms under independent data Godichon-Baggioni [32] gives a recursive online estimator of the asymptotic variance with almost sure and L^2 rates, and Chen et al. [15], Godichon-Baggioni and Lu [33], Zhu et al. [102] supply plug-in and batch-means constructions. **Online estimation of the *instantaneous* sandwich under independent data is therefore not new either**; what is missing is the long-run object under dependence, and the streaming second-order setting. Our contribution on this axis is (i) the analysis of the rank-one recursion when the accumulated terms are dependent, (ii) replacing Bernoulli thinning by a deterministic gap, which is strictly better at matched cost (Proposition 10), and (iii) turning the central limit theorem into usable intervals.

Online inference for stochastic approximation. Under independent sampling this is a mature area. Chen et al. [15] give the plug-in and increasing-batch-means variance estimators for averaged stochastic gradient descent; Zhu et al. [102] make batch means fully recursive with growing overlapping windows and obtain $\|\widehat{\Sigma} - \Sigma\| = O(N^{-1/8})$; Singh et al. [84] show equal batch sizes are preferable and apply the flat-top bias correction; Ni and Huo [69] improve the batch-means rate to $N^{-1/6}$ and establish the minimax rate $\Theta(N^{-(1-a)/2})$ for *Hessian-free* covariance estimation from the iterate path; Wei et al. [93] give a fully online de-biased estimator that needs no second-order information and reaches $N^{(a-1)/2}\sqrt{\log N}$, i.e. that minimax rate up to a logarithm, so the Hessian-free class is close to saturated. Fang [26], Fang et al. [27] bootstrap; Su and Zhu [85] split the trajectory; Chen et al. [16], Lee et al. [54], Li et al. [58] studentise by a random scaling matrix and estimate nothing. Leung and Chan [56] and Wu [99] design recursive long-run variance estimators directly. **None of these is a claim we duplicate**; what we take from them is the batch-means template and the flat-top correction, and what we add is the block-*gradient* construction and its consequence for the rate (Remark 1).

Two recent papers give online covariance estimation for streaming *second-order* methods: Kuang et al. [50] for a sketched Newton method and Wang et al. [91] for Nesterov-accelerated sketching. Both are under independent data, both target the instantaneous sandwich, and both require the gradient noise to be a martingale difference with respect to the algorithm's filtration (50, Assumption 3.2) — exactly the condition that temporal dependence destroys. Kuang et al. [50] remark that their analysis extends to conditioned stochastic gradient methods generally, which would include an independent-data version of our algorithm; the martingale assumption is what keeps that remark from covering the present setting.

Stochastic approximation on dependent data. Convergence of stochastic approximation under Markovian or mixing noise is classical [4, 5, 7, 29, 44, 52], and modern rates are available for Markov-chain gradient descent [1, 20, 23, 24, 49, 66, 86]. Godichon-Baggioni et al. [35] analyse streaming stochastic gradient methods with time-dependent and biased gradients, and identify blocked or growing mini-batches as the device that breaks dependence; they give L^2 rates but no central limit theorem and no inference. **Blocked gradients as a dependence-breaking device are theirs.**

On the inference side, Liu et al. [60] give a multiplier bootstrap for ϕ-mixing streams using alternating growing blocks, and establish the long-run sandwich limit and, qualitatively, the failure of dependence-blind intervals; Li et al. [57] give a functional central limit theorem and the semiparametric efficiency bound $H^{-1}SH^{-1}$ under Markovian sampling, with random scaling for inference; Samsonov et al. [82] give Berry–Esseen bounds and an overlapping-batch-means variance for *linear* stochastic approximation with Markovian noise; Roy and Balasubramanian [81] give an online batch-means estimator of the Markovian long-run sandwich; Li et al. [58] treat controlled Markov chains with random scaling. **The long-run sandwich limit, the efficiency bound, and the existence of online long-run covariance estimators under dependence are all prior art** — for first-order methods. Every one of these works is first-order; none maintains a curvature inverse, and the two that estimate a long-run covariance do so from the iterate path, at $O(N^{-1/8})$, with block lengths that must grow like $N^{3/4}$ (82, Prop. 2, Cor. 2). Remark 1 explains why block gradients escape that constraint, and Theorem 9 quantifies the gain.

Long-run variance estimation. The statistical device is heteroskedasticity- and autocorrelation-consistent covariance estimation [2, 3, 67, 68, 94, 96], its fixed-b alternative [47, 48, 87], and the batch-means literature from simulation output analysis and Markov chain Monte Carlo [17, 28, 30, 31, 43, 90]. The two-scale correction we use is the flat-top lag window of Politis and Romano [73], refined by Politis [72] and recast as "lugsail" lag windows by Vats and Flegal [89]; Singh et al. [84] bring it to stochastic gradient descent under independent data, where their Eqs. (15) and (17) coincide with our (11) and their Corollary 1 shows the convergence *rate* is unchanged. **We do not claim the correction.** We claim the bias order it attains at block-gradient resolution under geometric mixing, which is the property that makes

$b \asymp \log N$ admissible, and the streaming implementation. Block bootstrap and subsampling [51, 53, 74] are the resampling counterparts.

Asymptotics for dependent M-estimation. The population statements we use are standard: sandwich asymptotics for M-estimators and generalised method of moments [38, 41, 88, 95], misspecified models with dependent observations [21], marginal models for correlated responses [59, 101] — the setting our logistic design imitates — and central limit theorems for mixing sequences [10, 19, 22, 39, 42, 61, 77, 80, 97]. The technical inputs to our proofs are Davydov's covariance inequality [18, 76], the Bernstein-type inequality for weakly dependent sequences of Merlevède et al. [62], the almost sure behaviour of stochastic algorithms [25, 64, 70, 71, 79], and the functional central limit theorem of Herrndorf [39].

G Extended discussion of the limitations

We list the boundaries of the result plainly, because several of them are load-bearing.

Memory is $O(d^2)$, not $O(d)$. The $O(dN)$ claim is about *time*. The method stores a $d \times d$ inverse curvature estimate and two $d \times d$ covariance accumulators, so memory is $O(d^2)$; this is inherited from the independent-data method we extend and is not something we improve. For d in the thousands this is the binding constraint, and a sketched or diagonal-plus-low-rank curvature representation—as in Godichon-Baggioni et al. [36], Kuang et al. [50], Wang et al. [91]—would be the natural remedy. Whether the block-gradient long-run covariance estimator survives such a compression is open.

Geometric mixing is a real restriction, and the block length has to respect it. Theorem 9 buys $b \asymp \log N$ from the geometric decay in Lemma 11. Under algebraic mixing the flat-top bias is polynomial rather than exponential and the admissible b grows accordingly; long-memory streams are outside the theory altogether. This is not merely formal: our air-quality stream has a residual lag-one autocorrelation of 0.91 and a median variance-inflation factor of 9.9, and there the block length needed for calibrated intervals is an order of magnitude larger than for the traffic stream (Table D.2). The free diagnostic $r_j = |u_j^\top(\widehat{S}^{\mathrm{FT}} - \widehat{S}^{\mathrm{BM}})u_j| / (u_j^\top \widehat{S}^{\mathrm{FT}} u_j)$ detects the problem—it estimates the relative block-length bias of plain batch means—but it does not remove it. It could be turned into an automatic rule by maintaining covariance accumulators at several block scales $b, 2b, 4b, \ldots$ at once (one extra $O(d^2)$ update per block per scale, all from the same block gradients) and reporting from the smallest scale at which r_j falls below a tolerance; we did not implement this, and we do not claim an automatic rule for b.

$\widehat{S}^{\mathrm{FT}}$ need not be positive semidefinite. This is intrinsic to flat-top and lugsail estimators [84, Remark 5]. Marginal intervals need only positive scalars $u_j^\top \widehat{S}^{\mathrm{FT}} u_j$. The documented fallback to $\widehat{S}^{\mathrm{BM}}$ never triggered in any simulation, but it does trigger on the real streams — at a rate of 0.0% of coordinate-replicate pairs — which is what one should expect, since there the two designs are ill-conditioned and the lag-one correction is comparable to the batch-means term. We report the rate rather than claiming it away, and a user who needs the full matrix (for a joint test, say) must project it.

The curvature form is required. Everything rests on $\nabla^2 F(\theta) = \mathbb{E}[\alpha\, \Psi\Psi^\top]$, which holds for generalised linear models, ridge and softmax regression, the geometric median and Gauss–Newton problems, but not for a general objective. Without it there is no rank-one recursion and no inversion-free update.

Warm-up is a real requirement, not an implementation detail. The $1/t$ schedule with a poor preconditioner leaves a slowly decaying component in the error. On the autoregressive design the effect is now mild — with a negligible warm-up, coverage of nominal 95% intervals is 0.928 rather than 0.949 — because the step-length safeguard absorbs most of what a bad preconditioner does. On the regime-switching design it is not mild: the relative variance is 2.19 at $c_{\mathrm{w}} = 50$ against 1.41 at $c_{\mathrm{w}} = 200$ (Table D.4(c′)). We give a scaling rule ($c_{\mathrm{w}} d$ curvature updates, with $c_{\mathrm{w}} = 200$ throughout) and show insensitivity above it, but this is a tuning constant, and the asymptotic theory is silent about it. What sets its size is not the number of observations but the number of effectively distinct design points they span: on a persistent regime-switching stream, $50d$ consecutive observations cover only about thirty regime visits, and the curvature estimate is still rank-starved (Table D.4(c′)). The warm-up also has a cost consequence we state rather than bury: it performs $c_{\mathrm{w}} d$ rank-one updates at $O(d^2)$ each, so the total time is $O(dN + c_{\mathrm{w}} d^3)$ and the advertised $O(dN)$ only bites once $N \gtrsim c_{\mathrm{w}} d^2$. With $c_{\mathrm{w}} = 200$ and $d = 40$ that means $N \gtrsim 3 \times 10^5$, so at the $N = 2 \times 10^5$ of Table D.5 the warm-up dominates the streaming pass for every $d \geq 40$ we report, which is why that table separates the streaming count from the total.

Blocking creates a stability condition, and our fix is not fully proved. The contraction condition $\gamma_t \|\bar{H}^{-1} \widehat{H}_t\|_{\mathrm{op}} < 2$ is a consequence of blocking the gradient, so it is a cost of our construction and not of the method we build on. The step-length safeguard in (6) removes the failure it causes, and Proposition 1 shows that on the event where the safeguard is eventually inactive it changes no asymptotic statement. What we do *not* prove is that this event has probability one: the implication we establish runs from convergence to eventual inactivity, not back, because the safeguard factor depends on the current block's gradient and so is not predictable, which breaks the martingale structure the proofs use. Appendix E says exactly what would close the gap — an expanding-truncation construction [4, 52], or a predictable variant that caps using the previous block's amplification at an extra $O(d^2)$ per block — and we did neither. Instead we report the activation counts, which are between 2 and 7 per run and flat in N across every design and sample size in Tables D.1 and D.4. Two further caveats. The condition itself comes from the linearised recursion (30), so it is a heuristic for non-quadratic objectives, exact only in the linear-model case. And c_D is a tuning constant with a genuine failure mode: Table D.4(i) shows it is irrelevant over a factor of sixteen on the designs that do not need the safeguard, but on the one that does, $c_D = 4$ is too loose to prevent the blow-up. The asymptotic theory says nothing about it, and we have no rule for choosing it beyond the value that worked across our designs.

The central limit theorem is proved for fixed b, and we recommend growing b. Theorem 6 is unconditional at each fixed block length, but Theorem 9 takes $b \asymp \log N$, so the interval we recommend sits in the triangular-array regime where our proof needs an extra hypothesis: that the finite random constants in Lemma 13 are bounded in probability uniformly in N along the sequence b_N. We state that hypothesis in Theorem 6 rather than hiding it, and we do not verify it. Closing the gap means redoing Lemma 13 with constants explicit in b; the naive route — bounding $\mathbb{E}\|H^{-1} - P_{t-1}\|_{\mathrm{op}}\|\varepsilon_t\|$ directly — is the one the appendix shows loses a factor $T^{1/2}$, which is why Gordin's decomposition is there in the first place. Two things make us believe the composition is sound in practice rather than merely convenient: growing b makes the block scores less dependent, not more, so the mechanism the constants come from is working in our favour; and coverage is flat in b over $b \in [10, 320]$ at $N = 200{,}000$ (Table D.4(b)), which is the statement the two theorems need to compose.

Rates and what we do not prove. We give an almost sure rate and a central limit theorem, not a Berry–Esseen bound; the first-order Markovian literature has the latter for linear stochastic approximation

[82], and we do not match it. We attain the semiparametric efficiency bound of Li et al. [57] rather than establishing it. Our $\widetilde{O}(N^{-1/2})$ rate for $\widehat{S}^{\mathrm{FT}}$ is not comparable to the minimax rate $\Theta(N^{-(1-a)/2})$ for Hessian-free covariance estimation from an iterate path [69], because we use more information (block gradients and a maintained curvature estimate) and because our step exponent is $a = 1$, outside the range that bound is stated for. Within the Hessian-free class the bound is close to attained [93], so the gap we report is about the information used, not about slack in their analysis. We make no claim of optimality within our own information set: we do not know the minimax rate for estimators that may use block gradients and a running curvature estimate.

Scope of the empirical evidence. All designs are convex M-estimation with $d \leq 160$ and $N \leq 4 \times 10^5$; there is no deep-learning experiment and the theory would not support one. The dependence in the simulated designs is stationary and geometrically decaying by construction. On the real streams we do not know the truth, which is why we run two protocols; the semi-synthetic protocol has an exact target but a synthetic (autoregressive) error process, and the fully real protocol has a real error process but a target that is itself estimated—we correct the nominal level for that exactly, but the correction assumes the segments are approximately exchangeable, which non-stationarity would violate. Neither protocol alone would be convincing; together they bracket the honest answer.

One thing we deliberately do not claim. Gapping improves the curvature estimate at matched cost (Proposition 10, Figure 3), but the curvature estimate enters the limit distribution only through a rate, so the end-to-end effect on the point estimate and on coverage is second order and we could not measure it reliably. The case for gapping is that it is free, deterministic, and provably better than the randomised alternative—not that it changes the headline numbers.